SOFTWARE REQUIREMENTS SPECIFICATION

Cinema — Web .NET

Version 3.0

2026-10-28

Table 1: Document metadata

Item	Value
Version	3.0
Baseline date	2026-10-28 (Asia/Saigon)
Language	English
Evidence status	Static code/document baseline; runtime verification not performed
Paired document	`Cinema_Web_NET_SRS_VI.md`

Contents

1 Introduction 1

2 Purpose 2

3 Scope 3

4 Overall Description 4

5 Product Perspective 5

6 User Classes / Actors 6

7 Role & Permission 7

8 Business Rules 8

9 Functional Requirements 10

10 Use Case Specifications 106

11 External Interface Requirements 113

12 `UI` / Screen Requirements 114

13 `API` Requirements 130

14 Data Requirements 144

15 Business Workflows 145

16 Authentication & Authorization 146

17 Error Handling 147

18 Non-Functional Requirements 148

19 Security Requirements 150

20 Testing Requirements 151

21 Acceptance Criteria 153

22 Requirements Traceability Matrix 154

23 Constraints **194**

24 Out of Scope **195**

25 Future Enhancements **196**

26 Conclusion **197**

27 Implemented vs Requirement **198**

28 Evidence and source reconciliation **206**

List of Tables

1 Document metadata . . . i
2 5. Product Perspective . . . 5
3 6. User Classes / Actors . . . 6
4 7. Role & Permission . . . 7
5 8. Business Rules . . . 8
6 `WEB-FR-AUTH-01` — Register . . . 10
7 `WEB-FR-AUTH-02` — Sign in . . . 10
8 `WEB-FR-AUTH-03` — `JWT` claims . . . 11
9 `WEB-FR-AUTH-04` — Validate `JWT` . . . 12
10 `WEB-FR-AUTH-05` — Authorize roles . . . 13
11 `WEB-FR-AUTH-06` — Unauthenticated and forbidden responses . . . 13
12 `WEB-FR-AUTH-07` — Verify email `OTP` . . . 14
13 `WEB-FR-AUTH-08` — Repeat unverified registration . . . 15
14 `WEB-FR-AUTH-09` — Recover password . . . 16
15 `WEB-FR-AUTH-10` — Refresh and sign out . . . 16
16 `WEB-FR-AUTH-11` — Current account . . . 17
17 `WEB-FR-AUTH-12` — Lock and role-change enforcement . . . 18
18 `WEB-FR-AUTH-13` — Account field validation . . . 19
19 `WEB-FR-AUTH-14` — Authentication rate limits . . . 20
20 `WEB-FR-AUTH-15` — Find member by phone . . . 20
21 `WEB-FR-AUTH-16` — Edit profile . . . 21
22 `WEB-FR-AUTH-17` — Change current password . . . 22
23 `WEB-FR-USER-01` — List accounts . . . 23
24 `WEB-FR-USER-02` — Account details . . . 23
25 `WEB-FR-USER-03` — Create account . . . 24
26 `WEB-FR-USER-04` — Change account roles . . . 25
27 `WEB-FR-USER-05` — Lock and unlock . . . 25
28 `WEB-FR-USER-06` — List roles . . . 26
29 `WEB-FR-MOV-01` — Paged catalogue . . . 26
30 `WEB-FR-MOV-02` — Movie details . . . 27
31 `WEB-FR-MOV-03` — Title search . . . 28
32 `WEB-FR-MOV-04` — Active catalogue . . . 29
33 `WEB-FR-MOV-05` — Create movie . . . 29
34 `WEB-FR-MOV-06` — Edit and reactivate movie . . . 30
35 `WEB-FR-MOV-07` — Deactivate movie . . . 31

36 WEB-FR-MOV-08 — Open trailer . . . 31
37 WEB-FR-MOV-09 — Age rating . . . 32
38 WEB-FR-MOV-10 — Now showing and coming soon . . . 33
39 WEB-FR-MOV-11 — Director and cast . . . 34
40 WEB-FR-MOV-12 — Movie reviews . . . 35
41 WEB-FR-ROOM-01 — Find rooms . . . 36
42 WEB-FR-ROOM-02 — Room details . . . 36
43 WEB-FR-ROOM-03 — Create room and seats . . . 37
44 WEB-FR-ROOM-04 — Rename room . . . 38
45 WEB-FR-ROOM-05 — Delete eligible room . . . 38
46 WEB-FR-SEAT-01 — Generate seats . . . 39
47 WEB-FR-SEAT-02 — Seat details . . . 40
48 WEB-FR-SEAT-03 — Read room map . . . 40
49 WEB-FR-SEAT-04 — Delete seat API . . . 41
50 WEB-FR-SEAT-05 — Change seat type . . . 42
51 WEB-FR-SEAT-06 — Paint seat types . . . 42
52 WEB-FR-SEAT-07 — Change capacity . . . 43
53 WEB-FR-SEAT-08 — Protect scheduled rooms . . . 44
54 WEB-FR-SEAT-09 — Refresh management maps . . . 44
55 WEB-FR-SHOW-01 — List showtimes . . . 45
56 WEB-FR-SHOW-02 — Create showtime . . . 46
57 WEB-FR-SHOW-03 — Showtime and prices . . . 46
58 WEB-FR-SHOW-04 — Edit showtime . . . 47
59 WEB-FR-SHOW-05 — Cancel showtime . . . 48
60 WEB-FR-SHOW-06 — Open future showtimes . . . 48
61 WEB-FR-SHOW-07 — Showtimes by movie . . . 49
62 WEB-FR-SHOW-08 — Bookable movie showtimes . . . 50
63 WEB-FR-SHOW-09 — Date range search . . . 50
64 WEB-FR-SHOW-10 — Open and close sales . . . 51
65 WEB-FR-SHOW-11 — Reject overlap . . . 51
66 WEB-FR-SHOW-12 — Showtime seat/customer map . . . 52
67 WEB-FR-SHOW-13 — Screening format . . . 53
68 WEB-FR-BOOK-01 — Hold booking . . . 54
69 WEB-FR-BOOK-02 — Booking details . . . 55
70 WEB-FR-BOOK-03 — Management booking search . . . 55
71 WEB-FR-BOOK-04 — Own ticket history . . . 56
72 WEB-FR-BOOK-05 — Confirm paid booking . . . 57

73 WEB-FR-BOOK-06 — Cancel booking . . . 58
74 WEB-FR-BOOK-07 — Occupied seat IDs . . . 58
75 WEB-FR-BOOK-08 — Concurrent seat conflict . . . 59
76 WEB-FR-BOOK-09 — Sweetbox pair . . . 60
77 WEB-FR-BOOK-10 — Expire unpaid hold . . . 61
78 WEB-FR-BOOK-11 — Refresh customer map . . . 61
79 WEB-FR-BOOK-12 — Showtime booking statistics . . . 62
80 WEB-FR-BOOK-13 — Concessions with tickets . . . 63
81 WEB-FR-BOOK-14 — Concessions administration . . . 64
82 WEB-FR-BOOK-15 — Promotion and voucher . . . 65
83 WEB-FR-BOOK-16 — Promotion administration . . . 66
84 WEB-FR-BOOK-17 — Membership points . . . 67
85 WEB-FR-BOOK-18 — Member administration . . . 68
86 WEB-FR-BOOK-19 — Share and calendar . . . 69
87 WEB-FR-PAY-01 — Create/reuse PayOS link . . . 70
88 WEB-FR-PAY-02 — Verify PayOS . . . 71
89 WEB-FR-PAY-03 — Payment details . . . 72
90 WEB-FR-PAY-04 — Own payment history . . . 72
91 WEB-FR-PAY-05 — Payments by booking . . . 73
92 WEB-FR-PAY-06 — Refund access . . . 74
93 WEB-FR-PAY-07 — Signed webhook . . . 74
94 WEB-FR-PAY-08 — Late payment . . . 75
95 WEB-FR-PAY-09 — Management payments . . . 76
96 WEB-FR-PAY-10 — Revenue summary . . . 77
97 WEB-FR-PAY-11 — Automatic payment polling . . . 78
98 WEB-FR-REF-01 — Create refund request . . . 78
99 WEB-FR-REF-02 — Management refund list . . . 79
100 WEB-FR-REF-03 — Record completed transfer . . . 80
101 WEB-FR-REF-04 — Reject refund . . . 81
102 WEB-FR-REF-05 — Reopen rejected refund . . . 81
103 WEB-FR-REF-06 — Own refund status . . . 82
104 WEB-FR-REF-07 — Unique refund per payment . . . 83
105 WEB-FR-POS-01 — Create counter sale . . . 84
106 WEB-FR-POS-02 — Counter checkout . . . 84
107 WEB-FR-POS-03 — Cash and change . . . 85
108 WEB-FR-POS-04 — Bank-transfer QR . . . 86
109 WEB-FR-POS-05 — Switch transfer to cash . . . 86

110 WEB-FR-POS-06 — Retain counter email . . . 87
111 WEB-FR-POS-07 — Print and admit . . . 88
112 WEB-FR-POS-08 — Reopen paid ticket . . . 89
113 WEB-FR-CHK-01 — Ticket QR . . . 89
114 WEB-FR-CHK-02 — Lookup ticket code . . . 90
115 WEB-FR-CHK-03 — Check-in validation . . . 91
116 WEB-FR-CHK-04 — Camera/manual scanner . . . 91
117 WEB-FR-CHK-05 — Display check-in time . . . 92
118 WEB-FR-NOTI-01 — Notification history . . . 93
119 WEB-FR-NOTI-02 — Compose booking message . . . 94
120 WEB-FR-NOTI-03 — Send/retry message . . . 94
121 WEB-FR-NOTI-04 — Notification details . . . 95
122 WEB-FR-NOTI-05 — Delete notification API . . . 96
123 WEB-FR-NOTI-06 — Event emails . . . 96
124 WEB-FR-NOTI-07 — OTP and welcome email . . . 97
125 WEB-FR-NOTI-08 — Showtime reminder . . . 98
126 WEB-FR-NOTI-09 — Read/unread notifications . . . 99
127 WEB-FR-DASH-01 — Today totals . . . 100
128 WEB-FR-DASH-02 — Live showtime board . . . 100
129 WEB-FR-DASH-03 — Staff attention queue . . . 101
130 WEB-FR-DASH-04 — Admin monthly overview . . . 102
131 WEB-FR-NAV-01 — Route guards and recovery . . . 103
132 WEB-FR-STATE-01 — Loading, error and empty states . . . 103
133 WEB-FR-MEDIA-01 — Platform-specific movie media . . . 104
134 UC-WEB-01 — Authentication and account session . . . 106
135 UC-WEB-02 — Account administration . . . 106
136 UC-WEB-03 — Movie catalogue and reviews . . . 107
137 UC-WEB-04 — Room management . . . 107
138 UC-WEB-05 — Seat map and availability . . . 108
139 UC-WEB-06 — Showtimes and prices . . . 108
140 UC-WEB-07 — Booking, concessions, promotions and loyalty . . . 109
141 UC-WEB-08 — Online payment and reconciliation . . . 109
142 UC-WEB-09 — Cancellation and manual refund . . . 110
143 UC-WEB-10 — Counter ticket sale . . . 110
144 UC-WEB-11 — Ticket lookup and check-in . . . 111
145 UC-WEB-12 — Notifications and transactional email . . . 111
146 UC-WEB-13 — Operations dashboards . . . 112

147 11. External Interface Requirements . . . 113
148 WEB-SCR-LOGIN — Sign in . . . 114
149 WEB-SCR-REGISTER — Registration . . . 114
150 WEB-SCR-VERIFY — Verify email . . . 115
151 WEB-SCR-FORGOT — Forgot password . . . 115
152 WEB-SCR-RESET — Reset password . . . 115
153 WEB-SCR-HOME — Customer home . . . 116
154 WEB-SCR-MOVIE — Movie details . . . 116
155 WEB-SCR-SEATS — Seat selection . . . 117
156 WEB-SCR-SUMMARY — Booking summary . . . 117
157 WEB-SCR-PAYMENT — Payment . . . 117
158 WEB-SCR-RESULT — Payment result . . . 118
159 WEB-SCR-TICKETS — My tickets . . . 118
160 WEB-SCR-TICKET — Ticket details . . . 119
161 WEB-SCR-PAYHISTORY — Payment history . . . 119
162 WEB-SCR-NOTIFICATIONS — Notifications . . . 119
163 WEB-SCR-NOTIFICATION — Notification details . . . 120
164 WEB-SCR-PROFILE — Account and membership . . . 120
165 WEB-SCR-CHECKIN — Ticket check-in . . . 121
166 WEB-SCR-STAFF-DASH — Staff dashboard . . . 121
167 WEB-SCR-ADMIN-DASH — Admin dashboard . . . 121
168 WEB-SCR-OPERATIONS — Operations workspace . . . 122
169 WEB-SCR-COUNTER — Counter booking . . . 122
170 WEB-SCR-CHECKOUT — Counter checkout . . . 123
171 WEB-SCR-RECORD-BOOKINGS — Booking record . . . 123
172 WEB-SCR-RECORD-PAYMENTS — Payment record . . . 123
173 WEB-SCR-RECORD-REFUNDS — Refund record . . . 124
174 WEB-SCR-RECORD-SHOWTIMES — Showtime record . . . 124
175 WEB-SCR-RECORD-ROOMS — Room record . . . 125
176 WEB-SCR-SHOW-EDITOR — Showtime editor . . . 125
177 WEB-SCR-COMPOSE — Compose booking email . . . 125
178 WEB-SCR-ADMIN-MOVIES — Movies . . . 126
179 WEB-SCR-ADMIN-ROOMS — Rooms . . . 126
180 WEB-SCR-ADMIN-SNACKS — Snacks . . . 127
181 WEB-SCR-ADMIN-PROMOTIONS — Promotions . . . 127
182 WEB-SCR-ADMIN-MEMBERS — Members . . . 127
183 WEB-SCR-ADMIN-USERS — Accounts . . . 128

184 WEB-SCR-USER-EDITOR — Account create/details . . . 128
185 WEB-SCR-MGT-PROFILE — Management profile . . . 129
186 WEB-SCR-STAFF-ONLY — Management access notice . . . 129
187 13. API Requirements . . . 130
188 Request field definitions . . . 140
189 14. Data Requirements . . . 144
190 15. Business Workflows . . . 145
191 State contract . . . 145
192 17. Error Handling . . . 147
193 18. Non-Functional Requirements . . . 148
194 19. Security Requirements . . . 150
195 20. Testing Requirements . . . 151
196 Existing test evidence . . . 151
197 21. Acceptance Criteria . . . 153
198 22. Requirements Traceability Matrix . . . 154
199 23. Constraints . . . 194
200 25. Future Enhancements . . . 196
201 27. Implemented vs Requirement . . . 198
202 27. Implemented vs Requirement . . . 205
203 28. Evidence and source reconciliation . . . 206

1. Introduction

CosmoQ Cinema is one cinema booking and management system with multiple clients. This course-specific SRS refines consolidated SRS 2.0 using the current checkout; it does not define a second independent product. Executable routes/controllers/services/models take precedence over dated prose. SRS 1.0 is historical intent.

H = mandatory; M = important; L = optional. Implemented means source exists, not that integration tests passed. Each EN/VI pair uses identical IDs, scope, priority and criteria; neither language is a summary.

2. Purpose

Provide the contract for development, test-case design, acceptance and course defence. Each `FR` identifies actors, conditions, trigger, flow, exceptions, result and acceptance criteria, and traces to use cases, screens, APIs and source/test evidence.

3. Scope

Include customer browser screens, Staff/Admin browser workspace and the shared ASP.NET Core 8 backend for authentication, catalogue, scheduling, bookings, payments/refunds and notifications. Customer web is included because main.dart is built for browsers as well as native devices.

Technology boundary: no MVC .cshtml, Razor/Blazor .razor or separate .NET UI project is present. “Web .NET” names the course scope: ASP.NET Core REST plus its existing Flutter Web clients. A mandatory .NET-rendered UI is [TO BE CONFIRMED], not implemented evidence.

4. Overall Description

```
Flutter main.dart (customer web) + main_admin.dart (admin web)
                | REST/JSON + JWT
        ApiGateway / YARP :5000
          |           |             |
     AuthService  MovieService  BookingService
       :5100        :5168          :5063
          |           |             |
       PostgreSQL (6 databases) + RabbitMQ/outbox
                                  | PayOS + SMTP
```

Three business services replace the initial six-service design: `AuthService` owns identity; `MovieService` owns movie/room/seat/showtime; `BookingService` owns bookings, concessions, promotions, loyalty, reviews, payment/refund and notifications. Cross-database references are logical IDs, not foreign keys.

5. Product Perspective

Table 2: 5. Product Perspective

Area	Responsibility
Shared core	Identity/`RBAC`/ownership; pricing; seat concurrency; financial states; `PayOS` verification; refunds; events and data integrity
Web workspace	Counter/cash/printing; movie/room/showtime/account/menu/promotion/member administration; refund settlement; operations/revenue
Native behaviour	External trailer, device share sheet, phone camera check-in, emulator/`LAN` gateway address and payment verification on resume
Shared presentation	Customer web/native share Dart widgets and repositories; preview/printing is platform conditional

6. User Classes / Actors

Table 3: 6. User Classes / Actors

Actor	Responsibility
Guest	Public catalogue/showtimes; start account flows
Customer	Own profile, tickets, payment, refund state, notifications, reviews and points
Staff/Admin	Staff operate sales/schedule/admission/refund; Admin also controls master data/users/revenue
Walk-in guest	Counter-sale beneficiary through staff, not a signed-in mobile actor
`PayOS` / `SMTP` / workers	External settlement/messages; expire holds and exchange committed events

7. Role & Permission

Table 4: 7. Role & Permission

Operation	Guest	Customer	Staff	Admin
Browse public movies/showtimes	✓	✓	✓	✓
Own customer booking/payment/ profile/review	—	✓	Role landing	Role landing
Lookup/check in tickets	—	—	✓	✓
Counter/cash, schedules, seat types, refund settlement	—	—	✓	✓
Movies, rooms, accounts, menu, promotions, members, revenue	—	—	—	✓

`UI` guards control navigation; controller policies and record-ownership checks are authoritative. Accounts may hold multiple roles. Customer accounts in the admin entry see a customer-site notice; management accounts in the customer entry see management guidance/check-in.

8. Business Rules

Table 5: 8. Business Rules

ID	Rule
WEB-BR-01	Customer accounts must verify their email with an OTP before signing in. Accounts created by an Admin need no OTP.
WEB-BR-02	Every protected API uses JWT; both the gateway and the service check it.
WEB-BR-03	Access depends on the role; Customers cannot use management APIs.
WEB-BR-04	Each seat belongs to one room and its position (row + number) is unique in that room.
WEB-BR-05	A showtime belongs to one movie and one room, and starts before it ends.
WEB-BR-06	A room never has two showtimes at overlapping times (409).
WEB-BR-07	A seat of a showtime belongs to at most one active booking (held or sold).
WEB-BR-08	A booking has one showtime and its seats, and belongs to a customer account, except walk-in tickets sold at the counter.
WEB-BR-09	A payment belongs to a booking; booking/payment/refund transitions must follow the state contract in section 15.
WEB-BR-10	A PayOS webhook must carry a valid HMAC-SHA256 signature before any of its data is used.
WEB-BR-11	Events are written to the outbox in the same transaction as the data and are sent only after commit.
WEB-BR-12	Calls to other services or PayOS use limited retries and a circuit breaker.
WEB-BR-13	An email failure or a stopped RabbitMQ never breaks a booking or payment.
WEB-BR-14	Seats are held for 10 minutes; an unpaid booking then becomes EXPIRED and its seats are released.
WEB-BR-15	Seat price: the showtime price is the standard seat price; VIP adds 20% and sweetbox 30% per seat; the surcharge is rounded to 1,000 VND. The server computes prices and never takes them from the client.
WEB-BR-16	Sweetbox seats are always booked, changed and sold as a pair.
WEB-BR-17	A customer can cancel a paid ticket for a 100% refund until 2 hours before the show; later they must contact the cinema. Staff and Admin can cancel at any time.
WEB-BR-18	A payment has at most one refund request. Refunds are paid by manual bank transfer and recorded with the transfer reference.
WEB-BR-19	When a showtime is cancelled, all its active bookings are cancelled; paid ones get a refund request.
WEB-BR-20	Late payment (booking already expired): if the seats are still free the ticket is confirmed; if they were sold to someone else, a 100% refund request is created.
WEB-BR-21	Only CONFIRMED tickets can be checked in, once each, from 3 hours before to 2 hours after the start time.
WEB-BR-22	Cash taken must cover the total. If the booking has an unpaid bank-transfer QR, its PayOS link is cancelled before the cash is recorded; if the customer already transferred, cash is refused.
WEB-BR-23	Cash and bank transfers both create payment records, so both count in revenue and refunds in the same way.
WEB-BR-24	Locking an account or changing its roles takes effect on the next request, even while the old token is still valid.
WEB-BR-25	Forgot-password returns 200 regardless of email existence; incorrect login email/password share a message. Registration/resend may reveal account state, so system-wide non-enumeration is not claimed.
WEB-BR-26	Stopping a movie only sets it INACTIVE; nothing is deleted and existing tickets stay. A room with showtimes cannot be deleted and its seat map cannot be changed.

Continued on next page

Table 5 — continued

ID	Rule
WEB-BR-27	A promotion reduces tickets only, up to 90%, rounded down to 1,000 VND; one best automatic promotion or an entered eligible code applies. Count usage on successful payment.
WEB-BR-28	Point value is 1,000 VND; redeem at most 50% after promotion; MEMBER/VIP/VVIP spend thresholds are 0/2,000,000/5,000,000 VND and earn rates 5/7/10%. Reverse/release point effects according to booking cancellation/expiry/refund.
WEB-BR-29	A booking has at most ten distinct active concession items, quantity 1–20 each; persist purchase-time name and unit price, and include them in payment and refund totals.
WEB-BR-30	Format price adds 0/20,000/40,000 VND for 2D/3D/IMAX before type surcharges; NORMAL/VIP/SWEETBOX prices are returned by the server.

External-operator rules specify shared-backend effects, not additional phone screens. Time limits are current defaults and may be overridden by deployment configuration.

9. Functional Requirements

AC-01 verifies the exact stated behaviour; AC-02 verifies applicable rejection/permission paths. Apply only actions specified by the individual FR; the wider use case supplies context. Criteria are specifications, not recorded test results.

9.1 WEB-FR-AUTH-01 — Register

Table 6: WEB-FR-AUTH-01 — Register

Field	Specification
Name	Register
Description	Customers register with email, password, full name and an optional phone number. The system sends a verification OTP and issues no token yet.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Register” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Customers register with email, password, full name and an optional phone number. The system sends a verification OTP and issues no token yet. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	An unverified registration and OTP delivery outcome are recorded; no access/refresh tokens are issued before verification.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-01-AC-01** — Given the preconditions/trigger, verify: Customers register with email, password, full name and an optional phone number. The system sends a verification OTP and issues no token yet. **WEB-FR-AUTH-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/register_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-REGISTER; WEB-API-033

9.2 WEB-FR-AUTH-02 — Sign in

Table 7: WEB-FR-AUTH-02 — Sign in

Field	Specification
Name	Sign in
Description	Sign in with email and password to get an access token (JWT, 60 minutes) and a refresh token (7 days). An unverified email gets 403 EMAIL_NOT_VERIFIED.

Continued on next page

Table 7 — continued

Field	Specification
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/`AuthService` are configured; no prior signed-in session is required for registration/login/`OTP`/recovery. `SMTP` is required when sending a code.
Trigger	Invoke “Sign in” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Sign in with email and password to get an access token (`JWT`, 60 minutes) and a refresh token (7 days). An unverified email gets `403 EMAIL_NOT_VERIFIED`. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	Successful verified authentication stores the returned user/tokens and selects the allowed role landing; failure grants no session.
Priority	H
Acceptance criteria	**`WEB-FR-AUTH-02-AC-01`** — Given the preconditions/trigger, verify: Sign in with email and password to get an access token (`JWT`, 60 minutes) and a refresh token (7 days). An unverified email gets `403 EMAIL_NOT_VERIFIED`. **`WEB-FR-AUTH-02-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs`; `Cinema_Mobile/lib/features/auth/screens/login_screen.dart`; `Cinema_BE/AuthService/AuthService/Service/AuthService.cs`; `Cinema_BE/ApiGateway/ApiGateway/Program.cs`
Business rules	`WEB-BR-01`, `WEB-BR-02`, `WEB-BR-03`, `WEB-BR-24`, `WEB-BR-25`
Use case / screens / APIs	`UC-WEB-01`; `WEB-SCR-LOGIN`; `WEB-API-027`

9.3 `WEB-FR-AUTH-03` — `JWT` claims

Table 8: `WEB-FR-AUTH-03` — `JWT` claims

Field	Specification
Name	`JWT` claims
Description	The `JWT` carries the `userId`, email and roles.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “`JWT` claims” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The `JWT` carries the `userId`, email and roles. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H

Continued on next page

Table 8 — continued

Field	Specification
Acceptance criteria	**WEB-FR-AUTH-03-AC-01** — Given the preconditions/trigger, verify: The JWT carries the user Id, email and roles. **WEB-FR-AUTH-03-AC-02** — Exercise applicable invalid/unauthorized/ conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_ Mobile/lib/features/auth/screens/login_screen.dart; Cinema_BE/AuthService/ AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/ Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-LOGIN; WEB-API-027, WEB-API-032

9.4 WEB-FR-AUTH-04 — Validate JWT

Table 9: WEB-FR-AUTH-04 — Validate JWT

Field	Specification
Name	Validate JWT
Description	The gateway and services check the JWT signature, expiry, issuer and audience before protected APIs.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs SMTP; protected operations require a valid account session.
Trigger	Invoke “Validate JWT” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The gateway and services check the JWT signature, expiry, issuer and audience before protected APIs. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/ payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-04-AC-01** — Given the preconditions/trigger, verify: The gateway and services check the JWT signature, expiry, issuer and audience before protected APIs. **WEB-FR-AUTH-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_BE/ AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/Api Gateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; ; WEB-API-029

9.5 WEB-FR-AUTH-05 — Authorize roles

Table 10: WEB-FR-AUTH-05 — Authorize roles

Field	Specification
Name	Authorize roles
Description	APIs are authorized by role Customer / Staff / Admin (policies `StaffOrAdmin`, `AdminOnly`, and signed-in by default).
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “Authorize roles” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: APIs are authorized by role Customer / Staff / Admin (policies `StaffOrAdmin`, `AdminOnly`, and signed-in by default). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-05-AC-01** — Given the preconditions/trigger, verify: APIs are authorized by role Customer / Staff / Admin (policies `StaffOrAdmin`, `AdminOnly`, and signed-in by default). **WEB-FR-AUTH-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs`; `Cinema_Mobile/lib/features/admin/screens/staff_only_screen.dart`; `Cinema_BE/AuthService/AuthService/Service/AuthService.cs`; `Cinema_BE/ApiGateway/ApiGateway/Program.cs`
Business rules	`WEB-BR-01`, `WEB-BR-02`, `WEB-BR-03`, `WEB-BR-24`, `WEB-BR-25`
Use case / screens / APIs	`UC-WEB-01`; `WEB-SCR-STAFF-ONLY`; `WEB-API-029`

9.6 WEB-FR-AUTH-06 — Unauthenticated and forbidden responses

Table 11: WEB-FR-AUTH-06 — Unauthenticated and forbidden responses

Field	Specification
Name	Unauthenticated and forbidden responses
Description	A missing or invalid token gets `401`; a signed-in user without the role gets `403`.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “Unauthenticated and forbidden responses” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A missing or invalid token gets `401`; a signed-in user without the role gets `403`. 3. Return/render the authoritative result and refresh affected data after success.

Continued on next page

Table 11 — continued

Field	Specification
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-06-AC-01** — Given the preconditions/trigger, verify: A missing or invalid token gets 401; a signed-in user without the role gets 403. **WEB-FR-AUTH-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/login_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-LOGIN; WEB-API-029

9.7 WEB-FR-AUTH-07 — Verify email OTP

Table 12: WEB-FR-AUTH-07 — Verify email OTP

Field	Specification
Name	Verify email OTP
Description	Email verification with a 6-digit OTP: expires after 5 minutes, at most 5 attempts, 60 seconds between sends, a new code replaces the old one. Verification must include the registered password; on success the user is signed in and gets a welcome email.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Verify email OTP” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Email verification with a 6-digit OTP: expires after 5 minutes, at most 5 attempts, 60 seconds between sends, a new code replaces the old one. Verification must include the registered password; on success the user is signed in and gets a welcome email. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	Successful verified authentication stores the returned user/tokens and selects the allowed role landing; failure grants no session.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-07-AC-01** — Given the preconditions/trigger, verify: Email verification with a 6-digit OTP: expires after 5 minutes, at most 5 attempts, 60 seconds between sends, a new code replaces the old one. Verification must include the registered password; on success the user is signed in and gets a welcome email. **WEB-FR-AUTH-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 12 — continued

Field	Specification
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/verify_email_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-VERIFY; WEB-API-034, WEB-API-042

9.8 WEB-FR-AUTH-08 — Repeat unverified registration

Table 13: WEB-FR-AUTH-08 — Repeat unverified registration

Field	Specification
Name	Repeat unverified registration
Description	Registering again with an unverified email replaces the old details and sends a new code. A verified email is reported as already registered.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Repeat unverified registration” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Registering again with an unverified email replaces the old details and sends a new code. A verified email is reported as already registered. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	An unverified registration and OTP delivery outcome are recorded; no access/refresh tokens are issued before verification.
Priority	M
Acceptance criteria	**WEB-FR-AUTH-08-AC-01** — Given the preconditions/trigger, verify: Registering again with an unverified email replaces the old details and sends a new code. A verified email is reported as already registered. **WEB-FR-AUTH-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/register_screen.dart; Cinema_Mobile/lib/features/auth/screens/verify_email_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-REGISTER, WEB-SCR-VERIFY; WEB-API-033

9.9 WEB-FR-AUTH-09 — Recover password

Table 14: WEB-FR-AUTH-09 — Recover password

Field	Specification
Name	Recover password
Description	Forgot password: an OTP is emailed (the API always returns 200); setting a new password with the OTP revokes every refresh token and marks the email as verified.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Recover password” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Forgot password: an OTP is emailed (the API always returns 200); setting a new password with the OTP revokes every refresh token and marks the email as verified. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-09-AC-01** — Given the preconditions/trigger, verify: Forgot password: an OTP is emailed (the API always returns 200); setting a new password with the OTP revokes every refresh token and marks the email as verified. **WEB-FR-AUTH-09-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/forgot_password_screen.dart; Cinema_Mobile/lib/features/auth/screens/reset_password_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-FORGOT, WEB-SCR-RESET; WEB-API-026, WEB-API-035

9.10 WEB-FR-AUTH-10 — Refresh and sign out

Table 15: WEB-FR-AUTH-10 — Refresh and sign out

Field	Specification
Name	Refresh and sign out
Description	Renew the session with the refresh token (rotated; the old one stops working); signing out revokes the refresh token. The app renews automatically when the access token expires.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs SMTP; protected operations require a valid account session.
Trigger	Invoke “Refresh and sign out” or its specified backend event/worker.

Continued on next page

Table 15 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Renew the session with the refresh token (rotated; the old one stops working); signing out revokes the refresh token. The app renews automatically when the access token expires. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-AUTH-10-AC-01`** — Given the preconditions/trigger, verify: Renew the session with the refresh token (rotated; the old one stops working); signing out revokes the refresh token. The app renews automatically when the access token expires. **`WEB-FR-AUTH-10-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs`; `Cinema_Mobile/lib/features/profile/screens/profile_screen.dart`; `Cinema_BE/AuthService/AuthService/Service/AuthService.cs`; `Cinema_BE/ApiGateway/ApiGateway/Program.cs`
Business rules	`WEB-BR-01`, `WEB-BR-02`, `WEB-BR-03`, `WEB-BR-24`, `WEB-BR-25`
Use case / screens / APIs	`UC-WEB-01`; `WEB-SCR-PROFILE`, `WEB-SCR-MGT-PROFILE`; `WEB-API-028`, `WEB-API-032`

9.11 `WEB-FR-AUTH-11` — Current account

Table 16: `WEB-FR-AUTH-11` — Current account

Field	Specification
Name	Current account
Description	Show the current account (`GET /api/v1/auth/me`).
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “Current account” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show the current account (`GET /api/v1/auth/me`). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-AUTH-11-AC-01`** — Given the preconditions/trigger, verify: Show the current account (`GET /api/v1/auth/me`). **`WEB-FR-AUTH-11-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 16 — continued

Field	Specification
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/profile/screens/profile_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE; WEB-API-029

9.12 WEB-FR-AUTH-12 — Lock and role-change enforcement

Table 17: WEB-FR-AUTH-12 — Lock and role-change enforcement

Field	Specification
Name	Lock and role-change enforcement
Description	The gateway checks the account on every protected request: locked returns 401 ACCOUNT_LOCKED; changed roles return 403 ROLE_CHANGED; an unreachable AuthService returns 503. The customer app shows the reason and returns to sign-in.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs SMTP; protected operations require a valid account session.
Trigger	Invoke “Lock and role-change enforcement” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The gateway checks the account on every protected request: locked returns 401 ACCOUNT_LOCKED; changed roles return 403 ROLE_CHANGED; an unreachable AuthService returns 503. The customer app shows the reason and returns to sign-in. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-12-AC-01** — Given the preconditions/trigger, verify: The gateway checks the account on every protected request: locked returns 401 ACCOUNT_LOCKED; changed roles return 403 ROLE_CHANGED; an unreachable AuthService returns 503. The customer app shows the reason and returns to sign-in. **WEB-FR-AUTH-12-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/login_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-LOGIN; WEB-API-029

9.13 WEB-FR-AUTH-13 — Account field validation

Table 18: WEB-FR-AUTH-13 — Account field validation

Field	Specification
Name	Account field validation
Description	The app and the API check account fields the same way (registration, sign-in, forgot / reset password, accounts created by Admin): email required, at most 150 characters, a domain with a suffix of at least 2 letters, no spaces; a new password of 6–72 characters counted in UTF-8 bytes; full name required, at most 150 characters; phone optional, 8–15 digits, at most 20 characters. Mistakes return 400 with an error per field. The sign-up form asks for the password twice.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Account field validation” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The app and the API check account fields the same way (registration, sign-in, forgot / reset password, accounts created by Admin): email required, at most 150 characters, a domain with a suffix of at least 2 letters, no spaces; a new password of 6–72 characters counted in UTF-8 bytes; full name required, at most 150 characters; phone optional, 8–15 digits, at most 20 characters. Mistakes return 400 with an error per field. The sign-up form asks for the password twice. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-AUTH-13-AC-01** — Given the preconditions/trigger, verify: The app and the API check account fields the same way (registration, sign-in, forgot / reset password, accounts created by Admin): email required, at most 150 characters, a domain with a suffix of at least 2 letters, no spaces; a new password of 6–72 characters counted in UTF-8 bytes; full name required, at most 150 characters; phone optional, 8–15 digits, at most 20 characters. Mistakes return 400 with an error per field. The sign-up form asks for the password twice. **WEB-FR-AUTH-13-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs; Cinema_Mobile/lib/features/auth/screens/login_screen.dart; Cinema_Mobile/lib/features/auth/screens/register_screen.dart; Cinema_Mobile/lib/features/auth/screens/forgot_password_screen.dart; Cinema_Mobile/lib/features/auth/screens/reset_password_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-LOGIN, WEB-SCR-REGISTER, WEB-SCR-FORGOT, WEB-SCR-RESET, WEB-SCR-USER-EDITOR; WEB-API-027, WEB-API-033, WEB-API-035, WEB-API-041

9.14 WEB-FR-AUTH-14 — Authentication rate limits

Table 19: WEB-FR-AUTH-14 — Authentication rate limits

Field	Specification
Name	Authentication rate limits
Description	Registration, sign-in, OTP, refresh and forgot / reset password are limited to 10 requests per minute per IP; beyond that the API returns 429 with Retry-After.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Authentication rate limits” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Registration, sign-in, OTP, refresh and forgot / reset password are limited to 10 requests per minute per IP; beyond that the API returns 429 with Retry-After. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-AUTH-14-AC-01** — Given the preconditions/trigger, verify: Registration, sign-in, OTP, refresh and forgot / reset password are limited to 10 requests per minute per IP; beyond that the API returns 429 with Retry-After. **WEB-FR-AUTH-14-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/auth/screens/verify_email_screen.dart; Cinema_Mobile/lib/features/auth/screens/forgot_password_screen.dart; Cinema_Mobile/lib/features/auth/screens/reset_password_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-VERIFY, WEB-SCR-FORGOT, WEB-SCR-RESET; WEB-API-027, WEB-API-032, WEB-API-034

9.15 WEB-FR-AUTH-15 — Find member by phone

Table 20: WEB-FR-AUTH-15 — Find member by phone

Field	Specification
Name	Find member by phone
Description	Staff/Admin find a customer account by phone number (digits only; spaces, dots and dashes are ignored) to sell tickets and give points at the counter. Fewer than 8 or more than 15 digits is refused (400); customers get 403.
Actor	Staff/Admin
Preconditions	The gateway is configured. Registration/recovery needs SMTP; protected operations require a valid account session.
Trigger	Invoke “Find member by phone” or its specified backend event/worker.

Continued on next page

Table 20 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin find a customer account by phone number (digits only; spaces, dots and dashes are ignored) to sell tickets and give points at the counter. Fewer than 8 or more than 15 digits is refused (`400`); customers get `403`. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-AUTH-15-AC-01`** — Given the preconditions/trigger, verify: Staff/Admin find a customer account by phone number (digits only; spaces, dots and dashes are ignored) to sell tickets and give points at the counter. Fewer than 8 or more than 15 digits is refused (`400`); customers get `403`. **`WEB-FR-AUTH-15-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/MembersController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart`; `Cinema_BE/AuthService/AuthService/Service/AuthService.cs`; `Cinema_BE/ApiGateway/ApiGateway/Program.cs`
Business rules	`WEB-BR-01`, `WEB-BR-02`, `WEB-BR-03`, `WEB-BR-24`, `WEB-BR-25`
Use case / screens / APIs	`UC-WEB-01`; `WEB-SCR-COUNTER`; `WEB-API-030`

9.16 `WEB-FR-AUTH-16` — Edit profile

Table 21: `WEB-FR-AUTH-16` — Edit profile

Field	Specification
Name	Edit profile
Description	Users **edit their profile** (Account → Edit profile): full name (required, at most 150 characters) and phone (optional, same rules as registration; empty removes it). The email is the sign-in name and cannot change.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “Edit profile” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Users **edit their profile** (Account → Edit profile): full name (required, at most 150 characters) and phone (optional, same rules as registration; empty removes it). The email is the sign-in name and cannot change. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 21 — continued

Field	Specification
Acceptance criteria	**WEB-FR-AUTH-16-AC-01** — Given the preconditions/trigger, verify: Users **edit their profile** (Account → Edit profile): full name (required, at most 150 characters) and phone (optional, same rules as registration; empty removes it). The email is the sign-in name and cannot change. **WEB-FR-AUTH-16-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs`; `Cinema_Mobile/lib/features/profile/screens/profile_screen.dart`; `Cinema_BE/AuthService/AuthService/Service/AuthService.cs`; `Cinema_BE/ApiGateway/ApiGateway/Program.cs`
Business rules	`WEB-BR-01`, `WEB-BR-02`, `WEB-BR-03`, `WEB-BR-24`, `WEB-BR-25`
Use case / screens / APIs	`UC-WEB-01`; `WEB-SCR-PROFILE`, `WEB-SCR-MGT-PROFILE`; `WEB-API-031`

9.17 `WEB-FR-AUTH-17` — Change current password

Table 22: `WEB-FR-AUTH-17` — Change current password

Field	Specification
Name	Change current password
Description	**Change the password with the current one** (Account → Change password): current password, new password twice. A wrong current password or the same password is refused (`400`). Afterwards other devices are signed out (refresh tokens revoked) and this device gets new tokens. A forgotten current password leads to the email `OTP` flow. Limited to 10 requests per minute like sign-in.
Actor	Guest; signed-in account; system
Preconditions	The gateway is configured. Registration/recovery needs `SMTP`; protected operations require a valid account session.
Trigger	Invoke “Change current password” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Change the password with the current one** (Account → Change password): current password, new password twice. A wrong current password or the same password is refused (`400`). Afterwards other devices are signed out (refresh tokens revoked) and this device gets new tokens. A forgotten current password leads to the email `OTP` flow. Limited to 10 requests per minute like sign-in. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (`400`), bad credentials (`401`), unverified email (`403`), stale/incorrect `OTP`, or resend/rate limits (`429`). Clear the session for account-lock/role-change responses; show `SMTP`/dependency failure (`503`).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-AUTH-17-AC-01** — Given the preconditions/trigger, verify: **Change the password with the current one** (Account → Change password): current password, new password twice. A wrong current password or the same password is refused (`400`). Afterwards other devices are signed out (refresh tokens revoked) and this device gets new tokens. A forgotten current password leads to the email `OTP` flow. Limited to 10 requests per minute like sign-in. **WEB-FR-AUTH-17-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 22 — continued

Field	Specification
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_Mobile/lib/features/profile/screens/profile_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE; WEB-API-025

9.18 WEB-FR-USER-01 — List accounts

Table 23: WEB-FR-USER-01 — List accounts

Field	Specification
Name	List accounts
Description	List accounts: search, filter by role and status, paging.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “List accounts” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: List accounts: search, filter by role and status, paging. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-USER-01-AC-01** — Given the preconditions/trigger, verify: List accounts: search, filter by role and status, paging. **WEB-FR-USER-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs; Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart
Business rules	WEB-BR-03, WEB-BR-24
Use case / screens / APIs	UC-WEB-02; WEB-SCR-ADMIN-USERS; WEB-API-040

9.19 WEB-FR-USER-02 — Account details

Table 24: WEB-FR-USER-02 — Account details

Field	Specification
Name	Account details
Description	Show an account.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.

Continued on next page

Table 24 — continued

Field	Specification
Trigger	Invoke “Account details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show an account. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-USER-02-AC-01** — Given the preconditions/trigger, verify: Show an account. **WEB-FR-USER-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs; Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart
Business rules	WEB-BR-03, WEB-BR-24
Use case / screens / APIs	UC-WEB-02; WEB-SCR-USER-EDITOR; WEB-API-039

9.20 WEB-FR-USER-03 — Create account

Table 25: WEB-FR-USER-03 — Create account

Field	Specification
Name	Create account
Description	Create an account with one or more roles; no OTP needed.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Create account” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Create an account with one or more roles; no OTP needed. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-USER-03-AC-01** — Given the preconditions/trigger, verify: Create an account with one or more roles; no OTP needed. **WEB-FR-USER-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs; Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart
Business rules	WEB-BR-03, WEB-BR-24
Use case / screens / APIs	UC-WEB-02; WEB-SCR-USER-EDITOR; WEB-API-041

9.21 WEB-FR-USER-04 — Change account roles

Table 26: WEB-FR-USER-04 — Change account roles

Field	Specification
Name	Change account roles
Description	Change roles: old refresh tokens are revoked and the user must sign in again; saving the same roles keeps the session.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Change account roles” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Change roles: old refresh tokens are revoked and the user must sign in again; saving the same roles keeps the session. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-USER-04-AC-01** — Given the preconditions/trigger, verify: Change roles: old refresh tokens are revoked and the user must sign in again; saving the same roles keeps the session. **WEB-FR-USER-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs; Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart
Business rules	WEB-BR-03, WEB-BR-24
Use case / screens / APIs	UC-WEB-02; WEB-SCR-USER-EDITOR; WEB-API-037

9.22 WEB-FR-USER-05 — Lock and unlock

Table 27: WEB-FR-USER-05 — Lock and unlock

Field	Specification
Name	Lock and unlock
Description	Lock / unlock an account; an Admin cannot lock their own account.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Lock and unlock” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Lock / unlock an account; an Admin cannot lock their own account. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H

Continued on next page

Table 27 — continued

Field	Specification
Acceptance criteria	**WEB-FR-USER-05-AC-01** — Given the preconditions/trigger, verify: Lock / unlock an account; an Admin cannot lock their own account. **WEB-FR-USER-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart`
Business rules	`WEB-BR-03`, `WEB-BR-24`
Use case / screens / APIs	`UC-WEB-02`; `WEB-SCR-USER-EDITOR`; `WEB-API-038`

9.23 `WEB-FR-USER-06` — List roles

Table 28: `WEB-FR-USER-06` — List roles

Field	Specification
Name	List roles
Description	List roles (`ROLE_CUSTOMER`, `ROLE_STAFF`, `ROLE_ADMIN`).
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “List roles” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: List roles (`ROLE_CUSTOMER`, `ROLE_STAFF`, `ROLE_ADMIN`). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-USER-06-AC-01** — Given the preconditions/trigger, verify: List roles (`ROLE_CUSTOMER`, `ROLE_STAFF`, `ROLE_ADMIN`). **WEB-FR-USER-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/RolesController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart`; `Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs`
Business rules	`WEB-BR-03`, `WEB-BR-24`
Use case / screens / APIs	`UC-WEB-02`; `WEB-SCR-ADMIN-USERS`, `WEB-SCR-USER-EDITOR`; `WEB-API-036`

9.24 `WEB-FR-MOV-01` — Paged catalogue

Table 29: `WEB-FR-MOV-01` — Paged catalogue

Field	Specification
Name	Paged catalogue

Continued on next page

Table 29 — continued

Field	Specification
Description	List movies with paging (Admin).
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Paged catalogue” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: List movies with paging (Admin). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns `404`; a ticket not checked in returns `400`. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-MOV-01-AC-01`** — Given the preconditions/trigger, verify: List movies with paging (Admin). **`WEB-FR-MOV-01-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs`
Business rules	`WEB-BR-26`
Use case / screens / APIs	`UC-WEB-03`; `WEB-SCR-ADMIN-MOVIES`; `WEB-API-070`

9.25 `WEB-FR-MOV-02` — Movie details

Table 30: `WEB-FR-MOV-02` — Movie details

Field	Specification
Name	Movie details
Description	Movie details: title, description, duration, genre, language, release date, poster, trailer, status.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Movie details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Movie details: title, description, duration, genre, language, release date, poster, trailer, status. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns `404`; a ticket not checked in returns `400`. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-MOV-02-AC-01`** — Given the preconditions/trigger, verify: Movie details: title, description, duration, genre, language, release date, poster, trailer, status. **`WEB-FR-MOV-02-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 30 — continued

Field	Specification
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-MOVIE, WEB-SCR-ADMIN-MOVIES; WEB-API-064

9.26 WEB-FR-MOV-03 — Title search

Table 31: WEB-FR-MOV-03 — Title search

Field	Specification
Name	Title search
Description	Search movies by title.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Title search” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Search movies by title. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-MOV-03-AC-01** — Given the preconditions/trigger, verify: Search movies by title. **WEB-FR-MOV-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/home/screens/home_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-HOME, WEB-SCR-ADMIN-MOVIES; WEB-API-066, WEB-API-068

9.27 WEB-FR-MOV-04 — Active catalogue

Table 32: WEB-FR-MOV-04 — Active catalogue

Field	Specification
Name	Active catalogue
Description	Filter movies by status; the customer home page only shows ACTIVE movies.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Active catalogue” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Filter movies by status; the customer home page only shows ACTIVE movies. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-MOV-04-AC-01** — Given the preconditions/trigger, verify: Filter movies by status; the customer home page only shows ACTIVE movies. **WEB-FR-MOV-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/home/screens/home_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-HOME, WEB-SCR-ADMIN-MOVIES; WEB-API-066, WEB-API-069

9.28 WEB-FR-MOV-05 — Create movie

Table 33: WEB-FR-MOV-05 — Create movie

Field	Specification
Name	Create movie
Description	Admin adds a movie.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Create movie” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin adds a movie. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.

Continued on next page

Table 33 — continued

Field	Specification
Priority	H
Acceptance criteria	**WEB-FR-MOV-05-AC-01** — Given the preconditions/trigger, verify: Admin adds a movie. **WEB-FR-MOV-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-ADMIN-MOVIES; WEB-API-071

9.29 WEB-FR-MOV-06 — Edit and reactivate movie

Table 34: WEB-FR-MOV-06 — Edit and reactivate movie

Field	Specification
Name	Edit and reactivate movie
Description	Admin edits a movie, including bringing a stopped movie back (**Show again**).
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Edit and reactivate movie” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin edits a movie, including bringing a stopped movie back (**Show again**). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-MOV-06-AC-01** — Given the preconditions/trigger, verify: Admin edits a movie, including bringing a stopped movie back (**Show again**). **WEB-FR-MOV-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-ADMIN-MOVIES; WEB-API-065

9.30 WEB-FR-MOV-07 — Deactivate movie

Table 35: WEB-FR-MOV-07 — Deactivate movie

Field	Specification
Name	Deactivate movie
Description	Admin stops a movie (**Stop showing**): it becomes INACTIVE and disappears from the customer app; its data and tickets stay.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Deactivate movie” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin stops a movie (**Stop showing**): it becomes INACTIVE and disappears from the customer app; its data and tickets stay. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-MOV-07-AC-01** — Given the preconditions/trigger, verify: Admin stops a movie (**Stop showing**): it becomes INACTIVE and disappears from the customer app; its data and tickets stay. **WEB-FR-MOV-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-ADMIN-MOVIES; WEB-API-063

9.31 WEB-FR-MOV-08 — Open trailer

Table 36: WEB-FR-MOV-08 — Open trailer

Field	Specification
Name	Open trailer
Description	The customer app opens the movie trailer (trailerUrl).
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Open trailer” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The customer app opens the movie trailer (trailerUrl). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.

Continued on next page

Table 36 — continued

Field	Specification
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	L
Acceptance criteria	**WEB-FR-MOV-08-AC-01** — Given the preconditions/trigger, verify: The customer app opens the movie trailer (`trailerUrl`). **WEB-FR-MOV-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs`; `Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs`
Business rules	`WEB-BR-26`
Use case / screens / APIs	`UC-WEB-03`; `WEB-SCR-MOVIE`; `WEB-API-064`

9.32 `WEB-FR-MOV-09` — Age rating

Table 37: `WEB-FR-MOV-09` — Age rating

Field	Specification
Name	Age rating
Description	Support age ratings P, K, T13, T16 and T18. Admin selects a rating (P by default); cards, movie details, booking summary, counter and check-in display it. Before holding seats for a non-P film the customer acknowledges the age condition; staff are reminded to check identity at admission. This specifies implemented labels, not a legal compliance certification.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	A movie is available; changes need Admin; a review needs the owner's checked-in ticket.
Trigger	Invoke "Age rating" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Support age ratings P, K, T13, T16 and T18. Admin selects a rating (P by default); cards, movie details, booking summary, counter and check-in display it. Before holding seats for a non-P film the customer acknowledges the age condition; staff are reminded to check identity at admission. This specifies implemented labels, not a legal compliance certification. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user's ticket returns `404`; a ticket not checked in returns `400`. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-MOV-09-AC-01** — Given the preconditions/trigger, verify: Support age ratings P, K, T13, T16 and T18. Admin selects a rating (P by default); cards, movie details, booking summary, counter and check-in display it. Before holding seats for a non-P film the customer acknowledges the age condition; staff are reminded to check identity at admission. This specifies implemented labels, not a legal compliance certification. **WEB-FR-MOV-09-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 37 — continued

Field	Specification
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-MOVIE, WEB-SCR-SUMMARY, WEB-SCR-CHECKIN, WEB-SCR-ADMIN-MOVIES; WEB-API-064, WEB-API-065, WEB-API-071

9.33 WEB-FR-MOV-10 — Now showing and coming soon

Table 38: WEB-FR-MOV-10 — Now showing and coming soon

Field	Specification
Name	Now showing and coming soon
Description	The customer home has two tabs: **Now showing** (ACTIVE movies already released or without a release date, newest first) and **Coming soon** (ACTIVE movies released after today, earliest first, the release date on the poster). Movies can be filtered by genre (chips built from the genres of the movies on offer) and searched by title through the API, ignoring case. A movie coming soon shows its release date; without showtimes the page says they will be added closer to the release.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Now showing and coming soon” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The customer home has two tabs: **Now showing** (ACTIVE movies already released or without a release date, newest first) and **Coming soon** (ACTIVE movies released after today, earliest first, the release date on the poster). Movies can be filtered by genre (chips built from the genres of the movies on offer) and searched by title through the API, ignoring case. A movie coming soon shows its release date; without showtimes the page says they will be added closer to the release. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-MOV-10-AC-01** — Given the preconditions/trigger, verify: The customer home has two tabs: **Now showing** (ACTIVE movies already released or without a release date, newest first) and **Coming soon** (ACTIVE movies released after today, earliest first, the release date on the poster). Movies can be filtered by genre (chips built from the genres of the movies on offer) and searched by title through the API, ignoring case. A movie coming soon shows its release date; without showtimes the page says they will be added closer to the release. **WEB-FR-MOV-10-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 38 — continued

Field	Specification
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs`; `Cinema_Mobile/lib/features/home/screens/home_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs`
Business rules	`WEB-BR-26`
Use case / screens / APIs	`UC-WEB-03`; `WEB-SCR-HOME`; `WEB-API-066`, `WEB-API-067`

9.34 `WEB-FR-MOV-11` — Director and cast

Table 39: `WEB-FR-MOV-11` — Director and cast

Field	Specification
Name	Director and cast
Description	Movies have a **director** and **main cast** (optional, at most 150 / `500` characters). Admins fill them in when adding / editing a movie; the movie page shows them.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	The public catalogue is reachable; the movie exists for detail/trailer operations. Empty search results are valid.
Trigger	Invoke “Director and cast” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Movies have a **director** and **main cast** (optional, at most 150 / `500` characters). Admins fill them in when adding / editing a movie; the movie page shows them. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns `404`; a ticket not checked in returns `400`. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	L
Acceptance criteria	**`WEB-FR-MOV-11-AC-01`** — Given the preconditions/trigger, verify: Movies have a **director** and **main cast** (optional, at most 150 / `500` characters). Admins fill them in when adding / editing a movie; the movie page shows them. **`WEB-FR-MOV-11-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs`; `Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart`; `Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs`
Business rules	`WEB-BR-26`
Use case / screens / APIs	`UC-WEB-03`; `WEB-SCR-MOVIE`, `WEB-SCR-ADMIN-MOVIES`; `WEB-API-064`, `WEB-API-065`, `WEB-API-071`

9.35 WEB-FR-MOV-12 — Movie reviews

Table 40: WEB-FR-MOV-12 — Movie reviews

Field	Specification
Name	Movie reviews
Description	**Movie reviews**: customers whose ticket was checked in (they watched the movie) rate it 1–5 stars with a comment (at most 1,000 characters) from the ticket page. One review per customer per movie: sending again edits it, and it can be deleted. A ticket not checked in is refused (400) with the reason; someone else's ticket is treated as missing (404). The movie page shows everyone the average, the count per star and the newest reviews; the reviewer name is the account's full name (or a masked email). Admins remove unsuitable reviews from Admin → Movies → Reviews.
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Preconditions	A movie is available; changes need Admin; a review needs the owner's checked-in ticket.
Trigger	Invoke "Movie reviews" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Movie reviews**: customers whose ticket was checked in (they watched the movie) rate it 1–5 stars with a comment (at most 1,000 characters) from the ticket page. One review per customer per movie: sending again edits it, and it can be deleted. A ticket not checked in is refused (400) with the reason; someone else's ticket is treated as missing (404). The movie page shows everyone the average, the count per star and the newest reviews; the reviewer name is the account's full name (or a masked email). Admins remove unsuitable reviews from Admin → Movies → Reviews. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user's ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-MOV-12-AC-01** — Given the preconditions/trigger, verify: **Movie reviews**: customers whose ticket was checked in (they watched the movie) rate it 1–5 stars with a comment (at most 1,000 characters) from the ticket page. One review per customer per movie: sending again edits it, and it can be deleted. A ticket not checked in is refused (400) with the reason; someone else's ticket is treated as missing (404). The movie page shows everyone the average, the count per star and the newest reviews; the reviewer name is the account's full name (or a masked email). Admins remove unsuitable reviews from Admin → Movies → Reviews. **WEB-FR-MOV-12-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs; Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-MOVIE, WEB-SCR-TICKET, WEB-SCR-ADMIN-MOVIES; WEB-API-102, WEB-API-103, WEB-API-105, WEB-API-106

9.36 WEB-FR-ROOM-01 — Find rooms

Table 41: WEB-FR-ROOM-01 — Find rooms

Field	Specification
Name	Find rooms
Description	List rooms and search them by name (Staff/Admin).
Actor	Staff/Admin read; Admin writes
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Invoke “Find rooms” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: List rooms and search them by name (Staff/Admin). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-ROOM-01-AC-01** — Given the preconditions/trigger, verify: List rooms and search them by name (Staff/Admin). **WEB-FR-ROOM-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs; Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_rooms_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/RoomService.cs
Business rules	WEB-BR-04, WEB-BR-26
Use case / screens / APIs	UC-WEB-04; WEB-SCR-OPERATIONS, WEB-SCR-ADMIN-ROOMS; WEB-API-004, WEB-API-005

9.37 WEB-FR-ROOM-02 — Room details

Table 42: WEB-FR-ROOM-02 — Room details

Field	Specification
Name	Room details
Description	Room details with the seat map by row, coloured by seat type.
Actor	Staff/Admin read; Admin writes
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Invoke “Room details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Room details with the seat map by row, coloured by seat type. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.

Continued on next page

Table 42 — continued

Field	Specification
Priority	H
Acceptance criteria	**WEB-FR-ROOM-02-AC-01** — Given the preconditions/trigger, verify: Room details with the seat map by row, coloured by seat type. **WEB-FR-ROOM-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs; Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_rooms_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/RoomService.cs
Business rules	WEB-BR-04, WEB-BR-26
Use case / screens / APIs	UC-WEB-04; WEB-SCR-RECORD-ROOMS, WEB-SCR-ADMIN-ROOMS; WEB-API-002, WEB-API-013

9.38 WEB-FR-ROOM-03 — Create room and seats

Table 43: WEB-FR-ROOM-03 — Create room and seats

Field	Specification
Name	Create room and seats
Description	Admin creates a room from rows × seats per row; its seats are created at once.
Actor	Staff/Admin read; Admin writes
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Invoke “Create room and seats” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin creates a room from rows × seats per row; its seats are created at once. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-ROOM-03-AC-01** — Given the preconditions/trigger, verify: Admin creates a room from rows × seats per row; its seats are created at once. **WEB-FR-ROOM-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs; Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/admin/screens/admin_rooms_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/RoomService.cs
Business rules	WEB-BR-04, WEB-BR-26
Use case / screens / APIs	UC-WEB-04; WEB-SCR-ADMIN-ROOMS; WEB-API-006, WEB-API-010

9.39 WEB-FR-ROOM-04 — Rename room

Table 44: WEB-FR-ROOM-04 — Rename room

Field	Specification
Name	Rename room
Description	Admin renames a room; its seats stay.
Actor	Staff/Admin read; Admin writes
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Invoke “Rename room” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin renames a room; its seats stay. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-ROOM-04-AC-01** — Given the preconditions/trigger, verify: Admin renames a room; its seats stay. **WEB-FR-ROOM-04-AC-02** — Exercise applicable invalid/unauthorized/ conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs; Cinema_ Mobile/lib/features/admin/screens/admin_rooms_screen.dart; Cinema_BE/Movie Service/MovieService/Service/Impl/RoomService.cs
Business rules	WEB-BR-04, WEB-BR-26
Use case / screens / APIs	UC-WEB-04; WEB-SCR-ADMIN-ROOMS; WEB-API-003

9.40 WEB-FR-ROOM-05 — Delete eligible room

Table 45: WEB-FR-ROOM-05 — Delete eligible room

Field	Specification
Name	Delete eligible room
Description	Admin deletes a room without showtimes (with its seats); a room with showtimes is refused.
Actor	Staff/Admin read; Admin writes
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Invoke “Delete eligible room” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin deletes a room without showtimes (with its seats); a room with showtimes is refused. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 45 — continued

Field	Specification
Acceptance criteria	**WEB-FR-ROOM-05-AC-01** — Given the preconditions/trigger, verify: Admin deletes a room without showtimes (with its seats); a room with showtimes is refused. **WEB-FR-ROOM-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs; Cinema_Mobile/lib/features/admin/screens/admin_rooms_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/RoomService.cs
Business rules	WEB-BR-04, WEB-BR-26
Use case / screens / APIs	UC-WEB-04; WEB-SCR-ADMIN-ROOMS; WEB-API-001

9.41 WEB-FR-SEAT-01 — Generate seats

Table 46: WEB-FR-SEAT-01 — Generate seats

Field	Specification
Name	Generate seats
Description	Generate a standard seat map for an empty room (Staff/Admin).
Actor	Staff/Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Generate seats” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Generate a standard seat map for an empty room (Staff/Admin). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-SEAT-01-AC-01** — Given the preconditions/trigger, verify: Generate a standard seat map for an empty room (Staff/Admin). **WEB-FR-SEAT-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-RECORD-ROOMS; WEB-API-010

9.42 WEB-FR-SEAT-02 — Seat details

Table 47: WEB-FR-SEAT-02 — Seat details

Field	Specification
Name	Seat details
Description	Show one seat.
Actor	Staff/Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Seat details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show one seat. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	L
Acceptance criteria	**WEB-FR-SEAT-02-AC-01** — Given the preconditions/trigger, verify: Show one seat. **WEB-FR-SEAT-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-RECORD-ROOMS; WEB-API-009

9.43 WEB-FR-SEAT-03 — Read room map

Table 48: WEB-FR-SEAT-03 — Read room map

Field	Specification
Name	Read room map
Description	Get the seat map of a room (signed-in users).
Actor	Signed-in Customer reads/selects; Staff/Admin manage types; Admin manages capacity/deletion
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Read room map” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Get the seat map of a room (signed-in users). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H

Continued on next page

Table 48 — continued

Field	Specification
Acceptance criteria	**WEB-FR-SEAT-03-AC-01** — Given the preconditions/trigger, verify: Get the seat map of a room (signed-in users). **WEB-FR-SEAT-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-SEATS, WEB-SCR-RECORD-ROOMS; WEB-API-013

9.44 WEB-FR-SEAT-04 — Delete seat API

Table 49: WEB-FR-SEAT-04 — Delete seat API

Field	Specification
Name	Delete seat API
Description	Admin deletes a seat (API only).
Actor	Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Delete seat API” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin deletes a seat (API only). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	L
Acceptance criteria	**WEB-FR-SEAT-04-AC-01** — Given the preconditions/trigger, verify: Admin deletes a seat (API only). **WEB-FR-SEAT-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; ; WEB-API-008

9.45 WEB-FR-SEAT-05 — Change seat type

Table 50: WEB-FR-SEAT-05 — Change seat type

Field	Specification
Name	Change seat type
Description	Change one seat's type: NORMAL / VIP / SWEETBOX. A sweetbox changes with its pair; a seat without a pair cannot become a sweetbox.
Actor	Staff/Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke "Change seat type" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Change one seat's type: NORMAL / VIP / SWEETBOX. A sweetbox changes with its pair; a seat without a pair cannot become a sweetbox. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-SEAT-05-AC-01** — Given the preconditions/trigger, verify: Change one seat's type: NORMAL / VIP / SWEETBOX. A sweetbox changes with its pair; a seat without a pair cannot become a sweetbox. **WEB-FR-SEAT-05-AC-02** — Exercise applicable invalid/unauthorized/ conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_ Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/Movie Service/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/ MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-RECORD-ROOMS; WEB-API-007

9.46 WEB-FR-SEAT-06 — Paint seat types

Table 51: WEB-FR-SEAT-06 — Paint seat types

Field	Specification
Name	Paint seat types
Description	Change many seats at once on the map: pick a type, then click, drag across seats or click a row letter to change the whole row. A row with an unpaired seat returns 400.
Actor	Staff/Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke "Paint seat types" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Change many seats at once on the map: pick a type, then click, drag across seats or click a row letter to change the whole row. A row with an unpaired seat returns 400. 3. Return/render the authoritative result and refresh affected data after success.

Continued on next page

Table 51 — continued

Field	Specification
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-SEAT-06-AC-01** — Given the preconditions/trigger, verify: Change many seats at once on the map: pick a type, then click, drag across seats or click a row letter to change the whole row. A row with an unpaired seat returns `400`. **WEB-FR-SEAT-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-04`, `WEB-BR-07`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-26`
Use case / screens / APIs	`UC-WEB-05`; `WEB-SCR-RECORD-ROOMS`; `WEB-API-012`

9.47 `WEB-FR-SEAT-07` — Change capacity

Table 52: `WEB-FR-SEAT-07` — Change capacity

Field	Specification
Name	Change capacity
Description	Admin changes a room's seat count: new seats fill the last row, then new rows (up to row Z); removing takes the last seats; a leftover half sweetbox becomes standard.
Actor	Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke "Change capacity" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin changes a room's seat count: new seats fill the last row, then new rows (up to row Z); removing takes the last seats; a leftover half sweetbox becomes standard. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-SEAT-07-AC-01** — Given the preconditions/trigger, verify: Admin changes a room's seat count: new seats fill the last row, then new rows (up to row Z); removing takes the last seats; a leftover half sweetbox becomes standard. **WEB-FR-SEAT-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`

Continued on next page

Table 52 — continued

Field	Specification
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-RECORD-ROOMS; WEB-API-011

9.48 WEB-FR-SEAT-08 — Protect scheduled rooms

Table 53: WEB-FR-SEAT-08 — Protect scheduled rooms

Field	Specification
Name	Protect scheduled rooms
Description	Seats cannot be generated, retyped or recounted in a room that has showtimes (sold tickets stay correct).
Actor	Staff/Admin
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Protect scheduled rooms” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Seats cannot be generated, retyped or recounted in a room that has showtimes (sold tickets stay correct). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-SEAT-08-AC-01** — Given the preconditions/trigger, verify: Seats cannot be generated, retyped or recounted in a room that has showtimes (sold tickets stay correct). **WEB-FR-SEAT-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26
Use case / screens / APIs	UC-WEB-05; WEB-SCR-RECORD-ROOMS; WEB-API-007, WEB-API-010, WEB-API-011, WEB-API-012

9.49 WEB-FR-SEAT-09 — Refresh management maps

Table 54: WEB-FR-SEAT-09 — Refresh management maps

Field	Specification
Name	Refresh management maps
Description	Seat maps on the admin website refresh every 5 seconds when someone else changes them.
Actor	Staff/Admin

Continued on next page

Table 54 — continued

Field	Specification
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Invoke “Refresh management maps” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Seat maps on the admin website refresh every 5 seconds when someone else changes them. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	L
Acceptance criteria	**WEB-FR-SEAT-09-AC-01** — Given the preconditions/trigger, verify: Seat maps on the admin website refresh every 5 seconds when someone else changes them. **WEB-FR-SEAT-09-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/SeatService.cs`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-04`, `WEB-BR-07`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-26`
Use case / screens / APIs	`UC-WEB-05`; `WEB-SCR-RECORD-ROOMS`; `WEB-API-013`

9.50 `WEB-FR-SHOW-01` — List showtimes

Table 55: `WEB-FR-SHOW-01` — List showtimes

Field	Specification
Name	List showtimes
Description	List showtimes with paging and filters.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “List showtimes” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: List showtimes with paging and filters. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-SHOW-01-AC-01** — Given the preconditions/trigger, verify: List showtimes with paging and filters. **WEB-FR-SHOW-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`

Continued on next page

Table 55 — continued

Field	Specification
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-OPERATIONS; WEB-API-023

9.51 WEB-FR-SHOW-02 — Create showtime

Table 56: WEB-FR-SHOW-02 — Create showtime

Field	Specification
Name	Create showtime
Description	Staff/Admin create a showtime: movie, room, start, end, standard seat price, status.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Create showtime” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin create a showtime: movie, room, start, end, standard seat price, status. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-SHOW-02-AC-01** — Given the preconditions/trigger, verify: Staff/Admin create a showtime: movie, room, start, end, standard seat price, status. **WEB-FR-SHOW-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/staff/screens/staff_showtime_editor_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-SHOW-EDITOR; WEB-API-024

9.52 WEB-FR-SHOW-03 — Showtime and prices

Table 57: WEB-FR-SHOW-03 — Showtime and prices

Field	Specification
Name	Showtime and prices
Description	Showtime details with seatPrices: the price of one seat of each type.
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Showtime and prices” or its specified backend event/worker.

Continued on next page

Table 57 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Showtime details with `seatPrices`: the price of one seat of each type. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-SHOW-03-AC-01`** — Given the preconditions/trigger, verify: Showtime details with `seatPrices`: the price of one seat of each type. **`WEB-FR-SHOW-03-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart`; `Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`
Use case / screens / APIs	`UC-WEB-06`; `WEB-SCR-MOVIE`, `WEB-SCR-SEATS`, `WEB-SCR-RECORD-SHOWTIMES`; `WEB-API-017`

9.53 `WEB-FR-SHOW-04` — Edit showtime

Table 58: `WEB-FR-SHOW-04` — Edit showtime

Field	Specification
Name	Edit showtime
Description	Staff/Admin edit a showtime; overlaps are still checked.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Edit showtime” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin edit a showtime; overlaps are still checked. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-SHOW-04-AC-01`** — Given the preconditions/trigger, verify: Staff/Admin edit a showtime; overlaps are still checked. **`WEB-FR-SHOW-04-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_showtime_editor_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`

Continued on next page

Table 58 — continued

Field	Specification
Use case / screens / APIs	UC-WEB-06; WEB-SCR-SHOW-EDITOR; WEB-API-018

9.54 WEB-FR-SHOW-05 — Cancel showtime

Table 59: WEB-FR-SHOW-05 — Cancel showtime

Field	Specification
Name	Cancel showtime
Description	Staff/Admin cancel a showtime (status CANCELLED, not deleted); its tickets are cancelled and refunded as in WEB-BR-19.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Cancel showtime” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin cancel a showtime (status CANCELLED, not deleted); its tickets are cancelled and refunded as in WEB-BR-19. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-SHOW-05-AC-01** — Given the preconditions/trigger, verify: Staff/Admin cancel a showtime (status CANCELLED, not deleted); its tickets are cancelled and refunded as in WEB-BR-19. **WEB-FR-SHOW-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-RECORD-SHOWTIMES; WEB-API-016

9.55 WEB-FR-SHOW-06 — Open future showtimes

Table 60: WEB-FR-SHOW-06 — Open future showtimes

Field	Specification
Name	Open future showtimes
Description	Get showtimes open for sale (OPEN, not started).
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Open future showtimes” or its specified backend event/worker.

Continued on next page

Table 60 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Get showtimes open for sale (`OPEN`, not started). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-SHOW-06-AC-01`** — Given the preconditions/trigger, verify: Get showtimes open for sale (`OPEN`, not started). **`WEB-FR-SHOW-06-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`
Use case / screens / APIs	`UC-WEB-06`; `WEB-SCR-MOVIE`; `WEB-API-022`

9.56 `WEB-FR-SHOW-07` — Showtimes by movie

Table 61: `WEB-FR-SHOW-07` — Showtimes by movie

Field	Specification
Name	Showtimes by movie
Description	Get showtimes of a movie.
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Showtimes by movie” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Get showtimes of a movie. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-SHOW-07-AC-01`** — Given the preconditions/trigger, verify: Get showtimes of a movie. **`WEB-FR-SHOW-07-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`
Use case / screens / APIs	`UC-WEB-06`; `WEB-SCR-MOVIE`; `WEB-API-021`

9.57 WEB-FR-SHOW-08 — Bookable movie showtimes

Table 62: WEB-FR-SHOW-08 — Bookable movie showtimes

Field	Specification
Name	Bookable movie showtimes
Description	Get open showtimes of a movie; the customer app shows those in the next 7 days.
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Bookable movie showtimes” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Get open showtimes of a movie; the customer app shows those in the next 7 days. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-SHOW-08-AC-01** — Given the preconditions/trigger, verify: Get open showtimes of a movie; the customer app shows those in the next 7 days. **WEB-FR-SHOW-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-MOVIE; WEB-API-020

9.58 WEB-FR-SHOW-09 — Date range search

Table 63: WEB-FR-SHOW-09 — Date range search

Field	Specification
Name	Date range search
Description	Find showtimes in a time range (Staff/Admin).
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Date range search” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Find showtimes in a time range (Staff/Admin). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H

Continued on next page

Table 63 — continued

Field	Specification
Acceptance criteria	**WEB-FR-SHOW-09-AC-01** — Given the preconditions/trigger, verify: Find showtimes in a time range (Staff/Admin). **WEB-FR-SHOW-09-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-OPERATIONS; WEB-API-019

9.59 WEB-FR-SHOW-10 — Open and close sales

Table 64: WEB-FR-SHOW-10 — Open and close sales

Field	Specification
Name	Open and close sales
Description	Open / close sales of a showtime (OPEN ↔ CLOSED).
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Open and close sales” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Open / close sales of a showtime (OPEN ↔ CLOSED). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-SHOW-10-AC-01** — Given the preconditions/trigger, verify: Open / close sales of a showtime (OPEN ↔ CLOSED). **WEB-FR-SHOW-10-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-RECORD-SHOWTIMES; WEB-API-015

9.60 WEB-FR-SHOW-11 — Reject overlap

Table 65: WEB-FR-SHOW-11 — Reject overlap

Field	Specification
Name	Reject overlap

Continued on next page

Table 65 — continued

Field	Specification
Description	Overlapping showtimes in one room return `409`.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Reject overlap” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Overlapping showtimes in one room return `409`. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-SHOW-11-AC-01`** — Given the preconditions/trigger, verify: Overlapping showtimes in one room return `409`. **`WEB-FR-SHOW-11-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_showtime_editor_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`
Use case / screens / APIs	`UC-WEB-06`; `WEB-SCR-SHOW-EDITOR`; `WEB-API-018`, `WEB-API-024`

9.61 `WEB-FR-SHOW-12` — Showtime seat/customer map

Table 66: `WEB-FR-SHOW-12` — Showtime seat/customer map

Field	Specification
Name	Showtime seat/customer map
Description	Staff see a showtime’s seat map (free / awaiting payment / sold / checked in), open a ticket by clicking its seat, and see the customer list (email, seats, booking code, check-in time); refreshed every 5 seconds.
Actor	Staff/Admin
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke “Showtime seat/customer map” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff see a showtime’s seat map (free / awaiting payment / sold / checked in), open a ticket by clicking its seat, and see the customer list (email, seats, booking code, check-in time); refreshed every 5 seconds. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M

Continued on next page

Table 66 — continued

Field	Specification
Acceptance criteria	**WEB-FR-SHOW-12-AC-01** — Given the preconditions/trigger, verify: Staff see a showtime's seat map (free / awaiting payment / sold / checked in), open a ticket by clicking its seat, and see the customer list (email, seats, booking code, check-in time); refreshed every 5 seconds. **WEB-FR-SHOW-12-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs`; `Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs`
Business rules	`WEB-BR-05`, `WEB-BR-06`, `WEB-BR-15`, `WEB-BR-19`, `WEB-BR-30`
Use case / screens / APIs	`UC-WEB-06`; `WEB-SCR-RECORD-SHOWTIMES`; `WEB-API-013`, `WEB-API-017`, `WEB-API-050`

9.62 `WEB-FR-SHOW-13` — Screening format

Table 67: `WEB-FR-SHOW-13` — Screening format

Field	Specification
Name	Screening format
Description	Every showtime has a **format: 2D, 3D or `IMAX`** (2D by default, kept when left out on edit, anything else refused with `400`). The stored price is the standard seat at 2D; 3D adds 20,000 `VND` and `IMAX` 40,000 `VND` to the standard seat before the `VIP` / sweetbox surcharges. Seat prices shown to customers and charged by `BookingService` include it. The app names the format next to the room (e.g. "Room 1 · `IMAX`").
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Invoke "Screening format" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Every showtime has a **format: 2D, 3D or `IMAX`** (2D by default, kept when left out on edit, anything else refused with `400`). The stored price is the standard seat at 2D; 3D adds 20,000 `VND` and `IMAX` 40,000 `VND` to the standard seat before the `VIP` / sweetbox surcharges. Seat prices shown to customers and charged by `BookingService` include it. The app names the format next to the room (e.g. "Room 1 · `IMAX`"). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject overlap with `409` and invalid time/format with `400`; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-SHOW-13-AC-01** — Given the preconditions/trigger, verify: Every showtime has a **format: 2D, 3D or `IMAX`** (2D by default, kept when left out on edit, anything else refused with `400`). The stored price is the standard seat at 2D; 3D adds 20,000 `VND` and `IMAX` 40,000 `VND` to the standard seat before the `VIP` / sweetbox surcharges. Seat prices shown to customers and charged by `BookingService` include it. The app names the format next to the room (e.g. "Room 1 · `IMAX`"). **WEB-FR-SHOW-13-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 67 — continued

Field	Specification
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart; Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_showtime_editor_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/ShowtimeService.cs
Business rules	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30
Use case / screens / APIs	UC-WEB-06; WEB-SCR-MOVIE, WEB-SCR-SEATS, WEB-SCR-SHOW-EDITOR; WEB-API-017, WEB-API-018, WEB-API-024

9.63 WEB-FR-BOOK-01 — Hold booking

Table 68: WEB-FR-BOOK-01 — Hold booking

Field	Specification
Name	Hold booking
Description	Create a booking from a showtime and seats: the showtime must be OPEN and not started, with at least one seat and no duplicates. The server takes the show time, movie title and prices from MovieService. The booking is PENDING and holds the seats for 10 minutes.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Hold booking” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Create a booking from a showtime and seats: the showtime must be OPEN and not started, with at least one seat and no duplicates. The server takes the show time, movie title and prices from MovieService. The booking is PENDING and holds the seats for 10 minutes. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	One PENDING booking with server total and expiresAt is committed; seats are HELD for ten minutes. A rejected transaction leaves no partial holds or point deduction.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-01-AC-01** — Given the preconditions/trigger, verify: Create a booking from a showtime and seats: the showtime must be OPEN and not started, with at least one seat and no duplicates. The server takes the show time, movie title and prices from MovieService. The booking is PENDING and holds the seats for 10 minutes. **WEB-FR-BOOK-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SUMMARY; WEB-API-055

9.64 WEB-FR-BOOK-02 — Booking details

Table 69: WEB-FR-BOOK-02 — Booking details

Field	Specification
Name	Booking details
Description	Show a booking: to its owner or to Staff/Admin.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	Customer is signed in and owns the relevant ticket; the requested action must be permitted by ticket state and time. Staff access applies only where stated.
Trigger	Invoke “Booking details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show a booking: to its owner or to Staff/Admin. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-02-AC-01** — Given the preconditions/trigger, verify: Show a booking: to its owner or to Staff/Admin. **WEB-FR-BOOK-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/payment/screens/payment_result_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-RESULT, WEB-SCR-TICKET, WEB-SCR-RECORD-BOOKINGS; WEB-API-046

9.65 WEB-FR-BOOK-03 — Management booking search

Table 70: WEB-FR-BOOK-03 — Management booking search

Field	Specification
Name	Management booking search
Description	Staff/Admin list bookings and filter by status and creation date.
Actor	Staff/Admin
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Management booking search” or its specified backend event/worker.

Continued on next page

Table 70 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin list bookings and filter by status and creation date. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-BOOK-03-AC-01`** — Given the preconditions/trigger, verify: Staff/Admin list bookings and filter by status and creation date. **`WEB-FR-BOOK-03-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; `WEB-SCR-OPERATIONS`; `WEB-API-048`, `WEB-API-052`, `WEB-API-054`

9.66 `WEB-FR-BOOK-04` — Own ticket history

Table 71: `WEB-FR-BOOK-04` — Own ticket history

Field	Specification
Name	Own ticket history
Description	Customers see their own tickets in **Active** and **History**.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	Customer is signed in and owns the relevant ticket; the requested action must be permitted by ticket state and time. Staff access applies only where stated.
Trigger	Invoke “Own ticket history” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Customers see their own tickets in **Active** and **History**. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-BOOK-04-AC-01`** — Given the preconditions/trigger, verify: Customers see their own tickets in **Active** and **History**. **`WEB-FR-BOOK-04-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 71 — continued

Field	Specification
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/booking/screens/my_bookings_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-TICKETS; WEB-API-053

9.67 WEB-FR-BOOK-05 — Confirm paid booking

Table 72: WEB-FR-BOOK-05 — Confirm paid booking

Field	Specification
Name	Confirm paid booking
Description	Confirm a booking after a successful payment (see PAY and POS).
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Confirm paid booking” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Confirm a booking after a successful payment (see PAY and POS). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-05-AC-01** — Given the preconditions/trigger, verify: Confirm a booking after a successful payment (see PAY and POS). **WEB-FR-BOOK-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/payment/screens/payment_screen.dart; Cinema_Mobile/lib/features/payment/screens/payment_result_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-PAYMENT, WEB-SCR-RESULT, WEB-SCR-TICKET; WEB-API-086, WEB-API-087, WEB-API-089

9.68 WEB-FR-BOOK-06 — Cancel booking

Table 73: WEB-FR-BOOK-06 — Cancel booking

Field	Specification
Name	Cancel booking
Description	Cancel a booking: an unpaid one is cancelled and its seats released; a paid one follows WEB-BR-17. A customer cancelling a paid ticket creates a refund request: the booking becomes REFUND_PENDING and keeps its seats until staff handle it.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	Customer is signed in and owns the relevant ticket; the requested action must be permitted by ticket state and time. Staff access applies only where stated.
Trigger	Invoke “Cancel booking” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Cancel a booking: an unpaid one is cancelled and its seats released; a paid one follows WEB-BR-17. A customer cancelling a paid ticket creates a refund request: the booking becomes REFUND_PENDING and keeps its seats until staff handle it. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-06-AC-01** — Given the preconditions/trigger, verify: Cancel a booking: an unpaid one is cancelled and its seats released; a paid one follows WEB-BR-17. A customer cancelling a paid ticket creates a refund request: the booking becomes REFUND_PENDING and keeps its seats until staff handle it. **WEB-FR-BOOK-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-TICKET, WEB-SCR-RECORD-BOOKINGS; WEB-API-043

9.69 WEB-FR-BOOK-07 — Occupied seat IDs

Table 74: WEB-FR-BOOK-07 — Occupied seat IDs

Field	Specification
Name	Occupied seat IDs
Description	Get the held / sold seats of a showtime.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration

Continued on next page

Table 74 — continued

Field	Specification
Preconditions	A signed-in customer has selected an `OPEN` future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Occupied seat IDs” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Get the held / sold seats of a showtime. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-BOOK-07-AC-01`** — Given the preconditions/trigger, verify: Get the held / sold seats of a showtime. **`WEB-FR-BOOK-07-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; `WEB-SCR-SEATS`; `WEB-API-049`

9.70 `WEB-FR-BOOK-08` — Concurrent seat conflict

Table 75: `WEB-FR-BOOK-08` — Concurrent seat conflict

Field	Specification
Name	Concurrent seat conflict
Description	A seat held or sold by another booking returns `409`.
Actor	Shared backend system; customer receives outcome
Preconditions	A signed-in customer has selected an `OPEN` future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Concurrent seat conflict” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A seat held or sold by another booking returns `409`. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H

Continued on next page

Table 75 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-08-AC-01** — Given the preconditions/trigger, verify: A seat held or sold by another booking returns 409. **WEB-FR-BOOK-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SEATS; WEB-API-055

9.71 WEB-FR-BOOK-09 — Sweetbox pair

Table 76: WEB-FR-BOOK-09 — Sweetbox pair

Field	Specification
Name	Sweetbox pair
Description	Sweetbox seats must be booked as a full pair (the server refuses half a pair).
Actor	Shared backend system; customer receives outcome
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Sweetbox pair” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Sweetbox seats must be booked as a full pair (the server refuses half a pair). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-09-AC-01** — Given the preconditions/trigger, verify: Sweetbox seats must be booked as a full pair (the server refuses half a pair). **WEB-FR-BOOK-09-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SEATS, WEB-SCR-SUMMARY; WEB-API-055

9.72 WEB-FR-BOOK-10 — Expire unpaid hold

Table 77: WEB-FR-BOOK-10 — Expire unpaid hold

Field	Specification
Name	Expire unpaid hold
Description	A background job runs every 30 seconds, marks bookings past their hold EXPIRED and releases their seats.
Actor	Shared backend system; customer receives outcome
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Expire unpaid hold” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A background job runs every 30 seconds, marks bookings past their hold EXPIRED and releases their seats. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-BOOK-10-AC-01** — Given the preconditions/trigger, verify: A background job runs every 30 seconds, marks bookings past their hold EXPIRED and releases their seats. **WEB-FR-BOOK-10-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/payment/screens/payment_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-PAYMENT; WEB-API-046

9.73 WEB-FR-BOOK-11 — Refresh customer map

Table 78: WEB-FR-BOOK-11 — Refresh customer map

Field	Specification
Name	Refresh customer map
Description	The seat picker refreshes every 5 seconds; a selected seat taken by someone else is unselected and the customer is told.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Refresh customer map” or its specified backend event/worker.

Continued on next page

Table 78 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The seat picker refreshes every 5 seconds; a selected seat taken by someone else is unselected and the customer is told. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-BOOK-11-AC-01`** — Given the preconditions/trigger, verify: The seat picker refreshes every 5 seconds; a selected seat taken by someone else is unselected and the customer is told. **`WEB-FR-BOOK-11-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; `WEB-SCR-SEATS`; `WEB-API-013`, `WEB-API-049`

9.74 `WEB-FR-BOOK-12` — Showtime booking statistics

Table 79: `WEB-FR-BOOK-12` — Showtime booking statistics

Field	Specification
Name	Showtime booking statistics
Description	Ticket statistics per showtime for Staff/Admin: bookings, seats sold, seats held, checked in (by a list of ids or by a start-time range of at most 31 days).
Actor	Staff/Admin
Preconditions	A signed-in customer has selected an `OPEN` future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Showtime booking statistics” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Ticket statistics per showtime for Staff/Admin: bookings, seats sold, seats held, checked in (by a list of ids or by a start-time range of at most 31 days). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M

Continued on next page

Table 79 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-12-AC-01** — Given the preconditions/trigger, verify: Ticket statistics per showtime for Staff/Admin: bookings, seats sold, seats held, checked in (by a list of ids or by a start-time range of at most 31 days). **WEB-FR-BOOK-12-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-OPERATIONS, WEB-SCR-RECORD-SHOWTIMES; WEB-API-051

9.75 WEB-FR-BOOK-13 — Concessions with tickets

Table 80: WEB-FR-BOOK-13 — Concessions with tickets

Field	Specification
Name	Concessions with tickets
Description	**Snacks with tickets**: customers (booking summary) and the counter (sale screen) pick items from the menu, 1–20 of each, at most 10 different items. Prices come from the menu on sale; the snacks are part of the total, so payment, revenue and refunds include them. The order keeps the name and price at the time of purchase. Snacks show on the ticket, the payment page, the ticket email and the printed ticket. Stopped items are refused (400).
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Concessions with tickets” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Snacks with tickets**: customers (booking summary) and the counter (sale screen) pick items from the menu, 1–20 of each, at most 10 different items. Prices come from the menu on sale; the snacks are part of the total, so payment, revenue and refunds include them. The order keeps the name and price at the time of purchase. Snacks show on the ticket, the payment page, the ticket email and the printed ticket. Stopped items are refused (400). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 80 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-13-AC-01** — Given the preconditions/trigger, verify: **Snacks with tickets**: customers (booking summary) and the counter (sale screen) pick items from the menu, 1–20 of each, at most 10 different items. Prices come from the menu on sale; the snacks are part of the total, so payment, revenue and refunds include them. The order keeps the name and price at the time of purchase. Snacks show on the ticket, the payment page, the ticket email and the printed ticket. Stopped items are refused (400). **WEB-FR-BOOK-13-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/ConcessionController.cs; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-COUNTER; WEB-API-055, WEB-API-057

9.76 WEB-FR-BOOK-14 — Concessions administration

Table 81: WEB-FR-BOOK-14 — Concessions administration

Field	Specification
Name	Concessions administration
Description	Admins manage the snacks menu (Admin → Snacks): add, edit name / price / description / image / order, stop or resume selling (items are never deleted). Customers only see items on sale.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Concessions administration” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admins manage the snacks menu (Admin → Snacks): add, edit name / price / description / image / order, stop or resume selling (items are never deleted). Customers only see items on sale. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-BOOK-14-AC-01** — Given the preconditions/trigger, verify: Admins manage the snacks menu (Admin → Snacks): add, edit name / price / description / image / order, stop or resume selling (items are never deleted). Customers only see items on sale. **WEB-FR-BOOK-14-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 81 — continued

Field	Specification
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/ConcessionController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_snacks_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/ConcessionService.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; `WEB-SCR-ADMIN-SNACKS`; `WEB-API-056`, `WEB-API-057`, `WEB-API-058`

9.77 `WEB-FR-BOOK-15` — Promotion and voucher

Table 82: `WEB-FR-BOOK-15` — Promotion and voucher

Field	Specification
Name	Promotion and voucher
Description	**Promotions**: only the ticket price is discounted (never snacks), by a percentage (optionally capped) or an amount, at most 90% of the tickets, rounded down to 1,000 `VND`. Conditions: running period, day of the week of the show, minimum tickets, maximum uses. A promotion without a code applies by itself (the biggest discount wins); a promo code typed by the customer or the counter replaces it — one promotion per booking. A wrong or unusable code is refused (`400`) with the reason. The server works the discount out again when the booking is made; a use is only counted once the ticket is paid. The discount shows on the ticket, the payment page, the ticket email and the printed ticket.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A signed-in customer has selected an `OPEN` future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Promotion and voucher” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Promotions**: only the ticket price is discounted (never snacks), by a percentage (optionally capped) or an amount, at most 90% of the tickets, rounded down to 1,000 `VND`. Conditions: running period, day of the week of the show, minimum tickets, maximum uses. A promotion without a code applies by itself (the biggest discount wins); a promo code typed by the customer or the counter replaces it — one promotion per booking. A wrong or unusable code is refused (`400`) with the reason. The server works the discount out again when the booking is made; a use is only counted once the ticket is paid. The discount shows on the ticket, the payment page, the ticket email and the printed ticket. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 82 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-15-AC-01** — Given the preconditions/trigger, verify: **Promotions**: only the ticket price is discounted (never snacks), by a percentage (optionally capped) or an amount, at most 90% of the tickets, rounded down to 1,000 VND. Conditions: running period, day of the week of the show, minimum tickets, maximum uses. A promotion without a code applies by itself (the biggest discount wins); a promo code typed by the customer or the counter replaces it — one promotion per booking. A wrong or unusable code is refused (400) with the reason. The server works the discount out again when the booking is made; a use is only counted once the ticket is paid. The discount shows on the ticket, the payment page, the ticket email and the printed ticket. **WEB-FR-BOOK-15-AC-02** — Exercise applicable invalid/unauthorized/ conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-COUNTER; WEB-API-055, WEB-API-099

9.78 WEB-FR-BOOK-16 — Promotion administration

Table 83: WEB-FR-BOOK-16 — Promotion administration

Field	Specification
Name	Promotion administration
Description	Admins manage promotions (Admin → Promotions): add, edit, pause / resume (never deleted), see how often each was used. Promo codes are unique (409). Guests only see the automatic promotions that are running.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Promotion administration” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admins manage promotions (Admin → Promotions): add, edit, pause / resume (never deleted), see how often each was used. Promo codes are unique (409). Guests only see the automatic promotions that are running. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 83 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-16-AC-01** — Given the preconditions/trigger, verify: Admins manage promotions (Admin → Promotions): add, edit, pause / resume (never deleted), see how often each was used. Promo codes are unique (409). Guests only see the automatic promotions that are running. **WEB-FR-BOOK-16-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs; Cinema_Mobile/lib/features/admin/screens/admin_promotions_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/PromotionService.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-ADMIN-PROMOTIONS; WEB-API-098, WEB-API-100, WEB-API-101

9.79 WEB-FR-BOOK-17 — Membership points

Table 84: WEB-FR-BOOK-17 — Membership points

Field	Specification
Name	Membership points
Description	**Member points**: customers with an account earn points once a ticket is paid, on the amount actually paid and by tier: Member 5%, VIP (from 2,000,000 VND spent) 7%, VVIP (from 5,000,000 VND) 10%; 1 point = 1,000 VND. When booking (app or counter) points can pay at most 50% of the price (after promotions). An unpaid hold that expires or is cancelled gives the used points back; a paid ticket that is cancelled or refunded gives the used points back, takes the earned points back and lowers the total spent. Walk-in guests have no points. Customers see their member card (points, tier, what is left to the next tier, history) on the account page.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Invoke “Membership points” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Member points**: customers with an account earn points once a ticket is paid, on the amount actually paid and by tier: Member 5%, VIP (from 2,000,000 VND spent) 7%, VVIP (from 5,000,000 VND) 10%; 1 point = 1,000 VND. When booking (app or counter) points can pay at most 50% of the price (after promotions). An unpaid hold that expires or is cancelled gives the used points back; a paid ticket that is cancelled or refunded gives the used points back, takes the earned points back and lowers the total spent. Walk-in guests have no points. Customers see their member card (points, tier, what is left to the next tier, history) on the account page. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 84 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-17-AC-01** — Given the preconditions/trigger, verify: **Member points**: customers with an account earn points once a ticket is paid, on the amount actually paid and by tier: Member 5%, VIP (from 2,000,000 VND spent) 7%, VVIP (from 5,000,000 VND) 10%; 1 point = 1,000 VND. When booking (app or counter) points can pay at most 50% of the price (after promotions). An unpaid hold that expires or is cancelled gives the used points back; a paid ticket that is cancelled or refunded gives the used points back, takes the earned points back and lowers the total spent. Walk-in guests have no points. Customers see their member card (points, tier, what is left to the next tier, history) on the account page. **WEB-FR-BOOK-17-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs; Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/profile/screens/profile_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/LoyaltyService.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-PROFILE, WEB-SCR-COUNTER; WEB-API-055, WEB-API-060

9.80 WEB-FR-BOOK-18 — Member administration

Table 85: WEB-FR-BOOK-18 — Member administration

Field	Specification
Name	Member administration
Description	Admins see customers with their points, tier and total spent (Admin → Members), open the history and add or take away points with a reason (never below zero). The counter sees the card of a customer found by phone.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Member administration” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admins see customers with their points, tier and total spent (Admin → Members), open the history and add or take away points with a reason (never below zero). The counter sees the card of a customer found by phone. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 85 — continued

Field	Specification
Acceptance criteria	**WEB-FR-BOOK-18-AC-01** — Given the preconditions/trigger, verify: Admins see customers with their points, tier and total spent (Admin → Members), open the history and add or take away points with a reason (never below zero). The counter sees the card of a customer found by phone. **WEB-FR-BOOK-18-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/MembersController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_members_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/LoyaltyService.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; `WEB-SCR-ADMIN-MEMBERS`; `WEB-API-030`, `WEB-API-059`, `WEB-API-061`, `WEB-API-062`

9.81 `WEB-FR-BOOK-19` — Share and calendar

Table 86: `WEB-FR-BOOK-19` — Share and calendar

Field	Specification
Name	Share and calendar
Description	Paid tickets have **Share ticket** (movie, show time, seats and ticket code sent to friends through the device's share sheet; copied to the clipboard where the browser cannot share) and **Add to calendar** (a Google Calendar event in Vietnam time ending when the show ends; only for shows still to come).
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	Customer is signed in and owns the relevant ticket; the requested action must be permitted by ticket state and time. Staff access applies only where stated.
Trigger	Invoke "Share and calendar" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Paid tickets have **Share ticket** (movie, show time, seats and ticket code sent to friends through the device's share sheet; copied to the clipboard where the browser cannot share) and **Add to calendar** (a Google Calendar event in Vietnam time ending when the show ends; only for shows still to come). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	L
Acceptance criteria	**WEB-FR-BOOK-19-AC-01** — Given the preconditions/trigger, verify: Paid tickets have **Share ticket** (movie, show time, seats and ticket code sent to friends through the device's share sheet; copied to the clipboard where the browser cannot share) and **Add to calendar** (a Google Calendar event in Vietnam time ending when the show ends; only for shows still to come). **WEB-FR-BOOK-19-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 86 — continued

Field	Specification
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs; Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs
Business rules	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-07; WEB-SCR-TICKET; WEB-API-017

9.82 WEB-FR-PAY-01 — Create/reuse PayOS link

Table 87: WEB-FR-PAY-01 — Create/reuse PayOS link

Field	Specification
Name	Create/reuse PayOS link
Description	Create a PayOS payment link for a PENDING booking. One payment per booking; asking again returns the existing link. The PayOS order code is the booking id × 1,000,000 plus the six random digits that end the booking code: stable per booking and never repeating an order of another environment on the shared PayOS account. The link expires with the seat hold.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Create/reuse PayOS link” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Create a PayOS payment link for a PENDING booking. One payment per booking; asking again returns the existing link. The PayOS order code is the booking id × 1,000,000 plus the six random digits that end the booking code: stable per booking and never repeating an order of another environment on the shared PayOS account. The link expires with the seat hold. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-PAY-01-AC-01** — Given the preconditions/trigger, verify: Create a PayOS payment link for a PENDING booking. One payment per booking; asking again returns the existing link. The PayOS order code is the booking id × 1,000,000 plus the six random digits that end the booking code: stable per booking and never repeating an order of another environment on the shared PayOS account. The link expires with the seat hold. **WEB-FR-PAY-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/payment/screens/payment_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs

Continued on next page

Table 87 — continued

Field	Specification
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-PAYMENT, WEB-SCR-CHECKOUT; WEB-API-088

9.83 WEB-FR-PAY-02 — Verify PayOS

Table 88: WEB-FR-PAY-02 — Verify PayOS

Field	Specification
Name	Verify PayOS
Description	Verify a payment by order code (or booking id): paid at PayOS means payment SUCCESS and booking CONFIRMED; cancelled on the PayOS page means payment FAILED, booking CANCELLED and seats released at once. Calling it again is safe.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Verify PayOS” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Verify a payment by order code (or booking id): paid at PayOS means payment SUCCESS and booking CONFIRMED; cancelled on the PayOS page means payment FAILED, booking CANCELLED and seats released at once. Calling it again is safe. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-PAY-02-AC-01** — Given the preconditions/trigger, verify: Verify a payment by order code (or booking id): paid at PayOS means payment SUCCESS and booking CONFIRMED; cancelled on the PayOS page means payment FAILED, booking CANCELLED and seats released at once. Calling it again is safe. **WEB-FR-PAY-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/payment/screens/payment_screen.dart; Cinema_Mobile/lib/features/payment/screens/payment_result_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-PAYMENT, WEB-SCR-RESULT, WEB-SCR-CHECKOUT; WEB-API-087

9.84 WEB-FR-PAY-03 — Payment details

Table 89: WEB-FR-PAY-03 — Payment details

Field	Specification
Name	Payment details
Description	Show a payment: to the ticket owner or to Staff/Admin.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Payment details” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show a payment: to the ticket owner or to Staff/Admin. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-PAY-03-AC-01** — Given the preconditions/trigger, verify: Show a payment: to the ticket owner or to Staff/Admin. **WEB-FR-PAY-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/payment/screens/payment_history_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-TICKET, WEB-SCR-PAYHISTORY, WEB-SCR-RECORD-PAYMENTS; WEB-API-084

9.85 WEB-FR-PAY-04 — Own payment history

Table 90: WEB-FR-PAY-04 — Own payment history

Field	Specification
Name	Own payment history
Description	Customers see their payment history.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Own payment history” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Customers see their payment history. 3. Return/render the authoritative result and refresh affected data after success.

Continued on next page

Table 90 — continued

Field	Specification
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and `503` without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-PAY-04-AC-01`** — Given the preconditions/trigger, verify: Customers see their payment history. **`WEB-FR-PAY-04-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/payment/screens/payment_history_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs`
Business rules	`WEB-BR-09`, `WEB-BR-10`, `WEB-BR-11`, `WEB-BR-12`, `WEB-BR-13`, `WEB-BR-14`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-08`; `WEB-SCR-PAYHISTORY`; `WEB-API-095`

9.86 `WEB-FR-PAY-05` — Payments by booking

Table 91: `WEB-FR-PAY-05` — Payments by booking

Field	Specification
Name	Payments by booking
Description	Show the payments of a booking.
Actor	Booking owner; Staff/Admin for permitted operations; `PayOS`/system
Preconditions	A payable booking exists; `PayOS` credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Payments by booking” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show the payments of a booking. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and `503` without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-PAY-05-AC-01`** — Given the preconditions/trigger, verify: Show the payments of a booking. **`WEB-FR-PAY-05-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs`
Business rules	`WEB-BR-09`, `WEB-BR-10`, `WEB-BR-11`, `WEB-BR-12`, `WEB-BR-13`, `WEB-BR-14`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-08`; `WEB-SCR-RECORD-PAYMENTS`; `WEB-API-085`

9.87 WEB-FR-PAY-06 — Refund access

Table 92: WEB-FR-PAY-06 — Refund access

Field	Specification
Name	Refund access
Description	The booking owner requests a refund by cancelling an eligible paid booking; Staff/Admin review and complete or reject the manual-refund workflow, and Admin may initiate a manual payment refund through the API. Refund rules and states are specified in the REF requirements.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Refund access” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The booking owner requests a refund by cancelling an eligible paid booking; Staff/Admin review and complete or reject the manual-refund workflow, and Admin may initiate a manual payment refund through the API. Refund rules and states are specified in the REF requirements. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-PAY-06-AC-01** — Given the preconditions/trigger, verify: The booking owner requests a refund by cancelling an eligible paid booking; Staff/Admin review and complete or reject the manual-refund workflow, and Admin may initiate a manual payment refund through the API. Refund rules and states are specified in the REF requirements. **WEB-FR-PAY-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-TICKET; WEB-API-043, WEB-API-082, WEB-API-083

9.88 WEB-FR-PAY-07 — Signed webhook

Table 93: WEB-FR-PAY-07 — Signed webhook

Field	Specification
Name	Signed webhook
Description	Receive the PayOS webhook: check the HMAC-SHA256 signature (400 when wrong); an unknown order or wrong amount returns 200 and is ignored; a repeated webhook confirms and emails nothing twice. The payer’s account (number, name, bank) is stored for refunds.
Actor	Shared backend system; customer receives outcome

Continued on next page

Table 93 — continued

Field	Specification
Preconditions	The corresponding committed event, trusted `PayOS` result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Signed webhook” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Receive the `PayOS` webhook: check the `HMAC-SHA256` signature (`400` when wrong); an unknown order or wrong amount returns 200 and is ignored; a repeated webhook confirms and emails nothing twice. The payer’s account (number, name, bank) is stored for refunds. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and `503` without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-PAY-07-AC-01`** — Given the preconditions/trigger, verify: Receive the `PayOS` webhook: check the `HMAC-SHA256` signature (`400` when wrong); an unknown order or wrong amount returns 200 and is ignored; a repeated webhook confirms and emails nothing twice. The payer’s account (number, name, bank) is stored for refunds. **`WEB-FR-PAY-07-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs`
Business rules	`WEB-BR-09`, `WEB-BR-10`, `WEB-BR-11`, `WEB-BR-12`, `WEB-BR-13`, `WEB-BR-14`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-08`; ; `WEB-API-089`

9.89 `WEB-FR-PAY-08` — Late payment

Table 94: `WEB-FR-PAY-08` — Late payment

Field	Specification
Name	Late payment
Description	When `PayOS` reports payment after the hold expired, confirm the booking only if all seats can be reclaimed; otherwise create a full refund request. Repeated verification/webhook must not duplicate financial effects.
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted `PayOS` result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Late payment” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: When `PayOS` reports payment after the hold expired, confirm the booking only if all seats can be reclaimed; otherwise create a full refund request. Repeated verification/webhook must not duplicate financial effects. 3. Return/render the authoritative result and refresh affected data after success.

Continued on next page

Table 94 — continued

Field	Specification
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and `503` without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**`WEB-FR-PAY-08-AC-01`** — Given the preconditions/trigger, verify: When `PayOS` reports payment after the hold expired, confirm the booking only if all seats can be reclaimed; otherwise create a full refund request. Repeated verification/webhook must not duplicate financial effects. **`WEB-FR-PAY-08-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/payment/screens/payment_result_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs`
Business rules	`WEB-BR-09`, `WEB-BR-10`, `WEB-BR-11`, `WEB-BR-12`, `WEB-BR-13`, `WEB-BR-14`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-08`; `WEB-SCR-RESULT`; `WEB-API-087`, `WEB-API-089`

9.90 `WEB-FR-PAY-09` — Management payments

Table 95: `WEB-FR-PAY-09` — Management payments

Field	Specification
Name	Management payments
Description	Staff/Admin list payments and re-verify a pending `PayOS` payment.
Actor	Booking owner; Staff/Admin for permitted operations; `PayOS`/system
Preconditions	A payable booking exists; `PayOS` credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Management payments” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin list payments and re-verify a pending `PayOS` payment. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and `503` without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**`WEB-FR-PAY-09-AC-01`** — Given the preconditions/trigger, verify: Staff/Admin list payments and re-verify a pending `PayOS` payment. **`WEB-FR-PAY-09-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 95 — continued

Field	Specification
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-OPERATIONS, WEB-SCR-RECORD-PAYMENTS; WEB-API-087, WEB-API-096

9.91 WEB-FR-PAY-10 — Revenue summary

Table 96: WEB-FR-PAY-10 — Revenue summary

Field	Specification
Name	Revenue summary
Description	Admin sees revenue for a period: payments received, revenue (SUCCESS + REFUND_PENDING), amount refunded (REFUNDED).
Actor	Admin
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke “Revenue summary” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Admin sees revenue for a period: payments received, revenue (SUCCESS + REFUND_PENDING), amount refunded (REFUNDED). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-PAY-10-AC-01** — Given the preconditions/trigger, verify: Admin sees revenue for a period: payments received, revenue (SUCCESS + REFUND_PENDING), amount refunded (REFUNDED). **WEB-FR-PAY-10-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; ; WEB-API-094

9.92 WEB-FR-PAY-11 — Automatic payment polling

Table 97: WEB-FR-PAY-11 — Automatic payment polling

Field	Specification
Name	Automatic payment polling
Description	While open, the app's payment page checks the booking every 5 seconds and asks PayOS every 30 seconds; there is no "I have paid" button.
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Invoke "Automatic payment polling" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: While open, the app's payment page checks the booking every 5 seconds and asks PayOS every 30 seconds; there is no "I have paid" button. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-PAY-11-AC-01** — Given the preconditions/trigger, verify: While open, the app's payment page checks the booking every 5 seconds and asks PayOS every 30 seconds; there is no "I have paid" button. **WEB-FR-PAY-11-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/payment/screens/payment_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/PaymentService.cs; Cinema_BE/BookingService/BookingService/Helpers/PayOsSignature.cs
Business rules	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20
Use case / screens / APIs	UC-WEB-08; WEB-SCR-PAYMENT; WEB-API-046, WEB-API-087

9.93 WEB-FR-REF-01 — Create refund request

Table 98: WEB-FR-REF-01 — Create refund request

Field	Specification
Name	Create refund request
Description	A refund request (PENDING) is created when: a customer cancels a paid ticket in time; Staff/Admin cancel a paid ticket; a showtime is cancelled; a late payment lost its seats; an Admin refunds a payment manually. The payment becomes REFUND_PENDING and the customer gets an email.
Actor	Customer requests/reads own refund; Staff/Admin process; system creates requests
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke "Create refund request" or its specified backend event/worker.

Continued on next page

Table 98 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A refund request (`PENDING`) is created when: a customer cancels a paid ticket in time; Staff/Admin cancel a paid ticket; a showtime is cancelled; a late payment lost its seats; an Admin refunds a payment manually. The payment becomes `REFUND_PENDING` and the customer gets an email. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (`409`), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-REF-01-AC-01`** — Given the preconditions/trigger, verify: A refund request (`PENDING`) is created when: a customer cancels a paid ticket in time; Staff/Admin cancel a paid ticket; a showtime is cancelled; a late payment lost its seats; an Admin refunds a payment manually. The payment becomes `REFUND_PENDING` and the customer gets an email. **`WEB-FR-REF-01-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`
Business rules	`WEB-BR-17`, `WEB-BR-18`, `WEB-BR-19`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-09`; `WEB-SCR-TICKET`; `WEB-API-043`, `WEB-API-083`, `WEB-API-089`

9.94 `WEB-FR-REF-02` — Management refund list

Table 99: `WEB-FR-REF-02` — Management refund list

Field	Specification
Name	Management refund list
Description	Staff/Admin list refund requests (filter by status) and see their details: amount, reason, payer account.
Actor	Staff/Admin
Preconditions	A paid booking/payment exists; customer cancellation is at least two hours before start; staff processing requires a pending request.
Trigger	Invoke “Management refund list” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin list refund requests (filter by status) and see their details: amount, reason, payer account. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (`409`), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H

Continued on next page

Table 99 — continued

Field	Specification
Acceptance criteria	**WEB-FR-REF-02-AC-01** — Given the preconditions/trigger, verify: Staff/Admin list refund requests (filter by status) and see their details: amount, reason, payer account. **WEB-FR-REF-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs
Business rules	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20
Use case / screens / APIs	UC-WEB-09; WEB-SCR-OPERATIONS, WEB-SCR-RECORD-REFUNDS; WEB-API-092, WEB-API-093

9.95 WEB-FR-REF-03 — Record completed transfer

Table 100: WEB-FR-REF-03 — Record completed transfer

Field	Specification
Name	Record completed transfer
Description	After transferring the money, staff record the transfer reference: refund COMPLETED, payment REFUNDED, booking CANCELLED with seats released, and an email with the reference.
Actor	Staff/Admin
Preconditions	A paid booking/payment exists; customer cancellation is at least two hours before start; staff processing requires a pending request.
Trigger	Invoke “Record completed transfer” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: After transferring the money, staff record the transfer reference: refund COMPLETED, payment REFUNDED, booking CANCELLED with seats released, and an email with the reference. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (409), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-REF-03-AC-01** — Given the preconditions/trigger, verify: After transferring the money, staff record the transfer reference: refund COMPLETED, payment REFUNDED, booking CANCELLED with seats released, and an email with the reference. **WEB-FR-REF-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs
Business rules	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20
Use case / screens / APIs	UC-WEB-09; WEB-SCR-RECORD-REFUNDS; WEB-API-090

9.96 WEB-FR-REF-04 — Reject refund

Table 101: WEB-FR-REF-04 — Reject refund

Field	Specification
Name	Reject refund
Description	Staff reject a request (reason required, only while PENDING): the payment returns to SUCCESS and the booking to CONFIRMED with its seats; the customer sees the reason on the ticket and gets an email. Refunds of tickets the cinema cancelled cannot be rejected.
Actor	Staff/Admin
Preconditions	A paid booking/payment exists; customer cancellation is at least two hours before start; staff processing requires a pending request.
Trigger	Invoke “Reject refund” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff reject a request (reason required, only while PENDING): the payment returns to SUCCESS and the booking to CONFIRMED with its seats; the customer sees the reason on the ticket and gets an email. Refunds of tickets the cinema cancelled cannot be rejected. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (409), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-REF-04-AC-01** — Given the preconditions/trigger, verify: Staff reject a request (reason required, only while PENDING): the payment returns to SUCCESS and the booking to CONFIRMED with its seats; the customer sees the reason on the ticket and gets an email. Refunds of tickets the cinema cancelled cannot be rejected. **WEB-FR-REF-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs
Business rules	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20
Use case / screens / APIs	UC-WEB-09; WEB-SCR-RECORD-REFUNDS; WEB-API-091

9.97 WEB-FR-REF-05 — Reopen rejected refund

Table 102: WEB-FR-REF-05 — Reopen rejected refund

Field	Specification
Name	Reopen rejected refund
Description	A rejected request is reopened if the cinema later cancels the ticket or the showtime.
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.

Continued on next page

Table 102 — continued

Field	Specification
Trigger	Invoke “Reopen rejected refund” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A rejected request is reopened if the cinema later cancels the ticket or the showtime. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (`409`), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**`WEB-FR-REF-05-AC-01`** — Given the preconditions/trigger, verify: A rejected request is reopened if the cinema later cancels the ticket or the showtime. **`WEB-FR-REF-05-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart`; `Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs`; `Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`
Business rules	`WEB-BR-17`, `WEB-BR-18`, `WEB-BR-19`, `WEB-BR-20`
Use case / screens / APIs	`UC-WEB-09`; `WEB-SCR-TICKET`, `WEB-SCR-RECORD-REFUNDS`; `WEB-API-016`, `WEB-API-043`

9.98 `WEB-FR-REF-06` — Own refund status

Table 103: `WEB-FR-REF-06` — Own refund status

Field	Specification
Name	Own refund status
Description	Customers see the refund status of their ticket.
Actor	Customer requests/reads own refund; Staff/Admin process; system creates requests
Preconditions	A paid booking/payment exists; customer cancellation is at least two hours before start; staff processing requires a pending request.
Trigger	Invoke “Own refund status” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Customers see the refund status of their ticket. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (`409`), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-REF-06-AC-01`** — Given the preconditions/trigger, verify: Customers see the refund status of their ticket. **`WEB-FR-REF-06-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 103 — continued

Field	Specification
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs
Business rules	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20
Use case / screens / APIs	UC-WEB-09; WEB-SCR-TICKET, WEB-SCR-RECORD-PAYMENTS; WEB-API-082

9.99 WEB-FR-REF-07 — Unique refund per payment

Table 104: WEB-FR-REF-07 — Unique refund per payment

Field	Specification
Name	Unique refund per payment
Description	Each payment has only one refund request, even when an event is delivered again.
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Unique refund per payment” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Each payment has only one refund request, even when an event is delivered again. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject too-late customer cancellation (409), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-REF-07-AC-01** — Given the preconditions/trigger, verify: Each payment has only one refund request, even when an event is delivered again. **WEB-FR-REF-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/RefundService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs
Business rules	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20
Use case / screens / APIs	UC-WEB-09; WEB-SCR-RECORD-REFUNDS; WEB-API-043, WEB-API-082

9.100 WEB-FR-POS-01 — Create counter sale

Table 105: WEB-FR-POS-01 — Create counter sale

Field	Specification
Name	Create counter sale
Description	Staff/Admin sell tickets to a **walk-in guest** (no account; email and phone optional) or a **member** (account ID + email, or found by phone number; members use and earn points): choose a movie, a showtime in the next 7 days and seats on the map (sweetbox in pairs, priced by type).
Actor	Staff/Admin
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Invoke “Create counter sale” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin sell tickets to a **walk-in guest** (no account; email and phone optional) or a **member** (account ID + email, or found by phone number; members use and earn points): choose a movie, a showtime in the next 7 days and seats on the map (sweetbox in pairs, priced by type). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid PayOS link before switching to cash; if transfer already succeeded return 409. Without email provide print/on-screen QR.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-POS-01-AC-01** — Given the preconditions/trigger, verify: Staff/Admin sell tickets to a **walk-in guest** (no account; email and phone optional) or a **member** (account ID + email, or found by phone number; members use and earn points): choose a movie, a showtime in the next 7 days and seats on the map (sweetbox in pairs, priced by type). **WEB-FR-POS-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs; Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_BE/AuthService/AuthService/Controllers/MembersController.cs; Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-COUNTER; WEB-API-013, WEB-API-023, WEB-API-030, WEB-API-055

9.101 WEB-FR-POS-02 — Counter checkout

Table 106: WEB-FR-POS-02 — Counter checkout

Field	Specification
Name	Counter checkout
Description	After the seats are held, the **Take payment** screen shows the ticket, the customer, the total, the hold countdown and an optional email for the ticket.
Actor	Staff/Admin
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Invoke “Counter checkout” or its specified backend event/worker.

Continued on next page

Table 106 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: After the seats are held, the **Take payment** screen shows the ticket, the customer, the total, the hold countdown and an optional email for the ticket. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid `PayOS` link before switching to cash; if transfer already succeeded return `409`. Without email provide print/on-screen `QR`.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**`WEB-FR-POS-02-AC-01`** — Given the preconditions/trigger, verify: After the seats are held, the **Take payment** screen shows the ticket, the customer, the total, the hold countdown and an optional email for the ticket. **`WEB-FR-POS-02-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart`
Business rules	`WEB-BR-08`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-22`, `WEB-BR-23`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-10`; `WEB-SCR-CHECKOUT`; `WEB-API-046`

9.102 `WEB-FR-POS-03` — Cash and change

Table 107: `WEB-FR-POS-03` — Cash and change

Field	Specification
Name	Cash and change
Description	**Cash**: enter the cash received (with round-amount buttons); the screen shows the change and refuses less than the total. The server records a `CASH` payment as `SUCCESS` (reference `CASH-<booking code>`) and confirms the ticket.
Actor	Staff/Admin
Preconditions	A management session is active; an `OPEN` future showtime has free seats.
Trigger	Invoke “Cash and change” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Cash**: enter the cash received (with round-amount buttons); the screen shows the change and refuses less than the total. The server records a `CASH` payment as `SUCCESS` (reference `CASH-<booking code>`) and confirms the ticket. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid `PayOS` link before switching to cash; if transfer already succeeded return `409`. Without email provide print/on-screen `QR`.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-POS-03-AC-01`** — Given the preconditions/trigger, verify: **Cash**: enter the cash received (with round-amount buttons); the screen shows the change and refuses less than the total. The server records a `CASH` payment as `SUCCESS` (reference `CASH-<booking code>`) and confirms the ticket. **`WEB-FR-POS-03-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 107 — continued

Field	Specification
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-CHECKOUT; WEB-API-086

9.103 WEB-FR-POS-04 — Bank-transfer QR

Table 108: WEB-FR-POS-04 — Bank-transfer QR

Field	Specification
Name	Bank-transfer QR
Description	**Bank transfer**: show the PayOS VietQR for the customer's banking app (for a link made earlier, a QR code of its checkout page); the screen checks every 3 seconds and switches to paid when the money arrives.
Actor	Staff/Admin
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Invoke "Bank-transfer QR" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Bank transfer**: show the PayOS VietQR for the customer's banking app (for a link made earlier, a QR code of its checkout page); the screen checks every 3 seconds and switches to paid when the money arrives. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid PayOS link before switching to cash; if transfer already succeeded return 409. Without email provide print/on-screen QR.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-POS-04-AC-01** — Given the preconditions/trigger, verify: **Bank transfer**: show the PayOS VietQR for the customer's banking app (for a link made earlier, a QR code of its checkout page); the screen checks every 3 seconds and switches to paid when the money arrives. **WEB-FR-POS-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-CHECKOUT; WEB-API-087, WEB-API-088

9.104 WEB-FR-POS-05 — Switch transfer to cash

Table 109: WEB-FR-POS-05 — Switch transfer to cash

Field	Specification
Name	Switch transfer to cash

Continued on next page

Table 109 — continued

Field	Specification
Description	Switching from bank transfer to cash follows `WEB-BR-22`: the `PayOS` link is cancelled and that payment becomes `FAILED`; if the customer already transferred, cash returns `409`.
Actor	Staff/Admin
Preconditions	A management session is active; an `OPEN` future showtime has free seats.
Trigger	Invoke “Switch transfer to cash” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Switching from bank transfer to cash follows `WEB-BR-22`: the `PayOS` link is cancelled and that payment becomes `FAILED`; if the customer already transferred, cash returns `409`. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid `PayOS` link before switching to cash; if transfer already succeeded return `409`. Without email provide print/on-screen `QR`.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**`WEB-FR-POS-05-AC-01`** — Given the preconditions/trigger, verify: Switching from bank transfer to cash follows `WEB-BR-22`: the `PayOS` link is cancelled and that payment becomes `FAILED`; if the customer already transferred, cash returns `409`. **`WEB-FR-POS-05-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart`
Business rules	`WEB-BR-08`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-22`, `WEB-BR-23`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-10`; `WEB-SCR-CHECKOUT`; `WEB-API-086`

9.105 `WEB-FR-POS-06` — Retain counter email

Table 110: `WEB-FR-POS-06` — Retain counter email

Field	Specification
Name	Retain counter email
Description	An email given at the counter is saved on the booking, so later cancellation and refund emails reach the customer too.
Actor	Staff/Admin
Preconditions	A management session is active; an `OPEN` future showtime has free seats.
Trigger	Invoke “Retain counter email” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: An email given at the counter is saved on the booking, so later cancellation and refund emails reach the customer too. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid `PayOS` link before switching to cash; if transfer already succeeded return `409`. Without email provide print/on-screen `QR`.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M

Continued on next page

Table 110 — continued

Field	Specification
Acceptance criteria	**WEB-FR-POS-06-AC-01** — Given the preconditions/trigger, verify: An email given at the counter is saved on the booking, so later cancellation and refund emails reach the customer too. **WEB-FR-POS-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-CHECKOUT; WEB-API-055, WEB-API-086

9.106 WEB-FR-POS-07 — Print and admit

Table 111: WEB-FR-POS-07 — Print and admit

Field	Specification
Name	Print and admit
Description	After payment: the ticket QR code; **Print ticket** (web, 80 mm receipt paper); **Check in now** when the show starts within 3 hours; **Sell another ticket**. Without an email, print the ticket or let the customer photograph the QR code.
Actor	Staff/Admin
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Invoke “Print and admit” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: After payment: the ticket QR code; **Print ticket** (web, 80 mm receipt paper); **Check in now** when the show starts within 3 hours; **Sell another ticket**. Without an email, print the ticket or let the customer photograph the QR code. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid PayOS link before switching to cash; if transfer already succeeded return 409. Without email provide print/on-screen QR.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-POS-07-AC-01** — Given the preconditions/trigger, verify: After payment: the ticket QR code; **Print ticket** (web, 80 mm receipt paper); **Check in now** when the show starts within 3 hours; **Sell another ticket**. Without an email, print the ticket or let the customer photograph the QR code. **WEB-FR-POS-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-CHECKOUT; WEB-API-044, WEB-API-046

9.107 WEB-FR-POS-08 — Reopen paid ticket

Table 112: WEB-FR-POS-08 — Reopen paid ticket

Field	Specification
Name	Reopen paid ticket
Description	Paid tickets have **Show ticket** on their booking page, to print again or check in.
Actor	Staff/Admin
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Invoke “Reopen paid ticket” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Paid tickets have **Show ticket** on their booking page, to print again or check in. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid PayOS link before switching to cash; if transfer already succeeded return 409. Without email provide print/on-screen QR.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-POS-08-AC-01** — Given the preconditions/trigger, verify: Paid tickets have **Show ticket** on their booking page, to print again or check in. **WEB-FR-POS-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart
Business rules	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29
Use case / screens / APIs	UC-WEB-10; WEB-SCR-CHECKOUT, WEB-SCR-RECORD-BOOKINGS; WEB-API-046

9.108 WEB-FR-CHK-01 — Ticket QR

Table 113: WEB-FR-CHK-01 — Ticket QR

Field	Specification
Name	Ticket QR
Description	A paid ticket has a QR code holding its booking code, shown in the customer app, in the confirmation email and on the printed ticket.
Actor	Customer
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Invoke “Ticket QR” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: A paid ticket has a QR code holding its booking code, shown in the customer app, in the confirmation email and on the printed ticket. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.

Continued on next page

Table 113 — continued

Field	Specification
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-CHK-01-AC-01** — Given the preconditions/trigger, verify: A paid ticket has a QR code holding its booking code, shown in the customer app, in the confirmation email and on the printed ticket. **WEB-FR-CHK-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Configuration/CheckInOptions.cs
Business rules	WEB-BR-21
Use case / screens / APIs	UC-WEB-11; WEB-SCR-TICKET, WEB-SCR-CHECKOUT; WEB-API-046

9.109 WEB-FR-CHK-02 — Lookup ticket code

Table 114: WEB-FR-CHK-02 — Lookup ticket code

Field	Specification
Name	Lookup ticket code
Description	Staff/Admin look a ticket up by code (case-insensitive): movie, time, room, seats, total, status.
Actor	Staff/Admin
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Invoke “Lookup ticket code” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin look a ticket up by code (case-insensitive): movie, time, room, seats, total, status. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-CHK-02-AC-01** — Given the preconditions/trigger, verify: Staff/Admin look a ticket up by code (case-insensitive): movie, time, room, seats, total, status. **WEB-FR-CHK-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Configuration/CheckInOptions.cs
Business rules	WEB-BR-21
Use case / screens / APIs	UC-WEB-11; WEB-SCR-CHECKIN; WEB-API-047

9.110 WEB-FR-CHK-03 — Check-in validation

Table 115: WEB-FR-CHK-03 — Check-in validation

Field	Specification
Name	Check-in validation
Description	Check-in follows WEB-BR-21; otherwise 409 with the reason (unpaid, cancelled, refund pending, too early, too late, already checked in at a given time). When two counters scan the same ticket, the second gets 409.
Actor	Staff/Admin
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Invoke “Check-in validation” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Check-in follows WEB-BR-21; otherwise 409 with the reason (unpaid, cancelled, refund pending, too early, too late, already checked in at a given time). When two counters scan the same ticket, the second gets 409. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.
Postconditions	Only one check-in timestamp is stored; check-in is an attribute, not another booking status.
Priority	H
Acceptance criteria	**WEB-FR-CHK-03-AC-01** — Given the preconditions/trigger, verify: Check-in follows WEB-BR-21; otherwise 409 with the reason (unpaid, cancelled, refund pending, too early, too late, already checked in at a given time). When two counters scan the same ticket, the second gets 409. **WEB-FR-CHK-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Configuration/CheckInOptions.cs
Business rules	WEB-BR-21
Use case / screens / APIs	UC-WEB-11; WEB-SCR-CHECKIN, WEB-SCR-CHECKOUT; WEB-API-044

9.111 WEB-FR-CHK-04 — Camera/manual scanner

Table 116: WEB-FR-CHK-04 — Camera/manual scanner

Field	Specification
Name	Camera/manual scanner
Description	Scan the QR code with a phone camera (app), a laptop webcam or USB camera (admin website), or type the code / use a handheld scanner (code + Enter).
Actor	Staff/Admin
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Invoke “Camera/manual scanner” or its specified backend event/worker.

Continued on next page

Table 116 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Scan the QR code with a phone camera (app), a laptop webcam or USB camera (admin website), or type the code / use a handheld scanner (code + Enter). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.
Postconditions	Only one check-in timestamp is stored; check-in is an attribute, not another booking status.
Priority	H
Acceptance criteria	**WEB-FR-CHK-04-AC-01** — Given the preconditions/trigger, verify: Scan the QR code with a phone camera (app), a laptop webcam or USB camera (admin website), or type the code / use a handheld scanner (code + Enter). **WEB-FR-CHK-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Configuration/CheckInOptions.cs
Business rules	WEB-BR-21
Use case / screens / APIs	UC-WEB-11; WEB-SCR-CHECKIN; WEB-API-044, WEB-API-047

9.112 WEB-FR-CHK-05 — Display check-in time

Table 117: WEB-FR-CHK-05 — Display check-in time

Field	Specification
Name	Display check-in time
Description	The check-in time shows on the customer’s ticket, the booking page, the seat map and the showtime’s customer list.
Actor	Staff/Admin
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Invoke “Display check-in time” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The check-in time shows on the customer’s ticket, the booking page, the seat map and the showtime’s customer list. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-CHK-05-AC-01** — Given the preconditions/trigger, verify: The check-in time shows on the customer’s ticket, the booking page, the seat map and the showtime’s customer list. **WEB-FR-CHK-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.

Continued on next page

Table 117 — continued

Field	Specification
Evidence	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart; Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs; Cinema_BE/BookingService/BookingService/Configuration/CheckInOptions.cs
Business rules	WEB-BR-21
Use case / screens / APIs	UC-WEB-11; WEB-SCR-TICKET, WEB-SCR-CHECKIN, WEB-SCR-RECORD-BOOKINGS; WEB-API-046, WEB-API-050

9.113 WEB-FR-NOTI-01 — Notification history

Table 118: WEB-FR-NOTI-01 — Notification history

Field	Specification
Name	Notification history
Description	Customers see their notifications and details and open the related ticket. Admins see all notifications.
Actor	Customer reads own; Staff/Admin send permitted messages; system dispatches events
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Invoke “Notification history” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Customers see their notifications and details and open the related ticket. Admins see all notifications. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user’s data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-NOTI-01-AC-01** — Given the preconditions/trigger, verify: Customers see their notifications and details and open the related ticket. Admins see all notifications. **WEB-FR-NOTI-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_Mobile/lib/features/notification/screens/notification_list_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; WEB-SCR-NOTIFICATIONS; WEB-API-078, WEB-API-079

9.114 WEB-FR-NOTI-02 — Compose booking message

Table 119: WEB-FR-NOTI-02 — Compose booking message

Field	Specification
Name	Compose booking message
Description	Staff/Admin write an email notification to a ticket's customer (walk-in guests with an email included).
Actor	Staff/Admin
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Invoke "Compose booking message" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Staff/Admin write an email notification to a ticket's customer (walk-in guests with an email included). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-NOTI-02-AC-01** — Given the preconditions/trigger, verify: Staff/Admin write an email notification to a ticket's customer (walk-in guests with an email included). **WEB-FR-NOTI-02-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_Mobile/lib/features/staff/screens/staff_notification_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; WEB-SCR-COMPOSE; WEB-API-080

9.115 WEB-FR-NOTI-03 — Send/retry message

Table 120: WEB-FR-NOTI-03 — Send/retry message

Field	Specification
Name	Send/retry message
Description	Send a notification; a failure is stored as FAILED with the error, and the same notification can be sent again.
Actor	Staff/Admin
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Invoke "Send/retry message" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Send a notification; a failure is stored as FAILED with the error, and the same notification can be sent again. 3. Return/render the authoritative result and refresh affected data after success.

Continued on next page

Table 120 — continued

Field	Specification
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains `FAILED` for retry and must not undo payment. Event deduplication reduces duplicates but `SMTP` plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**`WEB-FR-NOTI-03-AC-01`** — Given the preconditions/trigger, verify: Send a notification; a failure is stored as `FAILED` with the error, and the same notification can be sent again. **`WEB-FR-NOTI-03-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_notification_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs`; `Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs`
Business rules	`WEB-BR-11`, `WEB-BR-13`
Use case / screens / APIs	`UC-WEB-12`; `WEB-SCR-COMPOSE`; `WEB-API-073`

9.116 `WEB-FR-NOTI-04` — Notification details

Table 121: `WEB-FR-NOTI-04` — Notification details

Field	Specification
Name	Notification details
Description	Show a notification.
Actor	Customer reads own; Staff/Admin send permitted messages; system dispatches events
Preconditions	An account session exists for in-app history; `SMTP`/event processing is configured for actual delivery.
Trigger	Invoke "Notification details" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Show a notification. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains `FAILED` for retry and must not undo payment. Event deduplication reduces duplicates but `SMTP` plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-NOTI-04-AC-01`** — Given the preconditions/trigger, verify: Show a notification. **`WEB-FR-NOTI-04-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs`; `Cinema_Mobile/lib/features/notification/screens/notification_detail_screen.dart`; `Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs`; `Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs`
Business rules	`WEB-BR-11`, `WEB-BR-13`
Use case / screens / APIs	`UC-WEB-12`; `WEB-SCR-NOTIFICATION`; `WEB-API-075`

9.117 WEB-FR-NOTI-05 — Delete notification API

Table 122: WEB-FR-NOTI-05 — Delete notification API

Field	Specification
Name	Delete notification API
Description	Delete a notification: the recipient deletes their own, an Admin any (API only).
Actor	Customer reads own; Staff/Admin send permitted messages; system dispatches events
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Invoke “Delete notification API” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Delete a notification: the recipient deletes their own, an Admin any (API only). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user’s data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	L
Acceptance criteria	**WEB-FR-NOTI-05-AC-01** — Given the preconditions/trigger, verify: Delete a notification: the recipient deletes their own, an Admin any (API only). **WEB-FR-NOTI-05-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; ; WEB-API-074

9.118 WEB-FR-NOTI-06 — Event emails

Table 123: WEB-FR-NOTI-06 — Event emails

Field	Specification
Name	Event emails
Description	Consume committed booking/payment events to send confirmation with inline QR, cancellation, expiry and refund emails. Deduplicate by event ID and skip tickets without email. Delivery is retryable; SMTP and database commit do not guarantee absolute exactly-once delivery after a crash.
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Event emails” or its specified backend event/worker.

Continued on next page

Table 123 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Consume committed booking/payment events to send confirmation with inline QR, cancellation, expiry and refund emails. Deduplicate by event ID and skip tickets without email. Delivery is retryable; SMTP and database commit do not guarantee absolute exactly-once delivery after a crash. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	H
Acceptance criteria	**WEB-FR-NOTI-06-AC-01** — Given the preconditions/trigger, verify: Consume committed booking/payment events to send confirmation with inline QR, cancellation, expiry and refund emails. Deduplicate by event ID and skip tickets without email. Delivery is retryable; SMTP and database commit do not guarantee absolute exactly-once delivery after a crash. **WEB-FR-NOTI-06-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; ; WEB-API-078

9.119 WEB-FR-NOTI-07 — OTP and welcome email

Table 124: WEB-FR-NOTI-07 — OTP and welcome email

Field	Specification
Name	OTP and welcome email
Description	AuthService emails OTP codes (verification, password reset) and a welcome email.
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke "OTP and welcome email" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Auth Service emails OTP codes (verification, password reset) and a welcome email. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H

Continued on next page

Table 124 — continued

Field	Specification
Acceptance criteria	**WEB-FR-NOTI-07-AC-01** — Given the preconditions/trigger, verify: AuthService emails OTP codes (verification, password reset) and a welcome email. **WEB-FR-NOTI-07-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; ; WEB-API-026, WEB-API-033, WEB-API-042

9.120 WEB-FR-NOTI-08 — Showtime reminder

Table 125: WEB-FR-NOTI-08 — Showtime reminder

Field	Specification
Name	Showtime reminder
Description	**Showtime reminders**: a background job (every minute, ShowtimeReminder settings) emails and records a notification for paid tickets not checked in whose show starts within 2 hours; each ticket once. Tickets bought inside that window are skipped (the customer just got the confirmation).
Actor	Shared backend system; customer receives outcome
Preconditions	The corresponding committed event, trusted PayOS result or scheduled-worker condition exists; backend processing is configured. No customer session is required for a background worker or signed webhook.
Trigger	Invoke “Showtime reminder” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Showtime reminders**: a background job (every minute, ShowtimeReminder settings) emails and records a notification for paid tickets not checked in whose show starts within 2 hours; each ticket once. Tickets bought inside that window are skipped (the customer just got the confirmation). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user’s data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-NOTI-08-AC-01** — Given the preconditions/trigger, verify: **Showtime reminders**: a background job (every minute, ShowtimeReminder settings) emails and records a notification for paid tickets not checked in whose show starts within 2 hours; each ticket once. Tickets bought inside that window are skipped (the customer just got the confirmation). **WEB-FR-NOTI-08-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13

Continued on next page

Table 125 — continued

Field	Specification
Use case / screens / APIs	UC-WEB-12; ; WEB-API-078

9.121 WEB-FR-NOTI-09 — Read/unread notifications

Table 126: WEB-FR-NOTI-09 — Read/unread notifications

Field	Specification
Name	Read/unread notifications
Description	**Read / unread**: unopened notifications have a dot and bold title; opening one marks it read; **Mark all read** button. The app bell and the website Notifications link show how many are unread. Only the owner can mark them (403).
Actor	Customer reads own; Staff/Admin send permitted messages; system dispatches events
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Invoke “Read/unread notifications” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Read / unread**: unopened notifications have a dot and bold title; opening one marks it read; **Mark all read** button. The app bell and the website Notifications link show how many are unread. Only the owner can mark them (403). 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Owner checks reject another user’s data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**WEB-FR-NOTI-09-AC-01** — Given the preconditions/trigger, verify: **Read / unread**: unopened notifications have a dot and bold title; opening one marks it read; **Mark all read** button. The app bell and the website Notifications link show how many are unread. Only the owner can mark them (403). **WEB-FR-NOTI-09-AC-02** — Exercise applicable invalid/ unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs; Cinema_Mobile/lib/features/home/screens/home_screen.dart; Cinema_Mobile/lib/features/notification/screens/notification_list_screen.dart; Cinema_Mobile/lib/features/notification/screens/notification_detail_screen.dart; Cinema_BE/BookingService/BookingService/Services/Implementations/NotificationService.cs; Cinema_BE/BookingService/BookingService/Workers/ShowtimeReminderWorker.cs
Business rules	WEB-BR-11, WEB-BR-13
Use case / screens / APIs	UC-WEB-12; WEB-SCR-HOME, WEB-SCR-NOTIFICATIONS, WEB-SCR-NOTIFICATION; WEB-API-072, WEB-API-076, WEB-API-077

9.122 WEB-FR-DASH-01 — Today totals

Table 127: WEB-FR-DASH-01 — Today totals

Field	Specification
Name	Today totals
Description	**Today** block (Staff and Admin): bookings made today; today's showtimes with bookings (x / total); tickets for today's shows; how many checked in.
Actor	Staff/Admin
Preconditions	Management session and source records exist; date ranges use cinema local time.
Trigger	Invoke "Today totals" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: **Today** block (Staff and Admin): bookings made today; today's showtimes with bookings (x / total); tickets for today's shows; how many checked in. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show zero/empty periods accurately; a card can show its own error and retry. Staff must not acquire revenue/account APIs by changing the URL.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-DASH-01-AC-01** — Given the preconditions/trigger, verify: **Today** block (Staff and Admin): bookings made today; today's showtimes with bookings (x / total); tickets for today's shows; how many checked in. **WEB-FR-DASH-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs; Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs; Cinema_Mobile/lib/features/staff/screens/staff_dashboard_screen.dart; Cinema_Mobile/lib/features/admin/screens/admin_dashboard_screen.dart
Business rules	WEB-BR-09, WEB-BR-23
Use case / screens / APIs	UC-WEB-13; WEB-SCR-STAFF-DASH, WEB-SCR-ADMIN-DASH; WEB-API-019, WEB-API-048, WEB-API-051

9.123 WEB-FR-DASH-02 — Live showtime board

Table 128: WEB-FR-DASH-02 — Live showtime board

Field	Specification
Name	Live showtime board
Description	Showtime board: **Now showing** (every show playing, watching now = checked in, tickets sold / capacity, time left); **Up next** (next 3 shows: booked, awaiting payment, checked in early); **Just finished** (last 3 shows: tickets sold, people who came). Clicking a show opens its seat map and customer list. Refreshes every minute.
Actor	Staff/Admin
Preconditions	Management session and source records exist; date ranges use cinema local time.
Trigger	Invoke "Live showtime board" or its specified backend event/worker.

Continued on next page

Table 128 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Showtime board: **Now showing** (every show playing, watching now = checked in, tickets sold / capacity, time left); **Up next** (next 3 shows: booked, awaiting payment, checked in early); **Just finished** (last 3 shows: tickets sold, people who came). Clicking a show opens its seat map and customer list. Refreshes every minute. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show zero/empty periods accurately; a card can show its own error and retry. Staff must not acquire revenue/account APIs by changing the `URL`.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**`WEB-FR-DASH-02-AC-01`** — Given the preconditions/trigger, verify: Showtime board: **Now showing** (every show playing, watching now = checked in, tickets sold / capacity, time left); **Up next** (next 3 shows: booked, awaiting payment, checked in early); **Just finished** (last 3 shows: tickets sold, people who came). Clicking a show opens its seat map and customer list. Refreshes every minute. **`WEB-FR-DASH-02-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_dashboard_screen.dart`; `Cinema_Mobile/lib/features/admin/screens/admin_dashboard_screen.dart`
Business rules	`WEB-BR-09`, `WEB-BR-23`
Use case / screens / APIs	`UC-WEB-13`; `WEB-SCR-STAFF-DASH`, `WEB-SCR-ADMIN-DASH`; `WEB-API-019`, `WEB-API-051`

9.124 `WEB-FR-DASH-03` — Staff attention queue

Table 129: `WEB-FR-DASH-03` — Staff attention queue

Field	Specification
Name	Staff attention queue
Description	The Staff dashboard has **Needs attention**: unpaid bookings and pending refund requests; shortcuts to **Sell tickets** and **Check in**.
Actor	Staff/Admin
Preconditions	Management session and source records exist; date ranges use cinema local time.
Trigger	Invoke “Staff attention queue” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The Staff dashboard has **Needs attention**: unpaid bookings and pending refund requests; shortcuts to **Sell tickets** and **Check in**. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show zero/empty periods accurately; a card can show its own error and retry. Staff must not acquire revenue/account APIs by changing the `URL`.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M

Continued on next page

Table 129 — continued

Field	Specification
Acceptance criteria	**WEB-FR-DASH-03-AC-01** — Given the preconditions/trigger, verify: The Staff dashboard has **Needs attention**: unpaid bookings and pending refund requests; shortcuts to **Sell tickets** and **Check in**. **WEB-FR-DASH-03-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/staff/screens/staff_dashboard_screen.dart`
Business rules	`WEB-BR-09`, `WEB-BR-23`
Use case / screens / APIs	`UC-WEB-13`; `WEB-SCR-STAFF-DASH`; `WEB-API-052`, `WEB-API-093`

9.125 `WEB-FR-DASH-04` — Admin monthly overview

Table 130: `WEB-FR-DASH-04` — Admin monthly overview

Field	Specification
Name	Admin monthly overview
Description	The Admin dashboard: this month's bookings and revenue, amount refunded, movies now showing, today's showtimes, account statistics (total, active, locked). Each part loads on its own, so one failure does not break the others.
Actor	Staff/Admin
Preconditions	Management session and source records exist; date ranges use cinema local time.
Trigger	Invoke "Admin monthly overview" or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: The Admin dashboard: this month's bookings and revenue, amount refunded, movies now showing, today's showtimes, account statistics (total, active, locked). Each part loads on its own, so one failure does not break the others. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show zero/empty periods accurately; a card can show its own error and retry. Staff must not acquire revenue/account APIs by changing the `URL`.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	M
Acceptance criteria	**WEB-FR-DASH-04-AC-01** — Given the preconditions/trigger, verify: The Admin dashboard: this month's bookings and revenue, amount refunded, movies now showing, today's showtimes, account statistics (total, active, locked). Each part loads on its own, so one failure does not break the others. **WEB-FR-DASH-04-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs`; `Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs`; `Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs`; `Cinema_Mobile/lib/features/admin/screens/admin_dashboard_screen.dart`
Business rules	`WEB-BR-09`, `WEB-BR-23`
Use case / screens / APIs	`UC-WEB-13`; `WEB-SCR-ADMIN-DASH`; `WEB-API-040`, `WEB-API-048`, `WEB-API-069`, `WEB-API-094`

9.126 WEB-FR-NAV-01 — Route guards and recovery

Table 131: WEB-FR-NAV-01 — Route guards and recovery

Field	Specification
Name	Route guards and recovery
Description	Guard named routes: a signed-out protected destination opens login with its pending route; successful sign-in resumes an allowed destination. Unknown routes fall back to the role landing. A summary lacking its in-memory BookingDraft returns to seat selection.
Actor	Guest; signed-in account; system
Preconditions	The account form is reachable and gateway/AuthService are configured; no prior signed-in session is required for registration/login/OTP/recovery. SMTP is required when sending a code.
Trigger	Invoke “Route guards and recovery” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Guard named routes: a signed-out protected destination opens login with its pending route; successful sign-in resumes an allowed destination. Unknown routes fall back to the role landing. A summary lacking its in-memory BookingDraft returns to seat selection. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/dependency failure (503).
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	H
Acceptance criteria	**WEB-FR-NAV-01-AC-01** — Given the preconditions/trigger, verify: Guard named routes: a signed-out protected destination opens login with its pending route; successful sign-in resumes an allowed destination. Unknown routes fall back to the role landing. A summary lacking its in-memory BookingDraft returns to seat selection. **WEB-FR-NAV-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_Mobile/lib/features/auth/screens/login_screen.dart; Cinema_Mobile/lib/features/admin/screens/staff_only_screen.dart; Cinema_BE/AuthService/AuthService/Service/AuthService.cs; Cinema_BE/ApiGateway/ApiGateway/Program.cs
Business rules	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25
Use case / screens / APIs	UC-WEB-01; WEB-SCR-LOGIN, WEB-SCR-STAFF-ONLY; Client behaviour; no dedicated endpoint

9.127 WEB-FR-STATE-01 — Loading, error and empty states

Table 132: WEB-FR-STATE-01 — Loading, error and empty states

Field	Specification
Name	Loading, error and empty states
Description	Provide loading, error/retry and empty states on data screens; disable repeated submission while a mutation is running and preserve an honest pending/failed result when the network fails.
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Preconditions	A data screen starts loading or a request is in progress.
Trigger	Invoke “Loading, error and empty states” or its specified backend event/worker.

Continued on next page

Table 132 — continued

Field	Specification
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Provide loading, error/retry and empty states on data screens; disable repeated submission while a mutation is running and preserve an honest pending/failed result when the network fails. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	A concurrent hold returns `409` and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	The successful observable result matches this requirement; any specified state change is persisted. Rejection does not grant access or leave an unauthorized partial change.
Priority	M
Acceptance criteria	**`WEB-FR-STATE-01-AC-01`** — Given the preconditions/trigger, verify: Provide loading, error/retry and empty states on data screens; disable repeated submission while a mutation is running and preserve an honest pending/failed result when the network fails. **`WEB-FR-STATE-01-AC-02`** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	`Cinema_BE/BookingService/BookingService/Services/Implementations/BookingService.cs`; `Cinema_BE/BookingService/BookingService/Helpers/PromotionRules.cs`; `Cinema_BE/BookingService/BookingService/Helpers/LoyaltyRules.cs`
Business rules	`WEB-BR-07`, `WEB-BR-08`, `WEB-BR-14`, `WEB-BR-15`, `WEB-BR-16`, `WEB-BR-17`, `WEB-BR-27`, `WEB-BR-28`, `WEB-BR-29`
Use case / screens / APIs	`UC-WEB-07`; ; Client behaviour; no dedicated endpoint

9.128 `WEB-FR-MEDIA-01` — Platform-specific movie media

Table 133: `WEB-FR-MEDIA-01` — Platform-specific movie media

Field	Specification
Name	Platform-specific movie media
Description	Web movie cards may preview a direct `HTTP`(S) `MP4`/`WebM` after 450 ms hover; stop on pointer exit and fall back to the poster, with a 12-second deadline and reduced-motion support. Native mobile has `supportsTrailerPreview`=false and opens the full trailer externally.
Actor	Admin
Preconditions	An Admin session is active; a target record exists for edit/deactivate/adjust operations.
Trigger	Invoke “Platform-specific movie media” or its specified backend event/worker.
Main flow	1. Read current input/resource and the applicable session. 2. Execute this operation: Web movie cards may preview a direct `HTTP`(S) `MP4`/`WebM` after 450 ms hover; stop on pointer exit and fall back to the poster, with a 12-second deadline and reduced-motion support. Native mobile has `supportsTrailerPreview`=false and opens the full trailer externally. 3. Return/render the authoritative result and refresh affected data after success.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user’s ticket returns `404`; a ticket not checked in returns `400`. Failed media must retain a poster/fallback.
Postconditions	The requested authorized data/state is returned or displayed; a read does not alter booking/payment state. Notification read markers are changed only by their separate write operation.
Priority	L

Continued on next page

Table 133 — continued

Field	Specification
Acceptance criteria	**WEB-FR-MEDIA-01-AC-01** — Given the preconditions/trigger, verify: Web movie cards may preview a direct HTTP(S) MP4/WebM after 450 ms hover; stop on pointer exit and fall back to the poster, with a 12-second deadline and reduced-motion support. Native mobile has supportsTrailerPreview=false and opens the full trailer externally. **WEB-FR-MEDIA-01-AC-02** — Exercise applicable invalid/unauthorized/conflicting input and verify the stated rejection, no unauthorized changes and no false success.
Evidence	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs; Cinema_Mobile/lib/features/home/screens/home_screen.dart; Cinema_BE/MovieService/MovieService/Service/Impl/MovieService.cs; Cinema_BE/BookingService/BookingService/Services/Implementations/MovieReviewService.cs
Business rules	WEB-BR-26
Use case / screens / APIs	UC-WEB-03; WEB-SCR-HOME; WEB-API-064

10. Use Case Specifications

10.1 UC-WEB-01 — Authentication and account session

Table 134: UC-WEB-01 — Authentication and account session

Field	Use-case specification
Actor	Guest; signed-in account; system
Goal	Authentication and account session
Preconditions	The gateway is configured. Registration/recovery needs SMTP; protected operations require a valid account session.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Enter account fields or request the session operation. 2. Validate fields; submit to AuthService through the gateway. 3. For verification/recovery, enter the emailed OTP; for sign-in/refresh, store returned tokens and user. 4. Route by role and resume an allowed pending destination.
Alternative / exception flow	Reject invalid fields (400), bad credentials (401), unverified email (403), stale/incorrect OTP, or resend/rate limits (429). Clear the session for account-lock/role-change responses; show SMTP/ dependency failure (503).
Postconditions	Only a successful token-issuing operation authenticates the account; rejected requests do not grant access.
Requirements / priority	WEB-FR-AUTH-01 (H), WEB-FR-AUTH-02 (H), WEB-FR-AUTH-03 (H), WEB-FR-AUTH-04 (H), WEB-FR-AUTH-05 (H), WEB-FR-AUTH-06 (H), WEB-FR-AUTH-07 (H), WEB-FR-AUTH-08 (M), WEB-FR-AUTH-09 (H), WEB-FR-AUTH-10 (H), WEB-FR-AUTH-11 (M), WEB-FR-AUTH-12 (H), WEB-FR-AUTH-13 (H), WEB-FR-AUTH-14 (M), WEB-FR-AUTH-15 (M), WEB-FR-AUTH-16 (M), WEB-FR-AUTH-17 (M), WEB-FR-NAV-01 (H)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.2 UC-WEB-02 — Account administration

Table 135: UC-WEB-02 — Account administration

Field	Use-case specification
Actor	Admin
Goal	Account administration
Preconditions	An Admin session is active; the target user exists for changes.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Search/filter the paged accounts or open a record. 2. Create an account, select roles, or change enabled status. 3. Validate and confirm the change. 4. Reload the account and apply session revocation where required.
Alternative / exception flow	Reject non-Admin callers; reject invalid fields, duplicate email and self-locking. Saving unchanged roles must not force reauthentication.
Postconditions	The saved account/roles/status is visible; locked or changed-role tokens cannot retain previous privileges.
Requirements / priority	WEB-FR-USER-01 (H), WEB-FR-USER-02 (H), WEB-FR-USER-03 (H), WEB-FR-USER-04 (H), WEB-FR-USER-05 (H), WEB-FR-USER-06 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.3 UC-WEB-03 — Movie catalogue and reviews

Table 136: UC-WEB-03 — Movie catalogue and reviews

Field	Use-case specification
Actor	Guest/Customer for browsing; Admin for catalogue changes and moderation
Goal	Movie catalogue and reviews
Preconditions	A movie is available; changes need Admin; a review needs the owner's checked-in ticket.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Browse/search/filter movies or open their details. 2. Load metadata, trailer, reviews and open showtimes; Admin may save metadata or deactivate a film. 3. For a review, load eligibility from the ticket, submit 1–5 stars and at most 1,000 characters, or delete the existing review. 4. Refresh the affected view.
Alternative / exception flow	Show empty results for no matching films/showtimes. Review of another user's ticket returns 404; a ticket not checked in returns 400. Failed media must retain a poster/fallback.
Postconditions	The catalogue/review reflects the authoritative response; deactivation preserves existing tickets.
Requirements / priority	WEB-FR-MOV-01 (H), WEB-FR-MOV-02 (H), WEB-FR-MOV-03 (M), WEB-FR-MOV-04 (H), WEB-FR-MOV-05 (H), WEB-FR-MOV-06 (H), WEB-FR-MOV-07 (H), WEB-FR-MOV-08 (L), WEB-FR-MOV-09 (H), WEB-FR-MOV-10 (M), WEB-FR-MOV-11 (L), WEB-FR-MOV-12 (M), WEB-FR-MEDIA-01 (L)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.4 UC-WEB-04 — Room management

Table 137: UC-WEB-04 — Room management

Field	Use-case specification
Actor	Staff/Admin read; Admin writes
Goal	Room management
Preconditions	A management session is active; layout changes require no showtimes in that room.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Search/list rooms and open the map. 2. Admin enters name and dimensions when creating, or renames/deletes the selected room. 3. Create the room then generate seats; refresh room and seat counts.
Alternative / exception flow	Reject invalid capacity, unknown room, forbidden role, and deletion/layout changes when showtimes reference the room. Room creation and seat generation are separate HTTP calls; report failure of the second call.
Postconditions	Room data is saved or a permitted room is deleted; existing seat labels survive rename.
Requirements / priority	WEB-FR-ROOM-01 (H), WEB-FR-ROOM-02 (H), WEB-FR-ROOM-03 (H), WEB-FR-ROOM-04 (M), WEB-FR-ROOM-05 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.5 UC-WEB-05 — Seat map and availability

Table 138: UC-WEB-05 — Seat map and availability

Field	Use-case specification
Actor	Signed-in Customer reads/selects; Staff/Admin manage types; Admin manages capacity/deletion
Goal	Seat map and availability
Preconditions	The room/showtime exists. Selecting seats requires sign-in; modifying layout requires management rights and no showtimes.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Load room seats and occupied IDs separately. 2. Display rows, type legend and authoritative prices; select free seats with sweetbox partners. 3. Refresh availability every five seconds; management changes types/capacity only through authorized APIs.
Alternative / exception flow	Remove selections newly occupied by another booking and notify the customer. Reject half sweetboxes, invalid room seats and prohibited layout edits.
Postconditions	The client shows the refreshed map; selecting seats alone does not reserve them.
Requirements / priority	WEB-FR-SEAT-01 (H), WEB-FR-SEAT-02 (L), WEB-FR-SEAT-03 (H), WEB-FR-SEAT-04 (L), WEB-FR-SEAT-05 (H), WEB-FR-SEAT-06 (M), WEB-FR-SEAT-07 (M), WEB-FR-SEAT-08 (H), WEB-FR-SEAT-09 (L)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.6 UC-WEB-06 — Showtimes and prices

Table 139: UC-WEB-06 — Showtimes and prices

Field	Use-case specification
Actor	Guest/Customer read public showtimes; Staff/Admin manage
Goal	Showtimes and prices
Preconditions	Movie and room exist; management changes need Staff/Admin.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Load showtimes by movie/date and select a day. 2. Show room, format, start/end and seat prices. 3. Management saves the schedule or opens/closes/cancels sales. 4. Validate overlap and persist; cancellation publishes an outbox event for affected bookings.
Alternative / exception flow	Reject overlap with 409 and invalid time/format with 400; cancelled showtimes are final. Customer selection excludes closed or started shows.
Postconditions	The schedule and computed seat prices are authoritative; cancellations propagate asynchronously.
Requirements / priority	WEB-FR-SHOW-01 (H), WEB-FR-SHOW-02 (H), WEB-FR-SHOW-03 (H), WEB-FR-SHOW-04 (H), WEB-FR-SHOW-05 (H), WEB-FR-SHOW-06 (H), WEB-FR-SHOW-07 (M), WEB-FR-SHOW-08 (H), WEB-FR-SHOW-09 (H), WEB-FR-SHOW-10 (M), WEB-FR-SHOW-11 (H), WEB-FR-SHOW-12 (M), WEB-FR-SHOW-13 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.7 UC-WEB-07 — Booking, concessions, promotions and loyalty

Table 140: UC-WEB-07 — Booking, concessions, promotions and loyalty

Field	Use-case specification
Actor	Customer owns online bookings; Staff/Admin operate counter and administration
Goal	Booking, concessions, promotions and loyalty
Preconditions	A signed-in customer has selected an OPEN future showtime and seats; counter guests are represented by Staff/Admin.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Review seats; optionally select active concessions, a promotion code and redeemable points. 2. Confirm age-rating acknowledgement where required. 3. Submit IDs/quantities; the server recalculates all prices, validates pairs, locks reservations and creates a PENDING booking with a ten-minute expiry. 4. Open payment or read/cancel an owned booking according to its state.
Alternative / exception flow	A concurrent hold returns 409 and must not charge money or leave partial seats/points. Invalid concessions/code/points return validation errors. Expiry releases seats and restores reserved points.
Postconditions	Successful creation returns the persisted booking, breakdown and expiry; unsuccessful creation rolls back its transaction.
Requirements / priority	WEB-FR-BOOK-01 (H), WEB-FR-BOOK-02 (H), WEB-FR-BOOK-03 (H), WEB-FR-BOOK-04 (H), WEB-FR-BOOK-05 (H), WEB-FR-BOOK-06 (H), WEB-FR-BOOK-07 (H), WEB-FR-BOOK-08 (H), WEB-FR-BOOK-09 (H), WEB-FR-BOOK-10 (H), WEB-FR-BOOK-11 (M), WEB-FR-BOOK-12 (M), WEB-FR-BOOK-13 (M), WEB-FR-BOOK-14 (M), WEB-FR-BOOK-15 (M), WEB-FR-BOOK-16 (M), WEB-FR-BOOK-17 (M), WEB-FR-BOOK-18 (M), WEB-FR-BOOK-19 (L), WEB-FR-STATE-01 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.8 UC-WEB-08 — Online payment and reconciliation

Table 141: UC-WEB-08 — Online payment and reconciliation

Field	Use-case specification
Actor	Booking owner; Staff/Admin for permitted operations; PayOS/system
Goal	Online payment and reconciliation
Preconditions	A payable booking exists; PayOS credentials and callback/return settings are configured for live payment.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Request/reuse a PayOS checkout for the booking. 2. Open checkout and display the hold countdown. 3. Query booking every five seconds and verify PayOS every thirty seconds while the customer payment page is active; also reconcile when returning to the app. 4. Signed webhook or verified PayOS data updates payment and booking; load the result from the server.
Alternative / exception flow	Never trust redirect query parameters as proof of payment. Handle cancellation, timeout and 503 without showing false success; late payment confirms only if seats can be reclaimed, otherwise creates a refund.
Postconditions	Payment status is authoritative; SUCCESS confirms the booking and schedules ticket notification without duplicate business effects.
Requirements / priority	WEB-FR-PAY-01 (H), WEB-FR-PAY-02 (H), WEB-FR-PAY-03 (H), WEB-FR-PAY-04 (M), WEB-FR-PAY-05 (M), WEB-FR-PAY-06 (H), WEB-FR-PAY-07 (H), WEB-FR-PAY-08 (M), WEB-FR-PAY-09 (M), WEB-FR-PAY-10 (M), WEB-FR-PAY-11 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.9 UC-WEB-09 — Cancellation and manual refund

Table 142: UC-WEB-09 — Cancellation and manual refund

Field	Use-case specification
Actor	Customer requests/reads own refund; Staff/Admin process; system creates requests
Goal	Cancellation and manual refund
Preconditions	A paid booking/payment exists; customer cancellation is at least two hours before start; staff processing requires a pending request.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Confirm cancellation and send reason. 2. The backend creates/reuses one refund per payment and notifies the customer. 3. Staff transfer money outside the application then record transfer reference, or reject an eligible request with a reason. 4. Customer reloads the ticket/refund status and reason.
Alternative / exception flow	Reject too-late customer cancellation (409), completion without transfer reference and rejection without reason; cinema-cancelled refunds cannot be rejected. Duplicate events must not create another refund.
Postconditions	Completion yields COMPLETED/REFUNDED and releases cancelled seats; allowed rejection restores SUCCESS/CONFIRMED; customer-requested pending refunds retain seats.
Requirements / priority	WEB-FR-REF-01 (H), WEB-FR-REF-02 (H), WEB-FR-REF-03 (H), WEB-FR-REF-04 (H), WEB-FR-REF-05 (M), WEB-FR-REF-06 (M), WEB-FR-REF-07 (H)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.10 UC-WEB-10 — Counter ticket sale

Table 143: UC-WEB-10 — Counter ticket sale

Field	Use-case specification
Actor	Staff/Admin; walk-in guest or member as beneficiary
Goal	Counter ticket sale
Preconditions	A management session is active; an OPEN future showtime has free seats.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Select walk-in or find a member by phone; choose showtime, seats and optional concessions/promotion/points. 2. Hold seats and open counter checkout. 3. Take sufficient cash and display change, or show VietQR and verify every three seconds. 4. After server confirmation show/print the QR ticket, optionally email it and check in within the permitted window.
Alternative / exception flow	Reject insufficient cash; cancel an unpaid PayOS link before switching to cash; if transfer already succeeded return 409. Without email provide print/on-screen QR.
Postconditions	A single confirmed ticket and successful payment are recorded and included in reporting.
Requirements / priority	WEB-FR-POS-01 (H), WEB-FR-POS-02 (H), WEB-FR-POS-03 (H), WEB-FR-POS-04 (H), WEB-FR-POS-05 (H), WEB-FR-POS-06 (M), WEB-FR-POS-07 (H), WEB-FR-POS-08 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.11 UC-WEB-11 — Ticket lookup and check-in

Table 144: UC-WEB-11 — Ticket lookup and check-in

Field	Use-case specification
Actor	Staff/Admin scan and admit; Customer displays their ticket
Goal	Ticket lookup and check-in
Preconditions	Staff/Admin is signed in; the ticket is CONFIRMED, unused and within 180 minutes before to 120 minutes after start.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Scan QR or type the booking code and press Enter. 2. Lookup ticket; enrich room and age rating when catalogue calls succeed. 3. Check identity/rating as prompted, then explicitly check in. 4. Store checkedInAt, show confirmation and prepare for the next code.
Alternative / exception flow	Keep manual entry available if camera is unavailable/denied. Unknown code returns 404; unpaid, cancelled, pending refund, out-of-window or already-used ticket returns 409. Concurrent second check-in fails. Optional metadata failure must not hide the ticket.
Postconditions	Only one check-in timestamp is stored; check-in is an attribute, not another booking status.
Requirements / priority	WEB-FR-CHK-01 (H), WEB-FR-CHK-02 (H), WEB-FR-CHK-03 (H), WEB-FR-CHK-04 (H), WEB-FR-CHK-05 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.12 UC-WEB-12 — Notifications and transactional email

Table 145: UC-WEB-12 — Notifications and transactional email

Field	Use-case specification
Actor	Customer reads own; Staff/Admin send permitted messages; system dispatches events
Goal	Notifications and transactional email
Preconditions	An account session exists for in-app history; SMTP/event processing is configured for actual delivery.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Load notifications and unread count, or create a management message for a booking. 2. Open detail and mark as read, mark all read, or follow the related ticket. 3. The server sends transactional email from committed events and reminder jobs; records SENT/FAILED and delivery error.
Alternative / exception flow	Owner checks reject another user's data; show loading/empty/error-retry states. Sending failure retains FAILED for retry and must not undo payment. Event deduplication reduces duplicates but SMTP plus database commit is not a guaranteed exactly-once transaction.
Postconditions	Read state/badge matches the response; successful delivery is recorded separately from booking/payment state.
Requirements / priority	WEB-FR-NOTI-01 (M), WEB-FR-NOTI-02 (M), WEB-FR-NOTI-03 (M), WEB-FR-NOTI-04 (M), WEB-FR-NOTI-05 (L), WEB-FR-NOTI-06 (H), WEB-FR-NOTI-07 (H), WEB-FR-NOTI-08 (M), WEB-FR-NOTI-09 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

10.13 UC-WEB-13 — Operations dashboards

Table 146: UC-WEB-13 — Operations dashboards

Field	Use-case specification
Actor	Staff/Admin; revenue and account statistics require Admin
Goal	Operations dashboards
Preconditions	Management session and source records exist; date ranges use cinema local time.
Trigger	Open the relevant screen or invoke the stated operation/event.
Main flow	1. Load today’s bookings, showtimes and check-in statistics. 2. Group now-playing, next three and last three shows; refresh each minute. 3. Open a show to inspect seat/customer details. 4. Admin loads monthly revenue/refunds/accounts independently so one failed card does not erase others.
Alternative / exception flow	Show zero/empty periods accurately; a card can show its own error and retry. Staff must not acquire revenue/account APIs by changing the URL.
Postconditions	Displayed totals use the documented date and state definitions and link to operational records.
Requirements / priority	WEB-FR-DASH-01 (M), WEB-FR-DASH-02 (M), WEB-FR-DASH-03 (M), WEB-FR-DASH-04 (M)
Acceptance	Execute AC-01/AC-02 for every included FR with permitted actors.

11. External Interface Requirements

Table 147: 11. External Interface Requirements

Interface	Contract
Client/gateway	REST/JSON UTF-8; Accept application/json; Content-Type application/json for request bodies; Bearer JWT on protected calls. Clients never connect to PostgreSQL or private service ports.
PayOS	Checkout URL/VietQR through backend; signed webhook; client verifies server state rather than trusting a redirect. No client card-data collection.
SMTP / RabbitMQ	Server-only OTP, ticket/refund delivery and committed events; clients read notification APIs, not the broker.
Device/browser	Hash/URL routes; HTTPS/localhost camera; share/clipboard/calendar; 80 mm browser printing; HTML preview with poster fallback.

12. UI / Screen Requirements

Screen IDs represent routed pages or role landings. Scanner overlays, review editors, profile/password dialogs and seat-type painters are components, not invented routes. Related FR/API lists are functional-area associations; §13 gives exact endpoint contracts.

12.1 WEB-SCR-LOGIN — Sign in

Table 148: WEB-SCR-LOGIN — Sign in

Field	Screen requirement
Route	/login
Actor	Guest/all account roles
Purpose	Sign in
Main components	Email, password, field errors, recovery/register links
Actions	Sign in; recover password; register on customer site
Navigation	Role landing or pending destination; /forgot-password; /register (customer only)
Related FRs	WEB-FR-AUTH-02, WEB-FR-AUTH-03, WEB-FR-AUTH-06, WEB-FR-AUTH-12, WEB-FR-AUTH-13, WEB-FR-NAV-01
Related API area	WEB-API-027, WEB-API-029, WEB-API-032, WEB-API-033, WEB-API-035
Source	Cinema_Mobile/lib/features/auth/screens/login_screen.dart

12.2 WEB-SCR-REGISTER — Registration

Table 149: WEB-SCR-REGISTER — Registration

Field	Screen requirement
Route	/register
Actor	Guest
Purpose	Registration
Main components	Name, email, phone, password and confirmation
Actions	Validate and register
Navigation	/login → /register → /verify-email
Related FRs	WEB-FR-AUTH-01, WEB-FR-AUTH-08, WEB-FR-AUTH-13
Related API area	WEB-API-027, WEB-API-033, WEB-API-035
Source	Cinema_Mobile/lib/features/auth/screens/register_screen.dart

12.3 WEB-SCR-VERIFY — Verify email

Table 150: WEB-SCR-VERIFY — Verify email

Field	Screen requirement
Route	/verify-email
Actor	Guest
Purpose	Verify email
Main components	OTP input; resend countdown; registered password context
Actions	Verify or resend
Navigation	Registration/login → OTP → role landing
Related FRs	WEB-FR-AUTH-07, WEB-FR-AUTH-08, WEB-FR-AUTH-14
Related API area	WEB-API-027, WEB-API-032, WEB-API-033, WEB-API-034, WEB-API-042
Source	Cinema_Mobile/lib/features/auth/screens/verify_email_screen.dart

12.4 WEB-SCR-FORGOT — Forgot password

Table 151: WEB-SCR-FORGOT — Forgot password

Field	Screen requirement
Route	/forgot-password
Actor	Guest/account owner
Purpose	Forgot password
Main components	Email field and submission status
Actions	Request reset OTP
Navigation	Login/Profile → reset request → /reset-password
Related FRs	WEB-FR-AUTH-09, WEB-FR-AUTH-13, WEB-FR-AUTH-14
Related API area	WEB-API-026, WEB-API-027, WEB-API-032, WEB-API-033, WEB-API-034, WEB-API-035
Source	Cinema_Mobile/lib/features/auth/screens/forgot_password_screen.dart

12.5 WEB-SCR-RESET — Reset password

Table 152: WEB-SCR-RESET — Reset password

Field	Screen requirement
Route	/reset-password
Actor	OTP recipient
Purpose	Reset password
Main components	OTP, new password and confirmation
Actions	Reset using emailed code
Navigation	Forgot-password → reset → sign-in

Continued on next page

Table 152 — continued

Field	Screen requirement
Related FRs	WEB-FR-AUTH-09, WEB-FR-AUTH-13, WEB-FR-AUTH-14
Related API area	WEB-API-026, WEB-API-027, WEB-API-032, WEB-API-033, WEB-API-034, WEB-API-035
Source	Cinema_Mobile/lib/features/auth/screens/reset_password_screen.dart

12.6 WEB-SCR-HOME — Customer home

Table 153: WEB-SCR-HOME — Customer home

Field	Screen requirement
Route	/home
Actor	Guest/Customer
Purpose	Customer home
Main components	Now/soon tabs; search; genre chips; movie cards; account/notification navigation
Actions	Search/filter; open movie
Navigation	/home → /movies/{id}; customer navigation to bookings, payments, notifications, profile
Related FRs	WEB-FR-MOV-03, WEB-FR-MOV-04, WEB-FR-MOV-10, WEB-FR-NOTI-09, WEB-FR-MEDIA-01
Related API area	WEB-API-064, WEB-API-066, WEB-API-067, WEB-API-069, WEB-API-072, WEB-API-076, WEB-API-077
Source	Cinema_Mobile/lib/features/home/screens/home_screen.dart

12.7 WEB-SCR-MOVIE — Movie details

Table 154: WEB-SCR-MOVIE — Movie details

Field	Screen requirement
Route	/movies/{id}
Actor	Guest/Customer
Purpose	Movie details
Main components	Poster, rating, metadata, cast/director, trailer, review summary, seven-day showtimes
Actions	Play trailer; select showtime; read reviews
Navigation	Home → movie → /showtimes/{id}/seats
Related FRs	WEB-FR-MOV-02, WEB-FR-MOV-08, WEB-FR-MOV-09, WEB-FR-MOV-11, WEB-FR-MOV-12, WEB-FR-SHOW-03, WEB-FR-SHOW-06, WEB-FR-SHOW-07, WEB-FR-SHOW-08, WEB-FR-SHOW-13
Related API area	WEB-API-017, WEB-API-020, WEB-API-064, WEB-API-102, WEB-API-103, WEB-API-105, WEB-API-106
Source	Cinema_Mobile/lib/features/movie/screens/movie_detail_screen.dart

12.8 WEB-SCR-SEATS — Seat selection

Table 155: WEB-SCR-SEATS — Seat selection

Field	Screen requirement
Route	/showtimes/{id}/seats
Actor	Customer
Purpose	Seat selection
Main components	Rows, type/occupied legend, selection, pair pricing, total
Actions	Select/unselect free seats; refresh
Navigation	Movie → seats → /showtimes/{id}/summary
Related FRs	WEB-FR-SEAT-03, WEB-FR-SHOW-03, WEB-FR-SHOW-13, WEB-FR-BOOK-07, WEB-FR-BOOK-08, WEB-FR-BOOK-09, WEB-FR-BOOK-11
Related API area	WEB-API-013, WEB-API-017, WEB-API-049, WEB-API-055
Source	Cinema_Mobile/lib/features/seat/screens/seat_selection_screen.dart

12.9 WEB-SCR-SUMMARY — Booking summary

Table 156: WEB-SCR-SUMMARY — Booking summary

Field	Screen requirement
Route	/showtimes/{id}/summary
Actor	Customer
Purpose	Booking summary
Main components	BookingDraft, seats, concessions picker, promotion quote, points, age acknowledgement
Actions	Adjust extras; hold seats
Navigation	Seats → summary → booking payment; missing draft → seats
Related FRs	WEB-FR-MOV-09, WEB-FR-BOOK-01, WEB-FR-BOOK-09, WEB-FR-BOOK-13, WEB-FR-BOOK-15, WEB-FR-BOOK-17
Related API area	WEB-API-055, WEB-API-057, WEB-API-060, WEB-API-064, WEB-API-099
Source	Cinema_Mobile/lib/features/booking/screens/booking_summary_screen.dart

12.10 WEB-SCR-PAYMENT — Payment

Table 157: WEB-SCR-PAYMENT — Payment

Field	Screen requirement
Route	/bookings/{id}/payment
Actor	Customer
Purpose	Payment
Main components	Booking/price breakdown; expiry countdown; PayOS link; automatic polling
Actions	Open/continue PayOS; return and reconcile

Continued on next page

Table 157 — continued

Field	Screen requirement
Navigation	Summary/ticket → payment → external PayOS → result
Related FRs	WEB-FR-BOOK-05, WEB-FR-BOOK-10, WEB-FR-PAY-01, WEB-FR-PAY-02, WEB-FR-PAY-11
Related API area	WEB-API-046, WEB-API-087, WEB-API-088
Source	Cinema_Mobile/lib/features/payment/screens/payment_screen.dart

12.11 WEB-SCR-RESULT — Payment result

Table 158: WEB-SCR-RESULT — Payment result

Field	Screen requirement
Route	/bookings/{id}/result
Actor	Customer
Purpose	Payment result
Main components	Verification/loading, successful/pending/failed result
Actions	Reload server result; open ticket
Navigation	PayOS/payment → result → /bookings/{id}
Related FRs	WEB-FR-BOOK-02, WEB-FR-BOOK-05, WEB-FR-PAY-02, WEB-FR-PAY-08
Related API area	WEB-API-046, WEB-API-087
Source	Cinema_Mobile/lib/features/payment/screens/payment_result_screen.dart

12.12 WEB-SCR-TICKETS — My tickets

Table 159: WEB-SCR-TICKETS — My tickets

Field	Screen requirement
Route	/bookings
Actor	Customer
Purpose	My tickets
Main components	Active/History tabs, ticket cards, refresh and empty/error states
Actions	Open ticket; refresh
Navigation	Customer navigation → tickets → /bookings/{id}
Related FRs	WEB-FR-BOOK-04
Related API area	WEB-API-053
Source	Cinema_Mobile/lib/features/booking/screens/my_bookings_screen.dart

12.13 WEB-SCR-TICKET — Ticket details

Table 160: WEB-SCR-TICKET — Ticket details

Field	Screen requirement
Route	/bookings/{id}
Actor	Customer
Purpose	Ticket details
Main components	Seats, extras, totals, QR, checkedInAt, cancel/refund, review and share/calendar
Actions	Pay pending ticket; cancel/request refund; review after check-in; share/calendar
Navigation	Tickets/notification → ticket → payment or return
Related FRs	WEB-FR-MOV-12, WEB-FR-BOOK-02, WEB-FR-BOOK-05, WEB-FR-BOOK-06, WEB-FR-BOOK-13, WEB-FR-BOOK-15, WEB-FR-BOOK-17, WEB-FR-BOOK-19, WEB-FR-PAY-03, WEB-FR-PAY-06, WEB-FR-REF-01, WEB-FR-REF-05, WEB-FR-REF-06, WEB-FR-CHK-01, WEB-FR-CHK-05
Related API area	WEB-API-017, WEB-API-043, WEB-API-046, WEB-API-055, WEB-API-057, WEB-API-060, WEB-API-082, WEB-API-087, WEB-API-099, WEB-API-102, WEB-API-103, WEB-API-105, WEB-API-106
Source	Cinema_Mobile/lib/features/booking/screens/booking_detail_screen.dart

12.14 WEB-SCR-PAYHISTORY — Payment history

Table 161: WEB-SCR-PAYHISTORY — Payment history

Field	Screen requirement
Route	/payments
Actor	Customer
Purpose	Payment history
Main components	Owned transaction rows, amounts, statuses
Actions	Refresh; inspect related booking where provided
Navigation	Customer navigation → payments → related ticket
Related FRs	WEB-FR-PAY-03, WEB-FR-PAY-04
Related API area	WEB-API-095
Source	Cinema_Mobile/lib/features/payment/screens/payment_history_screen.dart

12.15 WEB-SCR-NOTIFICATIONS — Notifications

Table 162: WEB-SCR-NOTIFICATIONS — Notifications

Field	Screen requirement
Route	/notifications
Actor	Customer
Purpose	Notifications
Main components	Notification list, unread badge, mark-all-read

Continued on next page

Table 162 — continued

Field	Screen requirement
Actions	Open notification; mark all read
Navigation	Customer bell/navigation → /notifications/{id}
Related FRs	WEB-FR-NOTI-01, WEB-FR-NOTI-09
Related API area	WEB-API-072, WEB-API-076, WEB-API-077, WEB-API-078
Source	Cinema_Mobile/lib/features/notification/screens/notification_list_screen.dart

12.16 WEB-SCR-NOTIFICATION — Notification details

Table 163: WEB-SCR-NOTIFICATION — Notification details

Field	Screen requirement
Route	/notifications/{id}
Actor	Customer
Purpose	Notification details
Main components	Subject, content, read state, related booking
Actions	Mark read on open; open ticket
Navigation	Notifications → detail → /bookings/{id}
Related FRs	WEB-FR-NOTI-04, WEB-FR-NOTI-09
Related API area	WEB-API-072, WEB-API-075, WEB-API-076, WEB-API-077
Source	Cinema_Mobile/lib/features/notification/screens/notification_detail_screen.dart

12.17 WEB-SCR-PROFILE — Account and membership

Table 164: WEB-SCR-PROFILE — Account and membership

Field	Screen requirement
Route	/profile
Actor	Customer
Purpose	Account and membership
Main components	Identity, edit-profile/password dialogs, membership card/history, logout
Actions	Edit name/phone; change/recover password; sign out
Navigation	Customer navigation → profile; password recovery → forgot/reset
Related FRs	WEB-FR-AUTH-10, WEB-FR-AUTH-11, WEB-FR-AUTH-16, WEB-FR-AUTH-17, WEB-FR-BOOK-17
Related API area	WEB-API-025, WEB-API-028, WEB-API-029, WEB-API-031, WEB-API-032, WEB-API-055, WEB-API-060
Source	Cinema_Mobile/lib/features/profile/screens/profile_screen.dart

12.18 WEB-SCR-CHECKIN — Ticket check-in

Table 165: WEB-SCR-CHECKIN — Ticket check-in

Field	Screen requirement
Route	/staff/check-in
Actor	Staff/Admin
Purpose	Ticket check-in
Main components	Code field; camera scanner overlay; ticket details; age reminder; admission button
Actions	Scan/type; lookup; check in once
Navigation	Management landing/navigation → check-in → next ticket
Related FRs	WEB-FR-MOV-09, WEB-FR-CHK-02, WEB-FR-CHK-03, WEB-FR-CHK-04, WEB-FR-CHK-05
Related API area	WEB-API-044, WEB-API-046, WEB-API-047, WEB-API-050, WEB-API-064, WEB-API-065, WEB-API-071
Source	Cinema_Mobile/lib/features/staff/screens/staff_check_in_screen.dart

12.19 WEB-SCR-STAFF-DASH — Staff dashboard

Table 166: WEB-SCR-STAFF-DASH — Staff dashboard

Field	Screen requirement
Route	/staff/dashboard
Actor	Staff/Admin
Purpose	Staff dashboard
Main components	Today totals, live board, pending bookings/refunds
Actions	Open operations; sell; check in
Navigation	Login → dashboard → operations/counter/check-in
Related FRs	WEB-FR-DASH-01, WEB-FR-DASH-02, WEB-FR-DASH-03
Related API area	WEB-API-019, WEB-API-048, WEB-API-051, WEB-API-052, WEB-API-093
Source	Cinema_Mobile/lib/features/staff/screens/staff_dashboard_screen.dart

12.20 WEB-SCR-ADMIN-DASH — Admin dashboard

Table 167: WEB-SCR-ADMIN-DASH — Admin dashboard

Field	Screen requirement
Route	/admin/dashboard
Actor	Admin
Purpose	Admin dashboard
Main components	Monthly revenue/refunds, today, movie and account cards
Actions	Open corresponding management area
Navigation	Login → dashboard → administration/operations

Continued on next page

Table 167 — continued

Field	Screen requirement
Related FRs	WEB-FR-DASH-01, WEB-FR-DASH-02, WEB-FR-DASH-04
Related API area	WEB-API-019, WEB-API-040, WEB-API-048, WEB-API-051, WEB-API-069, WEB-API-094
Source	Cinema_Mobile/lib/features/admin/screens/admin_dashboard_screen.dart

12.21 WEB-SCR-OPERATIONS — Operations workspace

Table 168: WEB-SCR-OPERATIONS — Operations workspace

Field	Screen requirement
Route	/staff/operations
Actor	Staff/Admin
Purpose	Operations workspace
Main components	Bookings/showtimes/rooms/payments/refunds tabs; filters; paging
Actions	Filter; open record; sell; create showtime
Navigation	Navigation → operations → record/new sale/new showtime
Related FRs	WEB-FR-ROOM-01, WEB-FR-SHOW-01, WEB-FR-SHOW-09, WEB-FR-BOOK-03, WEB-FR-BOOK-12, WEB-FR-PAY-09, WEB-FR-REF-02
Related API area	WEB-API-004, WEB-API-005, WEB-API-019, WEB-API-023, WEB-API-048, WEB-API-051, WEB-API-052, WEB-API-054, WEB-API-087, WEB-API-092, WEB-API-093, WEB-API-096
Source	Cinema_Mobile/lib/features/staff/screens/staff_operations_screen.dart

12.22 WEB-SCR-COUNTER — Counter booking

Table 169: WEB-SCR-COUNTER — Counter booking

Field	Screen requirement
Route	/staff/bookings/new
Actor	Staff/Admin
Purpose	Counter booking
Main components	Guest/member selector, phone lookup, movie/day/showtime/map, extras
Actions	Select buyer/seats; quote; hold
Navigation	Dashboard/operations → counter → checkout
Related FRs	WEB-FR-AUTH-15, WEB-FR-BOOK-13, WEB-FR-BOOK-15, WEB-FR-BOOK-17, WEB-FR-POS-01
Related API area	WEB-API-013, WEB-API-023, WEB-API-030, WEB-API-055, WEB-API-057, WEB-API-060, WEB-API-099
Source	Cinema_Mobile/lib/features/staff/screens/staff_create_booking_screen.dart

12.23 WEB-SCR-CHECKOUT — Counter checkout

Table 170: WEB-SCR-CHECKOUT — Counter checkout

Field	Screen requirement
Route	/staff/bookings/{id}/checkout
Actor	Staff/Admin
Purpose	Counter checkout
Main components	Countdown, email, cash/change, VietQR, paid QR/print
Actions	Take cash; verify transfer; print; admit; sell again
Navigation	Counter/booking record → checkout → print/check-in/new sale
Related FRs	WEB-FR-PAY-01, WEB-FR-PAY-02, WEB-FR-POS-02, WEB-FR-POS-03, WEB-FR-POS-04, WEB-FR-POS-05, WEB-FR-POS-06, WEB-FR-POS-07, WEB-FR-POS-08, WEB-FR-CHK-01, WEB-FR-CHK-03
Related API area	WEB-API-044, WEB-API-046, WEB-API-055, WEB-API-086, WEB-API-087, WEB-API-088
Source	Cinema_Mobile/lib/features/staff/screens/staff_counter_checkout_screen.dart

12.24 WEB-SCR-RECORD-BOOKINGS — Booking record

Table 171: WEB-SCR-RECORD-BOOKINGS — Booking record

Field	Screen requirement
Route	/staff/bookings/{id}
Actor	Staff/Admin
Purpose	Booking record
Main components	Ticket, payment state, customer and actions
Actions	Inspect and execute permitted record actions
Navigation	Operations → record → associated records or editor
Related FRs	WEB-FR-BOOK-02, WEB-FR-BOOK-06, WEB-FR-POS-08, WEB-FR-CHK-05
Related API area	WEB-API-043, WEB-API-046, WEB-API-050
Source	Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart

12.25 WEB-SCR-RECORD-PAYMENTS — Payment record

Table 172: WEB-SCR-RECORD-PAYMENTS — Payment record

Field	Screen requirement
Route	/staff/payments/{id}
Actor	Staff/Admin
Purpose	Payment record
Main components	Amount, method, status, verification and refund
Actions	Inspect and execute permitted record actions
Navigation	Operations → record → associated records or editor

Continued on next page

Table 172 — continued

Field	Screen requirement
Related FRs	WEB-FR-PAY-03, WEB-FR-PAY-05, WEB-FR-PAY-09, WEB-FR-REF-06
Related API area	WEB-API-082, WEB-API-084, WEB-API-085, WEB-API-087, WEB-API-096
Source	Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart

12.26 WEB-SCR-RECORD-REFUNDS — Refund record

Table 173: WEB-SCR-RECORD-REFUNDS — Refund record

Field	Screen requirement
Route	/staff/refunds/{id}
Actor	Staff/Admin
Purpose	Refund record
Main components	Payer bank, reason, status, transfer reference
Actions	Inspect and execute permitted record actions
Navigation	Operations → record → associated records or editor
Related FRs	WEB-FR-REF-02, WEB-FR-REF-03, WEB-FR-REF-04, WEB-FR-REF-05, WEB-FR-REF-07
Related API area	WEB-API-016, WEB-API-043, WEB-API-082, WEB-API-090, WEB-API-091, WEB-API-092, WEB-API-093
Source	Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart

12.27 WEB-SCR-RECORD-SHOWTIMES — Showtime record

Table 174: WEB-SCR-RECORD-SHOWTIMES — Showtime record

Field	Screen requirement
Route	/staff/showtimes/{id}
Actor	Staff/Admin
Purpose	Showtime record
Main components	Schedule, seat occupancy map, customer list
Actions	Inspect and execute permitted record actions
Navigation	Operations → record → associated records or editor
Related FRs	WEB-FR-SHOW-03, WEB-FR-SHOW-05, WEB-FR-SHOW-10, WEB-FR-SHOW-12, WEB-FR-BOOK-12
Related API area	WEB-API-013, WEB-API-015, WEB-API-016, WEB-API-017, WEB-API-050, WEB-API-051
Source	Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart

12.28 WEB-SCR-RECORD-ROOMS — Room record

Table 175: WEB-SCR-RECORD-ROOMS — Room record

Field	Screen requirement
Route	/staff/rooms/{id}
Actor	Staff/Admin
Purpose	Room record
Main components	Room metadata, seat map/type painter and capacity
Actions	Inspect and execute permitted record actions
Navigation	Operations → record → associated records or editor
Related FRs	WEB-FR-ROOM-02, WEB-FR-SEAT-01, WEB-FR-SEAT-02, WEB-FR-SEAT-03, WEB-FR-SEAT-05, WEB-FR-SEAT-06, WEB-FR-SEAT-07, WEB-FR-SEAT-08, WEB-FR-SEAT-09
Related API area	WEB-API-002, WEB-API-007, WEB-API-009, WEB-API-010, WEB-API-011, WEB-API-012, WEB-API-013
Source	Cinema_Mobile/lib/features/staff/screens/staff_record_screen.dart

12.29 WEB-SCR-SHOW-EDITOR — Showtime editor

Table 176: WEB-SCR-SHOW-EDITOR — Showtime editor

Field	Screen requirement
Route	/staff/showtimes/new; /staff/showtimes/{id}/edit
Actor	Staff/Admin
Purpose	Showtime editor
Main components	Movie, room, start/end, format, base price, status
Actions	Validate and save
Navigation	Operations/showtime record → editor → record
Related FRs	WEB-FR-SHOW-02, WEB-FR-SHOW-04, WEB-FR-SHOW-11, WEB-FR-SHOW-13
Related API area	WEB-API-017, WEB-API-018, WEB-API-024
Source	Cinema_Mobile/lib/features/staff/screens/staff_showtime_editor_screen.dart

12.30 WEB-SCR-COMPOSE — Compose booking email

Table 177: WEB-SCR-COMPOSE — Compose booking email

Field	Screen requirement
Route	/staff/bookings/{id}/notify
Actor	Staff/Admin
Purpose	Compose booking email
Main components	Recipient email, subject, content, send result
Actions	Create and send/retry

Continued on next page

Table 177 — continued

Field	Screen requirement
Navigation	Booking record → notify → record
Related FRs	WEB-FR-NOTI-02, WEB-FR-NOTI-03
Related API area	WEB-API-073, WEB-API-080
Source	Cinema_Mobile/lib/features/staff/screens/staff_notification_screen.dart

12.31 WEB-SCR-ADMIN-MOVIES — Movies

Table 178: WEB-SCR-ADMIN-MOVIES — Movies

Field	Screen requirement
Route	/admin/movies
Actor	Admin
Purpose	Movies
Main components	Search, status, movie editor, reviews dialog
Actions	Create/edit; stop/resume; moderate
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-MOV-01, WEB-FR-MOV-02, WEB-FR-MOV-03, WEB-FR-MOV-04, WEB-FR-MOV-05, WEB-FR-MOV-06, WEB-FR-MOV-07, WEB-FR-MOV-09, WEB-FR-MOV-11, WEB-FR-MOV-12
Related API area	WEB-API-063, WEB-API-064, WEB-API-065, WEB-API-066, WEB-API-068, WEB-API-069, WEB-API-070, WEB-API-071, WEB-API-102, WEB-API-103, WEB-API-105, WEB-API-106
Source	Cinema_Mobile/lib/features/admin/screens/admin_movies_screen.dart

12.32 WEB-SCR-ADMIN-ROOMS — Rooms

Table 179: WEB-SCR-ADMIN-ROOMS — Rooms

Field	Screen requirement
Route	/admin/rooms
Actor	Admin
Purpose	Rooms
Main components	Room list, create/rename dialog, seat-map links
Actions	Create/rename/delete; open map
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-ROOM-01, WEB-FR-ROOM-02, WEB-FR-ROOM-03, WEB-FR-ROOM-04, WEB-FR-ROOM-05
Related API area	WEB-API-001, WEB-API-002, WEB-API-003, WEB-API-004, WEB-API-005, WEB-API-006, WEB-API-010, WEB-API-013
Source	Cinema_Mobile/lib/features/admin/screens/admin_rooms_screen.dart

12.33 WEB-SCR-ADMIN-SNACKS — Snacks

Table 180: WEB-SCR-ADMIN-SNACKS — Snacks

Field	Screen requirement
Route	/admin/snacks
Actor	Admin
Purpose	Snacks
Main components	Menu item editor, price, active toggle
Actions	Create/edit; stop/resume
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-BOOK-14
Related API area	WEB-API-056, WEB-API-057, WEB-API-058
Source	Cinema_Mobile/lib/features/admin/screens/admin_snacks_screen.dart

12.34 WEB-SCR-ADMIN-PROMOTIONS — Promotions

Table 181: WEB-SCR-ADMIN-PROMOTIONS — Promotions

Field	Screen requirement
Route	/admin/promotions
Actor	Admin
Purpose	Promotions
Main components	Code/automatic deals, dates, limits, use count
Actions	Create/edit; pause/resume
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-BOOK-16
Related API area	WEB-API-098, WEB-API-100, WEB-API-101
Source	Cinema_Mobile/lib/features/admin/screens/admin_promotions_screen.dart

12.35 WEB-SCR-ADMIN-MEMBERS — Members

Table 182: WEB-SCR-ADMIN-MEMBERS — Members

Field	Screen requirement
Route	/admin/members
Actor	Admin
Purpose	Members
Main components	Customer list, points/tier/spend, history/adjust dialog
Actions	Find member; adjust with reason
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-BOOK-18

Continued on next page

Table 182 — continued

Field	Screen requirement
Related API area	WEB-API-030, WEB-API-059, WEB-API-061, WEB-API-062
Source	Cinema_Mobile/lib/features/admin/screens/admin_members_screen.dart

12.36 WEB-SCR-ADMIN-USERS — Accounts

Table 183: WEB-SCR-ADMIN-USERS — Accounts

Field	Screen requirement
Route	/admin/users
Actor	Admin
Purpose	Accounts
Main components	Search, role/status filters, paging
Actions	List; create; open account
Navigation	Admin navigation → list → editor/details
Related FRs	WEB-FR-USER-01, WEB-FR-USER-06
Related API area	WEB-API-036, WEB-API-040
Source	Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart

12.37 WEB-SCR-USER-EDITOR — Account create/details

Table 184: WEB-SCR-USER-EDITOR — Account create/details

Field	Screen requirement
Route	/admin/users/new; /admin/users/{id}
Actor	Admin
Purpose	Account create/details
Main components	Identity fields, roles, enabled state
Actions	Create; change roles; lock/unlock
Navigation	Accounts → new/detail → accounts
Related FRs	WEB-FR-AUTH-13, WEB-FR-USER-02, WEB-FR-USER-03, WEB-FR-USER-04, WEB-FR-USER-05, WEB-FR-USER-06
Related API area	WEB-API-027, WEB-API-033, WEB-API-035, WEB-API-036, WEB-API-037, WEB-API-038, WEB-API-039, WEB-API-041
Source	Cinema_Mobile/lib/features/admin/screens/admin_users_screen.dart

12.38 WEB-SCR-MGT-PROFILE — Management profile

Table 185: WEB-SCR-MGT-PROFILE — Management profile

Field	Screen requirement
Route	/staff/profile; /admin/profile
Actor	Staff/Admin
Purpose	Management profile
Main components	Identity, edit/password dialogs, sign out
Actions	Edit own profile; change password; logout
Navigation	Navigation → profile → authentication
Related FRs	WEB-FR-AUTH-10, WEB-FR-AUTH-11, WEB-FR-AUTH-16, WEB-FR-AUTH-17
Related API area	WEB-API-025, WEB-API-028, WEB-API-029, WEB-API-031, WEB-API-032
Source	Cinema_Mobile/lib/features/profile/screens/profile_screen.dart

12.39 WEB-SCR-STAFF-ONLY — Management access notice

Table 186: WEB-SCR-STAFF-ONLY — Management access notice

Field	Screen requirement
Route	/home (Customer in admin site)
Actor	Customer
Purpose	Management access notice
Main components	Role explanation and customer site link
Actions	Open customer site; sign out
Navigation	Customer login to admin entry → notice → customer site
Related FRs	WEB-FR-AUTH-05, WEB-FR-NAV-01
Related API area	WEB-API-029
Source	Cinema_Mobile/lib/features/admin/screens/staff_only_screen.dart

Web authentication screens are reused between entry points, but admin login disables registration. Profile editing/password changes are dialogs. The phone management landing is selected by role, not a separate administrative URL.

13. API Requirements

Endpoints are extracted from controller HTTP attributes, removing :long constraints from parameter notation only. Direct client means matching method/path found in the scoped Dart services; API-only/shared means a controller exists without a detected dedicated client call. Controller policy must be combined with gateway policy and record ownership.

Table 187: 13. API Requirements

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-001	DELETE /api/rooms/{roomId}	Policy = AuthorizationPolicies.AdminOnly	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:80
WEB-API-002	GET /api/rooms/{roomId}	Policy = AuthorizationPolicies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:41
WEB-API-003	PUT /api/rooms/{roomId}	Policy = AuthorizationPolicies.AdminOnly	Body: RoomRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:67
WEB-API-004	GET /api/rooms/search	Policy = AuthorizationPolicies.StaffOrAdmin	Query: string keyword; Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:91
WEB-API-005	GET /api/rooms	Policy = AuthorizationPolicies.StaffOrAdmin	Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:24
WEB-API-006	POST /api/rooms	Policy = AuthorizationPolicies.AdminOnly	Body: RoomRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/RoomsController.cs:52
WEB-API-007	PUT /api/seats/{seatId}/type	Policy = AuthorizationPolicies.StaffOrAdmin	Body: UpdateSeatTypeRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:60
WEB-API-008	DELETE /api/seats/{seatId}	Policy = AuthorizationPolicies.AdminOnly	—	Shared backend/API-only; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:101

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-009	GET /api/seats/{seatId}	Policy = Authorization Policies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:23
WEB-API-010	POST /api/seats/generate	Policy = Authorization Policies.StaffOrAdmin	Body: GenerateSeatRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:48
WEB-API-011	PUT /api/seats/room/{roomId}/capacity	Policy = Authorization Policies.AdminOnly	Body: UpdateRoomSeatsRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:88
WEB-API-012	PUT /api/seats/room/{roomId}/types	Policy = Authorization Policies.StaffOrAdmin	Body: UpdateSeatTypeRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:74
WEB-API-013	GET /api/seats/room/{roomId}	authenticated	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/SeatsController.cs:36
WEB-API-014	POST /api/showtimes/{showtimeId}/seats	authenticated	Body: List<long> seatIds	Shared backend/API-only; implemented	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs:55
WEB-API-015	PATCH /api/showtimes/{showtimeId}/status	Policy = Authorization Policies.StaffOrAdmin	Body: UpdateShowtimeStatusRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs:108
WEB-API-016	DELETE /api/showtimes/{showtimeId}	Policy = Authorization Policies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs:97
WEB-API-017	GET /api/showtimes/{showtimeId}	public	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs:41
WEB-API-018	PUT /api/showtimes/{showtimeId}	Policy = Authorization Policies.StaffOrAdmin	Body: ShowtimeRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/ShowtimesController.cs:83

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-019	GET /api/showtimes/date-range	Policy = Authorization Policies.StaffOrAdmin	Query: DateTime start; Query: DateTime end; Query: int page Number; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/ API-only; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:180
WEB-API-020	GET /api/showtimes/movie/{movie Id}/open	public	Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:160
WEB-API-021	GET /api/showtimes/movie/{movie Id}	public	Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/ API-only; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:139
WEB-API-022	GET /api/showtimes/open	public	Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/ API-only; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:121
WEB-API-023	GET /api/showtimes	public	Query: int pageNumber; Query: int pageSize; Query: string sortBy; Query: string sortDir	Shared backend/ API-only; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:23
WEB-API-024	POST /api/showtimes	Policy = Authorization Policies.StaffOrAdmin	Body: ShowtimeRequest request	Direct client; implemented	Cinema_BE/MovieService/Movie Service/Controllers/Showtime sController.cs:68
WEB-API-025	POST /api/v1/auth/change-password	public	Body: ChangePassword Request request	Direct client; implemented	Cinema_BE/AuthService/Auth Service/Controllers/Auth Controller.cs:105
WEB-API-026	POST /api/v1/auth/forgot-password	public	Body: EmailRequest request	Direct client; implemented	Cinema_BE/AuthService/Auth Service/Controllers/Auth Controller.cs:54
WEB-API-027	POST /api/v1/auth/login	public	Body: LoginRequest request	Direct client; implemented	Cinema_BE/AuthService/Auth Service/Controllers/Auth Controller.cs:46
WEB-API-028	POST /api/v1/auth/logout	public	Body: RefreshTokenRequest request	Direct client; implemented	Cinema_BE/AuthService/Auth Service/Controllers/Auth Controller.cs:79

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-029	GET /api/v1/auth/me	public	—	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:88
WEB-API-030	GET /api/v1/auth/members	Roles = "ROLE_STAFF,ROLE_ADMIN"	Query: string? phone	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/MembersController.cs:22
WEB-API-031	PUT /api/v1/auth/me	public	Body: UpdateProfileRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:97
WEB-API-032	POST /api/v1/auth/refresh	public	Body: RefreshTokenRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:70
WEB-API-033	POST /api/v1/auth/register	public	Body: RegisterRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:22
WEB-API-034	POST /api/v1/auth/resend-verification	public	Body: EmailRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:38
WEB-API-035	POST /api/v1/auth/reset-password	public	Body: ResetPasswordRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:62
WEB-API-036	GET /api/v1/auth/roles	Roles = "ROLE_ADMIN"	—	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/RolesController.cs:20
WEB-API-037	PUT /api/v1/auth/users/{id}/roles	Roles = "ROLE_ADMIN"	Body: UpdateUserRolesRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs:51
WEB-API-038	PATCH /api/v1/auth/users/{id}/status	Roles = "ROLE_ADMIN"	Body: UpdateUserStatusRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs:58
WEB-API-039	GET /api/v1/auth/users/{id}	Roles = "ROLE_ADMIN"	—	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs:36

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-040	GET /api/v1/auth/users	Roles = "ROLE_ADMIN"	Query: string? keyword; Query: string? role; Query: bool? enabled; Query: int page; Query: int size	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs:24
WEB-API-041	POST /api/v1/auth/users	Roles = "ROLE_ADMIN"	Body: CreateUserRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/UsersController.cs:43
WEB-API-042	POST /api/v1/auth/verify-email	public	Body: VerifyEmailRequest request	Direct client; implemented	Cinema_BE/AuthService/AuthService/Controllers/AuthController.cs:30
WEB-API-043	POST /api/v1/bookings/{id}/cancel	Policy = AuthorizationPolicies.UserOrStaff	Query: string? reason	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:201
WEB-API-044	POST /api/v1/bookings/{id}/check-in	Policy = AuthorizationPolicies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:147
WEB-API-045	POST /api/v1/bookings/{id}/confirm	Policy = AuthorizationPolicies.StaffOrAdmin	Query: string paymentMethod; Query: string? recipientEmail	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:180
WEB-API-046	GET /api/v1/bookings/{id}	Policy = AuthorizationPolicies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:37
WEB-API-047	GET /api/v1/bookings/code/{bookingCode}	Policy = AuthorizationPolicies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:137
WEB-API-048	GET /api/v1/bookings/date-range	Policy = AuthorizationPolicies.StaffOrAdmin	Query: DateTime start; Query: DateTime end; Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:84
WEB-API-049	GET /api/v1/bookings/showtime/{showtimeId}/occupied-seats	Policy = AuthorizationPolicies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:100

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-050	GET /api/v1/bookings/showtime/{showtimeId}/seats	Policy = Authorization Policies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:111
WEB-API-051	GET /api/v1/bookings/showtimes/stats	Policy = Authorization Policies.StaffOrAdmin	Query: List<long>? showtimeIds; Query: DateTime? start; Query: DateTime? end	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:123
WEB-API-052	GET /api/v1/bookings/status/{status}	Policy = Authorization Policies.StaffOrAdmin	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:69
WEB-API-053	GET /api/v1/bookings/user/{userId}	Policy = Authorization Policies.AnyRole	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:50
WEB-API-054	GET /api/v1/bookings	Policy = Authorization Policies.StaffOrAdmin	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:23
WEB-API-055	POST /api/v1/bookings	Policy = Authorization Policies.AnyRole	Body: BookingRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/BookingController.cs:156
WEB-API-056	PUT /api/v1/concessions/{id}	Policy = Authorization Policies.AdminOnly	Body: ConcessionItemRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/ConcessionController.cs:41
WEB-API-057	GET /api/v1/concessions	public	Query: bool includeInactive	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/ConcessionController.cs:24
WEB-API-058	POST /api/v1/concessions	Policy = Authorization Policies.AdminOnly	Body: ConcessionItemRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/ConcessionController.cs:33
WEB-API-059	GET /api/v1/loyalty/accounts	Policy = Authorization Policies.StaffOrAdmin	Query: string? userIds	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs:38
WEB-API-060	GET /api/v1/loyalty/me	authenticated	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs:24

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-061	POST /api/v1/loyalty/users/{userId}/adjust	Policy = Authorization Policies.AdminOnly	Body: LoyaltyAdjustRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs:49
WEB-API-062	GET /api/v1/loyalty/users/{userId}	Policy = Authorization Policies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/LoyaltyController.cs:30
WEB-API-063	DELETE /api/v1/movies/{id}	Policy = Authorization Policies.AdminOnly	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:107
WEB-API-064	GET /api/v1/movies/{id}	public	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:34
WEB-API-065	PUT /api/v1/movies/{id}	Policy = Authorization Policies.AdminOnly	Body: MovieRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:99
WEB-API-066	GET /api/v1/movies/browse	public	Query: string? when; Query: string? genre; Query: string? keyword; Query: int page; Query: int size	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:57
WEB-API-067	GET /api/v1/movies/genres	public	—	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:70
WEB-API-068	GET /api/v1/movies/search	public	Query: string keyword; Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:77
WEB-API-069	GET /api/v1/movies/status/{status}	public	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:42
WEB-API-070	GET /api/v1/movies	public	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:22

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-071	POST /api/v1/movies	Policy = Authorization Policies.AdminOnly	Body: MovieRequest request	Direct client; implemented	Cinema_BE/MovieService/MovieService/Controllers/MoviesController.cs:90
WEB-API-072	PATCH /api/v1/notifications/{id}/read	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:77
WEB-API-073	POST /api/v1/notifications/{id}/send	Policy = Authorization Policies.StaffOrAdmin	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:104
WEB-API-074	DELETE /api/v1/notifications/{id}	Policy = Authorization Policies.AnyRole	—	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:112
WEB-API-075	GET /api/v1/notifications/{id}	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:36
WEB-API-076	POST /api/v1/notifications/read-all	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:89
WEB-API-077	GET /api/v1/notifications/unread-count	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:67
WEB-API-078	GET /api/v1/notifications/user/{userId}	Policy = Authorization Policies.AnyRole	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:49
WEB-API-079	GET /api/v1/notifications	Policy = Authorization Policies.AdminOnly	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:24
WEB-API-080	POST /api/v1/notifications	Policy = Authorization Policies.StaffOrAdmin	Body: NotificationRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/NotificationController.cs:96

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-081	POST /api/v1/payments/{id}/process	Policy = Authorization Policies.StaffOrAdmin	Query: string? recipientEmail	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:186
WEB-API-082	GET /api/v1/payments/{id}/refund	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:208
WEB-API-083	POST /api/v1/payments/{id}/refund	Policy = Authorization Policies.AdminOnly	Query: string? reason	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:198
WEB-API-084	GET /api/v1/payments/{id}	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:57
WEB-API-085	GET /api/v1/payments/booking/{bookingId}	Policy = Authorization Policies.AnyRole	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:88
WEB-API-086	POST /api/v1/payments/cash	Policy = Authorization Policies.StaffOrAdmin	Body: CashPaymentRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:136
WEB-API-087	POST /api/v1/payments/payos/{orderCode}/verify	Policy = Authorization Policies.AnyRole	Query: string? recipientEmail	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:147
WEB-API-088	POST /api/v1/payments/payos/checkout	Policy = Authorization Policies.AnyRole	Body: PayOsCreateRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:122
WEB-API-089	POST /api/v1/payments/payos/webhook	public	Body: JsonElement body	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:172
WEB-API-090	POST /api/v1/payments/refunds/{refundId}/complete	Policy = Authorization Policies.StaffOrAdmin	Body: CompleteRefundRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:255
WEB-API-091	POST /api/v1/payments/refunds/{refundId}/reject	Policy = Authorization Policies.StaffOrAdmin	Body: RejectRefundRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:245

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-092	GET /api/v1/payments/refunds/{refundId}	Policy = Authorization Policies.AnyRole	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:232
WEB-API-093	GET /api/v1/payments/refunds	Policy = Authorization Policies.StaffOrAdmin	Query: string? status; Query: int page; Query: int size	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:222
WEB-API-094	GET /api/v1/payments/summary	Policy = Authorization Policies.AdminOnly	Query: DateTime start; Query: DateTime end	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:46
WEB-API-095	GET /api/v1/payments/user/{userId}	Policy = Authorization Policies.AnyRole	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:70
WEB-API-096	GET /api/v1/payments	Policy = Authorization Policies.StaffOrAdmin	Query: int page; Query: int size; Query: string sortBy; Query: string sortDir	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:31
WEB-API-097	POST /api/v1/payments	Policy = Authorization Policies.AnyRole	Body: PaymentRequest request	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/PaymentController.cs:110
WEB-API-098	PUT /api/v1/promotions/{id}	Policy = Authorization Policies.AdminOnly	Body: PromotionRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs:47
WEB-API-099	POST /api/v1/promotions/quote	authenticated	Body: PromotionQuoteRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs:33
WEB-API-100	GET /api/v1/promotions	public	Query: bool all	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs:24
WEB-API-101	POST /api/v1/promotions	Policy = Authorization Policies.AdminOnly	Body: PromotionRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/PromotionController.cs:40
WEB-API-102	DELETE /api/v1/reviews/{id}	authenticated	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs:58

Continued on next page

Table 187 — continued

ID	Method / endpoint	Controller policy	Input contract	Usage / status	Evidence
WEB-API-103	GET /api/v1/reviews/booking/{bookingId}	authenticated	—	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs:43
WEB-API-104	GET /api/v1/reviews/mine	authenticated	—	Shared backend/API-only; implemented	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs:36
WEB-API-105	GET /api/v1/reviews/movie/{movieId}	public	Query: int page; Query: int size	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs:25
WEB-API-106	POST /api/v1/reviews	authenticated	Body: MovieReviewRequest request	Direct client; implemented	Cinema_BE/BookingService/BookingService/Controllers/MovieReviewController.cs:50

POST /api/showtimes/{showtimeId}/seats is called internally by BookingService with the forwarded token. Its controller permits authenticated callers while the gateway general showtime-write route requires StaffOrAdmin; do not treat it as a Customer client call. PayOS webhook is anonymous to JWT but authenticated by HMAC. Internal calls and webhook remain dependencies, not extra mobile screens.

Pagination is endpoint-specific: page/size for many catalogue/booking/payment APIs; pageNumber/pageSize for declared showtime/room APIs. Paged sizes are capped, usually at 50; genre/menu/seat/loyalty arrays are not all paginated. ApiClient uses a 15-second timeout and UTF-8 JSON/ProblemDetails parsing. Status/result shapes follow current controller DTOs.

13.1 Request field definitions

Table 188: Request field definitions

DTO	Fields / types	Source
ChangePasswordRequest	CurrentPassword: string; NewPassword: string	Cinema_BE/AuthService/AuthService/DTOs/Request/ChangePasswordRequest.cs

Continued on next page

Table 188 — continued

DTO	Fields / types	Source
CreateUserRequest	Email: string; Password: string; FullName: string; Phone: string?; Roles: List<string>	Cinema_BE/AuthService/AuthService/DTOs/Request/CreateUserRequest.cs
EmailRequest	Email: string	Cinema_BE/AuthService/AuthService/DTOs/Request/EmailRequest.cs
LoginRequest	Email: string; Password: string	Cinema_BE/AuthService/AuthService/DTOs/Request/LoginRequest.cs
RefreshTokenRequest	RefreshToken: string	Cinema_BE/AuthService/AuthService/DTOs/Request/RefreshTokenRequest.cs
RegisterRequest	Email: string; Password: string; FullName: string; Phone: string?	Cinema_BE/AuthService/AuthService/DTOs/Request/RegisterRequest.cs
ResetPasswordRequest	Email: string; Otp: string; NewPassword: string	Cinema_BE/AuthService/AuthService/DTOs/Request/ResetPasswordRequest.cs
UpdateProfileRequest	FullName: string; Phone: string?	Cinema_BE/AuthService/AuthService/DTOs/Request/UpdateProfileRequest.cs
UpdateUserRolesRequest	Roles: List<string>	Cinema_BE/AuthService/AuthService/DTOs/Request/UpdateUserRolesRequest.cs
UpdateUserStatusRequest	Enabled: bool?	Cinema_BE/AuthService/AuthService/DTOs/Request/UpdateUserStatusRequest.cs
VerifyEmailRequest	Email: string; Otp: string; Password: string	Cinema_BE/AuthService/AuthService/DTOs/Request/VerifyEmailRequest.cs
BookingRequest	UserId: long?; ShowtimeId: long?; MovieTitle: string?; CustomerEmail: string?; CustomerPhone: string?; ShowTime: DateTime?; Seats: List<SeatRequest>; PromotionCode: string?; UsePoints: int?; Concessions: List<BookingConcessionRequest>	Cinema_BE/BookingService/BookingService/DTOs/Requests/BookingRequest.cs

Continued on next page

Table 188 — continued

DTO	Fields / types	Source
CashPaymentRequest	BookingId: long?; AmountReceived: decimal?; RecipientEmail: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/CashPaymentRequest.cs
CompleteRefundRequest	TransactionRef: string?; Note: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/CompleteRefundRequest.cs
ConcessionItemRequest	Name: string?; Description: string?; Price: decimal?; ImageUrl: string?; Active: bool?; SortOrder: int?	Cinema_BE/BookingService/BookingService/DTOs/Requests/ConcessionItemRequest.cs
LoyaltyAdjustRequest	Points: int?; Note: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/LoyaltyAdjustRequest.cs
MovieReviewRequest	BookingId: long?; Rating: int?; Comment: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/MovieReviewRequest.cs
NotificationRequest	UserId: long?; BookingId: long?; RecipientEmail: string?; Subject: string?; Type: string?; Content: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/NotificationRequest.cs
PaymentRequest	BookingId: long?; UserId: long?; RecipientEmail: string?; Amount: decimal?; Method: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/PaymentRequest.cs
PayOsCreateRequest	BookingId: long?; UserId: long?; Amount: decimal?; Description: string?; ReturnUrl: string?; CancelUrl: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/PayOsCreateRequest.cs
PromotionQuoteRequest	ShowtimeId: long?; TicketAmount: decimal; Tickets: int; Code: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/PromotionQuoteRequest.cs
PromotionRequest	Code: string?; Name: string?; Description: string?; DiscountType: string?; DiscountValue: decimal?; MaxDiscount: decimal?; MinTickets: int?; DaysOfWeek: List<int>?; StartsAt: DateTime?; EndsAt: DateTime?; UsageLimit: int?; Active: bool?	Cinema_BE/BookingService/BookingService/DTOs/Requests/PromotionRequest.cs
RejectRefundRequest	Reason: string?	Cinema_BE/BookingService/BookingService/DTOs/Requests/RejectRefundRequest.cs

Continued on next page

Table 188 — continued

DTO	Fields / types	Source
GenerateSeatRequest	RoomId: long; Rows: int; SeatsPerRow: int; SeatType: string	Cinema_BE/MovieService/MovieService/DTOs/Request/GenerateSeatRequest.cs
MovieRequest	Title: string?; Description: string?; DurationMinutes: int; Genre: string?; Language: string?; ReleaseDate: DateTime; PosterUrl: string?; TrailerUrl: string?; PreviewUrl: string?; AgeRating: string?; Director: string?; Cast: string?; Status: string?	Cinema_BE/MovieService/MovieService/DTOs/Request/MovieRequest.cs
RoomRequest	Name: string; TotalSeats: int; Rows: int; SeatsPerRow: int	Cinema_BE/MovieService/MovieService/DTOs/Request/RoomRequest.cs
ShowtimeRequest	MovieId: long; RoomId: long; StartTime: DateTime; EndTime: DateTime; Price: decimal; Status: string?; Format: string?	Cinema_BE/MovieService/MovieService/DTOs/Request/ShowtimeRequest.cs
UpdateRoomSeatsRequest	TotalSeats: int; SeatsPerRow: int?; SeatType: string?	Cinema_BE/MovieService/MovieService/DTOs/Request/UpdateRoomSeatsRequest.cs
UpdateSeatTypeRequest	SeatIds: List<long>; SeatType: string	Cinema_BE/MovieService/MovieService/DTOs/Request/UpdateSeatTypeRequest.cs
UpdateShowtimeStatusRequest	Status: string	Cinema_BE/MovieService/MovieService/DTOs/Request/UpdateShowtimeStatusRequest.cs

14. Data Requirements

Project-Cinema-DB is uninitialized in this checkout (git submodule status begins with "-"). Its SQL dumps/seed/migrations are unavailable; exact DDL and live migration state are [TO BE CONFIRMED]. This contract uses local EF models/DbContexts and available backend SQL migrations. Model mappings do not prove a live DB has been migrated.

Table 189: 14. Data Requirements

Database	Entities / tables	Integrity and fields
cinema_auth_db	User/Role; users, roles, users_roles, refresh_tokens, otp_codes	Unique email/token; BCrypt hashes; role joins; OTP lifetime and token revocation
cinema_movie_db	Movie; movies	Title/duration/genre/language/rating/director/cast; poster/trailer/preview; ACTIVE/INACTIVE
cinema_showtime_db	Room/Seat/Showtime; rooms, seats, showtimes, outbox_messages	Unique room/row/number; format, time, price; room-local foreign keys
cinema_booking_db	Booking, BookingSeat, SeatReservation, ConcessionItem, BookingConcession, Promotion, LoyaltyAccount, LoyaltyTransaction, MovieReview; bookings, booking_seats, seat_reservations, concession_items, booking_concessions, promotions, loyalty_accounts, loyalty_transactions, movie_reviews, outbox_messages	Unique booking code; unique showtime/seat reservation; unique user/movie review; amount snapshots; nullable counter userId; checkedInAt/expiresAt/remindedAt
cinema_payment_db	Payment/Refund; payments, refunds, outbox_messages	Booking, amount, method/state, PayOS reference and payer bank; one refund per payment
cinema_notification_db	Notification; notifications	Recipient/booking/event; deduplicate event; sentAt/readAt; state and error

Client persistence is session tokens/user metadata via flutter_secure_storage and in-memory booking drafts, not six local databases. Server decimal totals snapshot seat/concession/discount/point values. Cinema time uses Vietnam local time and timestamp without time zone mappings. Cross-service IDs are application/event-validated logical references.

15. Business Workflows

Table 190: 15. Business Workflows

ID	Workflow
WEB-WF-01	Register → verify OTP → role landing; unverified login goes to OTP. A failed recovery never authenticates an account.
WEB-WF-02	Browse → movie → open showtime → seats → summary with extras/code/points/age → create PENDING hold → PayOS → server verify → CONFIRMED + ticket notification.
WEB-WF-03	expiresAt reached → expiry worker (default every 30 seconds) → EXPIRED and release seats/points. Late payment reclaims seats and confirms, or creates full refund.
WEB-WF-04	Paid customer cancellation ≥2 hours before show → REFUND_PENDING with seats retained → web staff manual transfer/record → REFUNDED/CANCELLED. Eligible rejection restores SUCCESS/CONFIRMED.
WEB-WF-05	1. Scan QR or type the booking code and press Enter. 2. Lookup ticket; enrich room and age rating when catalogue calls succeed. 3. Check identity/rating as prompted, then explicitly check in. 4. Store checkedInAt, show confirmation and prepare for the next code.
WEB-WF-06	1. Select walk-in or find a member by phone; choose showtime, seats and optional concessions/promotion/points. 2. Hold seats and open counter checkout. 3. Take sufficient cash and display change, or show VietQR and verify every three seconds. 4. After server confirmation show/print the QR ticket, optionally email it and check in within the permitted window.
WEB-WF-07	Cancel showtime → MovieService outbox → BookingService event handler → cancel bookings/create paid refunds → notifications. Propagation is eventual, not one transaction across databases.

15.1 State contract

Table 191: State contract

Object	Transitions
Booking	PENDING → CONFIRMED / CANCELLED / EXPIRED; EXPIRED → CONFIRMED (verified late payment and available seats); CONFIRMED → REFUND_PENDING / CANCELLED; REFUND_PENDING → CONFIRMED / CANCELLED
Payment	PENDING → SUCCESS / FAILED; FAILED → SUCCESS (verified late payment); SUCCESS → REFUND_PENDING → REFUNDED; REFUND_PENDING → SUCCESS
Refund	PENDING → COMPLETED / REJECTED; REJECTED → PENDING (cinema cancels later)
Showtime	OPEN ↔ CLOSED; OPEN/CLOSED → CANCELLED
Notification	PENDING → SENT/FAILED; FAILED → SENT; readAt independent of send state

checkedInAt is an attribute of a confirmed booking, not a separate booking status.

16. Authentication & Authorization

Gateway and services validate `JWT` signature/expiry/issuer/audience. Defaults: access 60 minutes, refresh seven days rotated on use; `OTP` six digits, five minutes, five attempts, 60-second resend. Admin-created accounts skip registration `OTP`. Password and `OTP` values are BCrypt-hashed.

`ApiClient` renews and retries a protected request once. Protected checks reject `ACCOUNT_LOCKED` (`401`), `ROLE_CHANGED` (`403`) or unavailable `AuthService` (`503`). Public catalogue GETs omit bearer token; a signed-in client probes /auth/me at most once per minute, ignores ordinary availability errors for browsing and expires the session for lock/role-change signals. Logout clears storage/revokes refresh; reset/change password revokes old refresh tokens. Refresh revocation alone does not certify immediate invalidation of every issued access token.

17. Error Handling

Table 192: 17. Error Handling

Code / condition	Handling
`400`	Show field/business validation; do not advance
`401`	Renew once if possible; otherwise clear session/sign in; retain lock/session reason
`403`	Deny role/ownership action; `EMAIL_NOT_VERIFIED` opens `OTP`; `ROLE_CHANGED` signs in again
`404`	Show missing record without leaking another owner's review ticket
`409`	Explain seat/state/time conflict and reload before retrying
`429`	Respect Retry-After/countdown; preserve inputs
`500` / `503`	Show retryable service error; no false booking/payment success
Timeout/offline/invalid `JSON`	Bounded error (15-second client timeout); retry and honest pending/failure state
Loading/empty/camera denial	Explicit state without fake records; preserve manual ticket-code entry

18. Non-Functional Requirements

Table 193: 18. Non-Functional Requirements

ID	Requirement	Allocation
WEB-NFR-SEC-01	Passwords and OTP codes are hashed with BCrypt; responses never include stack traces or internal errors.	Web/backend
WEB-NFR-SEC-02	JWTs are signed with a secret of at least 32 characters; access tokens last 60 minutes; refresh tokens last 7 days, rotate, and are revoked on sign-out, password reset and role change.	Web/backend
WEB-NFR-SEC-03	Authorization is checked at the gateway (per route) and in the service (per record, e.g. only one's own tickets).	Web/backend
WEB-NFR-SEC-04	Rate limits: 100 requests per minute per IP for all APIs, 10 for authentication APIs.	Web/backend
WEB-NFR-SEC-05	Only the gateway is reachable from the network; services, PostgreSQL, RabbitMQ and monitoring listen on 127.0.0.1.	Web/backend
WEB-NFR-SEC-06	Secrets are not in the source code; they are shared with SOPS + age.	Web/backend
WEB-NFR-SEC-07	The server validates every input (DataAnnotations + validators); clients validate first to report mistakes early.	Web/backend
WEB-NFR-PERF-01	Paged endpoints cap size, normally 50; genres/seats/menu arrays are exempt. Ticket UI currently loads only the first page. Response-time/concurrent-user SLA is [TO BE CONFIRMED], not a measured guarantee.	Web/backend
WEB-NFR-PERF-02	Holding seats, confirming tickets and taking payment lock the booking row (SELECT … FOR UPDATE) in a transaction, so seats are never sold twice and money is never taken twice under concurrent use.	Web/backend
WEB-NFR-REL-01	Calls to MovieService / PayOS retry transient errors at most twice (creating a PayOS link is never retried); a circuit breaker opens for 15 seconds after repeated failures and returns 503 at once.	Web/backend
WEB-NFR-REL-02	Outbox: no event is lost while RabbitMQ is down; it is sent when RabbitMQ is back.	Web/backend
WEB-NFR-REL-03	Consumers are idempotent; failing messages are retried 3 times and then moved to a DLQ.	Web/backend
WEB-NFR-REL-04	The gateway checks each service's /health/live every 10 seconds and returns 503 at once for a dead service.	Web/backend
WEB-NFR-OBS-01	Prometheus metrics for HTTP, runtime and business events (bookings, payments, refunds, outbox); Grafana dashboard "Cinema - Tổng quan".	Web/backend
WEB-NFR-OBS-02	OpenTelemetry traces to Jaeger across the gateway, services, SQL, PayOS and RabbitMQ.	Web/backend
WEB-NFR-OBS-03	Logs carry TraceId, SpanId and CorrelationId; every response has an X-Correlation-ID header.	Web/backend
WEB-NFR-MAINT-01	Layered services, DTOs, validators and a global exception handler everywhere; documentation in docs/.	Web/backend
WEB-NFR-MAINT-02	CI declares .NET tests, Flutter analyze/test and customer/admin web builds. Current Flutter workflow pins 3.47.4, superseding older prose; no current successful CI run is asserted.	Web/backend
WEB-NFR-DEP-01	One command, docker compose up -d --build, runs the whole backend; monitoring through the monitoring profile.	Web/backend
WEB-NFR-USE-01	The customer app supports Android, web and Windows; the admin website is designed for desktops and still works on narrow screens.	Web/backend
WEB-NFR-COMP-01	Web cameras need localhost or HTTPS; a real deployment must use HTTPS.	Web/backend

`NFR` mechanisms may be evidenced in code; availability, throughput, universal accessibility and compatibility require runtime validation and are not established by static inspection.

19. Security Requirements

Table 194: 19. Security Requirements

ID	Security requirement and verification
WEB-SEC-01	Reject forged/expired/wrong-issuer JWT and forbidden roles at gateway/service; test record-ID changes against ownership.
WEB-SEC-02	Recompute prices and validate seat pairs/menu/code/points on server; do not trust client totals or redirect parameters as payment.
WEB-SEC-03	Verify PayOS HMAC before effects; test repeat webhook, wrong signature, unknown order and amount mismatch. Keep merchant keys in backend.
WEB-SEC-04	Use deployment HTTPS and restricted CORS/private-service exposure. Development HTTP is not proof of production TLS.
WEB-SEC-05	Use platform secure-storage abstraction; never embed backend secrets in client. Clear/revoke credentials on logout/reset and handle unreadable storage as signed out.
WEB-SEC-06	Encode user content in email; suppress stack traces/secrets. Generic forgot-password does not imply universal resistance to email enumeration.

20. Testing Requirements

Derive tests from each FR AC-01/AC-02. Existing test files evidence coverage intent, not a newly run report. Backend fixtures use Testcontainers PostgreSQL/RabbitMQ and configured fakes for external PayOS/SMTP/ catalogue dependencies. Live PayOS/SMTP, camera/share/printing require separate controlled acceptance.

Table 195: 20. Testing Requirements

ID	Required test
WEB-TEST-01	OTP expiry/attempts/resend; UTF-8 password bounds; refresh reuse; lock/role changes; wrong owner/role.
WEB-TEST-02	Two simultaneous holds on one seat: one success, one 409; no partial seats/points. Reject half sweetbox.
WEB-TEST-03	Payment success/cancel/timeout, duplicate signed webhook, late payment with available/lost seats; never false paid UI.
WEB-TEST-04	Cancellation exactly at and around two hours; complete/reject/reopen refund; one refund, pending-seat retention and point reversal.
WEB-TEST-05	Check-in immediately before/at/after time limits; two simultaneous scans; manual lookup after camera denial.
WEB-TEST-06	Route restoration, missing draft, loading/empty/error, timeout/retry, unread counts and return from external app.
WEB-TEST-07	Cash underpayment/change, transfer-to-cash race, 80 mm print; schedule overlap; Admin-only master-data/revenue access.

20.1 Existing test evidence

Table 196: Existing test evidence

Area / use case	Files
UC-WEB-01	Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart; Cinema_Mobile/test/register_validation_test.dart
UC-WEB-02	Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
UC-WEB-03	Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart; Cinema_Mobile/test/liquid_glass_responsive_preview_test.dart; Cinema_Mobile/test/review_test.dart
UC-WEB-04	Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs; Cinema_Mobile/test/seat_map_test.dart; Cinema_Mobile/test/showtime_format_test.dart
UC-WEB-05	Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
UC-WEB-06	Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs; Cinema_Mobile/test/showtime_format_test.dart

Continued on next page

Table 196 — continued

Area / use case	Files
UC-WEB-07	Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PromotionTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/concession_test.dart; Cinema_Mobile/test/loyalty_test.dart; Cinema_Mobile/test/promotion_test.dart
UC-WEB-08	Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs; Cinema_BE/BookingService/BookingService.Tests/Unit/PayOsSignatureTests.cs; Cinema_Mobile/test/booking_flow_test.dart
UC-WEB-09	Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart; Cinema_Mobile/test/staff_booking_details_ui_test.dart
UC-WEB-10	Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
UC-WEB-11	Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart
UC-WEB-12	Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
UC-WEB-13	Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_Mobile/test/today_at_cinema_test.dart

21. Acceptance Criteria

All H requirements must meet both criteria; security, ownership, financial and concurrency tests must pass. Disclose every M/L omission and current implementation gap. This document does not certify a passing run.

Table 197: 21. Acceptance Criteria

ID	Acceptance gate
WEB-ACC-01	Demonstrate account → movie → showtime → seats → booking → payment → ticket/ notification with real backend or documented fixtures.
WEB-ACC-02	Reject cross-account unauthorized access and duplicate concurrent seat sale/check-in/payment effects.
WEB-ACC-03	Demonstrate refund and late-payment outcomes with correct seats/points and no false confirmed ticket.
WEB-ACC-04	Verify navigation, loading/error/empty and device fallbacks; disclose §27 limitations.

22. Requirements Traceability Matrix

Complete FR matrix. Source IDs provide backward mapping; NAV/STATE/MEDIA are code-derived refinements. API associations are by functional area, not a claim that every screen invokes every listed operation. Existing tests are coverage candidates, not measured test results.

Table 198: 22. Requirements Traceability Matrix

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-01 ← FR-AUTH-01	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-REGISTER	WEB-API-033	WEB-FR-AUTH-01-AC-01, WEB-FR-AUTH-01-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-02 ← FR-AUTH-02	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN	WEB-API-027	WEB-FR-AUTH-02-AC-01, WEB-FR-AUTH-02-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-03 ← FR-AUTH-03	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN	WEB-API-027, WEB-API-032	WEB-FR-AUTH-03-AC-01, WEB-FR-AUTH-03-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-04 ← FR-AUTH-04	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01		WEB-API-029	WEB-FR-AUTH-04-AC-01, WEB-FR-AUTH-04-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-05 ← FR-AUTH-05	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-STAFF-ONLY	WEB-API-029	WEB-FR-AUTH-05-AC-01, WEB-FR-AUTH-05-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-06 ← FR-AUTH-06	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN	WEB-API-029	WEB-FR-AUTH-06-AC-01, WEB-FR-AUTH-06-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-07 ← FR-AUTH-07	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-VERIFY	WEB-API-034, WEB-API-042	WEB-FR-AUTH-07-AC-01, WEB-FR-AUTH-07-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-08 ← FR-AUTH-08	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-REGISTER, WEB-SCR-VERIFY	WEB-API-033	WEB-FR-AUTH-08-AC-01, WEB-FR-AUTH-08-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-09 ← FR-AUTH-09	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-FORGOT, WEB-SCR-RESET	WEB-API-026, WEB-API-035	WEB-FR-AUTH-09-AC-01, WEB-FR-AUTH-09-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-10 ← FR-AUTH-10	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE	WEB-API-028, WEB-API-032	WEB-FR-AUTH-10-AC-01, WEB-FR-AUTH-10-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-11 ← FR-AUTH-11	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE	WEB-API-029	WEB-FR-AUTH-11-AC-01, WEB-FR-AUTH-11-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-12 ← FR-AUTH-12	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN	WEB-API-029	WEB-FR-AUTH-12-AC-01, WEB-FR-AUTH-12-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-13 ← FR-AUTH-13	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN, WEB-SCR-REGISTER, WEB-SCR-FORGOT, WEB-SCR-RESET, WEB-SCR-USER-EDITOR	WEB-API-027, WEB-API-033, WEB-API-035, WEB-API-041	WEB-FR-AUTH-13-AC-01, WEB-FR-AUTH-13-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-14 ← FR-AUTH-14	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-VERIFY, WEB-SCR-FORGOT, WEB-SCR-RESET	WEB-API-027, WEB-API-032, WEB-API-034	WEB-FR-AUTH-14-AC-01, WEB-FR-AUTH-14-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-15 ← FR-AUTH-15	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-COUNTER	WEB-API-030	WEB-FR-AUTH-15-AC-01, WEB-FR-AUTH-15-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-AUTH-16 ← FR-AUTH-16	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE	WEB-API-031	WEB-FR-AUTH-16-AC-01, WEB-FR-AUTH-16-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-AUTH-17 ← FR-AUTH-17	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-PROFILE, WEB-SCR-MGT-PROFILE	WEB-API-025	WEB-FR-AUTH-17-AC-01, WEB-FR-AUTH-17-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-USER-01 ← FR-USER-01	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-ADMIN-USERS	WEB-API-040	WEB-FR-USER-01-AC-01, WEB-FR-USER-01-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
WEB-FR-USER-02 ← FR-USER-02	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-USER-EDITOR	WEB-API-039	WEB-FR-USER-02-AC-01, WEB-FR-USER-02-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-USER-03 ← FR-USER-03	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-USER-EDITOR	WEB-API-041	WEB-FR-USER-03-AC-01, WEB-FR-USER-03-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
WEB-FR-USER-04 ← FR-USER-04	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-USER-EDITOR	WEB-API-037	WEB-FR-USER-04-AC-01, WEB-FR-USER-04-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
WEB-FR-USER-05 ← FR-USER-05	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-USER-EDITOR	WEB-API-038	WEB-FR-USER-05-AC-01, WEB-FR-USER-05-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
WEB-FR-USER-06 ← FR-USER-06	WEB-BR-03, WEB-BR-24	UC-WEB-02	WEB-SCR-ADMIN-USERS, WEB-SCR-USER-EDITOR	WEB-API-036	WEB-FR-USER-06-AC-01, WEB-FR-USER-06-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/UserManagementTests.cs; Cinema_Mobile/test/admin_user_roles_test.dart
WEB-FR-MOV-01 ← FR-MOV-01	WEB-BR-26	UC-WEB-03	WEB-SCR-ADMIN-MOVIES	WEB-API-070	WEB-FR-MOV-01-AC-01, WEB-FR-MOV-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-MOV-02 ← FR-MOV-02	WEB-BR-26	UC-WEB-03	WEB-SCR-MOVIE, WEB-SCR-ADMIN-MOVIES	WEB-API-064	WEB-FR-MOV-02-AC-01, WEB-FR-MOV-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-03 ← FR-MOV-03	WEB-BR-26	UC-WEB-03	WEB-SCR-HOME, WEB-SCR-ADMIN-MOVIES	WEB-API-066, WEB-API-068	WEB-FR-MOV-03-AC-01, WEB-FR-MOV-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-04 ← FR-MOV-04	WEB-BR-26	UC-WEB-03	WEB-SCR-HOME, WEB-SCR-ADMIN-MOVIES	WEB-API-066, WEB-API-069	WEB-FR-MOV-04-AC-01, WEB-FR-MOV-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-05 ← FR-MOV-05	WEB-BR-26	UC-WEB-03	WEB-SCR-ADMIN-MOVIES	WEB-API-071	WEB-FR-MOV-05-AC-01, WEB-FR-MOV-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-MOV-06 ← FR-MOV-06	WEB-BR-26	UC-WEB-03	WEB-SCR-ADMIN-MOVIES	WEB-API-065	WEB-FR-MOV-06-AC-01, WEB-FR-MOV-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-07 ← FR-MOV-07	WEB-BR-26	UC-WEB-03	WEB-SCR-ADMIN-MOVIES	WEB-API-063	WEB-FR-MOV-07-AC-01, WEB-FR-MOV-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-08 ← FR-MOV-08	WEB-BR-26	UC-WEB-03	WEB-SCR-MOVIE	WEB-API-064	WEB-FR-MOV-08-AC-01, WEB-FR-MOV-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-09 ← FR-MOV-09	WEB-BR-26	UC-WEB-03	WEB-SCR-MOVIE, WEB-SCR-SUMMARY, WEB-SCR-CHECKIN, WEB-SCR-ADMIN-MOVIES	WEB-API-064, WEB-API-065, WEB-API-071	WEB-FR-MOV-09-AC-01, WEB-FR-MOV-09-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-MOV-10 ← FR-MOV-10	WEB-BR-26	UC-WEB-03	WEB-SCR-HOME	WEB-API-066, WEB-API-067	WEB-FR-MOV-10-AC-01, WEB-FR-MOV-10-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-11 ← FR-MOV-11	WEB-BR-26	UC-WEB-03	WEB-SCR-MOVIE, WEB-SCR-ADMIN-MOVIES	WEB-API-064, WEB-API-065, WEB-API-071	WEB-FR-MOV-11-AC-01, WEB-FR-MOV-11-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart
WEB-FR-MOV-12 ← FR-MOV-12	WEB-BR-26	UC-WEB-03	WEB-SCR-MOVIE, WEB-SCR-TICKET, WEB-SCR-ADMIN-MOVIES	WEB-API-102, WEB-API-103, WEB-API-105, WEB-API-106	WEB-FR-MOV-12-AC-01, WEB-FR-MOV-12-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-ROOM-01 ← FR-ROOM-01	WEB-BR-04, WEB-BR-26	UC-WEB-04	WEB-SCR-OPERATIONS, WEB-SCR-ADMIN-ROOMS	WEB-API-004, WEB-API-005	WEB-FR-ROOM-01-AC-01, WEB-FR-ROOM-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs
WEB-FR-ROOM-02 ← FR-ROOM-02	WEB-BR-04, WEB-BR-26	UC-WEB-04	WEB-SCR-RECORD-ROOMS, WEB-SCR-ADMIN-ROOMS	WEB-API-002, WEB-API-013	WEB-FR-ROOM-02-AC-01, WEB-FR-ROOM-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs
WEB-FR-ROOM-03 ← FR-ROOM-03	WEB-BR-04, WEB-BR-26	UC-WEB-04	WEB-SCR-ADMIN-ROOMS	WEB-API-006, WEB-API-010	WEB-FR-ROOM-03-AC-01, WEB-FR-ROOM-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-ROOM-04 ← FR-ROOM-04	WEB-BR-04, WEB-BR-26	UC-WEB-04	WEB-SCR-ADMIN-ROOMS	WEB-API-003	WEB-FR-ROOM-04-AC-01, WEB-FR-ROOM-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs
WEB-FR-ROOM-05 ← FR-ROOM-05	WEB-BR-04, WEB-BR-26	UC-WEB-04	WEB-SCR-ADMIN-ROOMS	WEB-API-001	WEB-FR-ROOM-05-AC-01, WEB-FR-ROOM-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs
WEB-FR-SEAT-01 ← FR-SEAT-01	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-010	WEB-FR-SEAT-01-AC-01, WEB-FR-SEAT-01-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SEAT-02 ← FR-SEAT-02	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-009	WEB-FR-SEAT-02-AC-01, WEB-FR-SEAT-02-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-03 ← FR-SEAT-03	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-SEATS, WEB-SCR-RECORD-ROOMS	WEB-API-013	WEB-FR-SEAT-03-AC-01, WEB-FR-SEAT-03-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-04 ← FR-SEAT-04	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05		WEB-API-008	WEB-FR-SEAT-04-AC-01, WEB-FR-SEAT-04-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-05 ← FR-SEAT-05	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-007	WEB-FR-SEAT-05-AC-01, WEB-FR-SEAT-05-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SEAT-06 ← FR-SEAT-06	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-012	WEB-FR-SEAT-06-AC-01, WEB-FR-SEAT-06-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-07 ← FR-SEAT-07	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-011	WEB-FR-SEAT-07-AC-01, WEB-FR-SEAT-07-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-08 ← FR-SEAT-08	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-007, WEB-API-010, WEB-API-011, WEB-API-012	WEB-FR-SEAT-08-AC-01, WEB-FR-SEAT-08-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart
WEB-FR-SEAT-09 ← FR-SEAT-09	WEB-BR-04, WEB-BR-07, WEB-BR-15, WEB-BR-16, WEB-BR-26	UC-WEB-05	WEB-SCR-RECORD-ROOMS	WEB-API-013	WEB-FR-SEAT-09-AC-01, WEB-FR-SEAT-09-AC-02; Cinema_BE/MovieService/MovieService.Tests/Integration/SeatTests.cs; Cinema_Mobile/test/booking_flow_test.dart; Cinema_Mobile/test/seat_map_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SHOW-01 ← FR-SHOW-01	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-OPERATIONS	WEB-API-023	WEB-FR-SHOW-01-AC-01, WEB-FR-SHOW-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-02 ← FR-SHOW-02	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-SHOW-EDITOR	WEB-API-024	WEB-FR-SHOW-02-AC-01, WEB-FR-SHOW-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-03 ← FR-SHOW-03	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-MOVIE, WEB-SCR-SEATS, WEB-SCR-RECORD-SHOWTIMES	WEB-API-017	WEB-FR-SHOW-03-AC-01, WEB-FR-SHOW-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SHOW-04 ← FR-SHOW-04	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-SHOW-EDITOR	WEB-API-018	WEB-FR-SHOW-04-AC-01, WEB-FR-SHOW-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-05 ← FR-SHOW-05	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-RECORD-SHOWTIMES	WEB-API-016	WEB-FR-SHOW-05-AC-01, WEB-FR-SHOW-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-06 ← FR-SHOW-06	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-MOVIE	WEB-API-022	WEB-FR-SHOW-06-AC-01, WEB-FR-SHOW-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SHOW-07 ← FR-SHOW-07	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-MOVIE	WEB-API-021	WEB-FR-SHOW-07-AC-01, WEB-FR-SHOW-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-08 ← FR-SHOW-08	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-MOVIE	WEB-API-020	WEB-FR-SHOW-08-AC-01, WEB-FR-SHOW-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-09 ← FR-SHOW-09	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-OPERATIONS	WEB-API-019	WEB-FR-SHOW-09-AC-01, WEB-FR-SHOW-09-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SHOW-10 ← FR-SHOW-10	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-RECORD-SHOWTIMES	WEB-API-015	WEB-FR-SHOW-10-AC-01, WEB-FR-SHOW-10-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-11 ← FR-SHOW-11	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-SHOW-EDITOR	WEB-API-018, WEB-API-024	WEB-FR-SHOW-11-AC-01, WEB-FR-SHOW-11-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-SHOW-12 ← FR-SHOW-12	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-RECORD-SHOWTIMES	WEB-API-013, WEB-API-017, WEB-API-050	WEB-FR-SHOW-12-AC-01, WEB-FR-SHOW-12-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-SHOW-13 ← FR-SHOW-13	WEB-BR-05, WEB-BR-06, WEB-BR-15, WEB-BR-19, WEB-BR-30	UC-WEB-06	WEB-SCR-MOVIE, WEB-SCR-SEATS, WEB-SCR-SHOW-EDITOR	WEB-API-017, WEB-API-018, WEB-API-024	WEB-FR-SHOW-13-AC-01, WEB-FR-SHOW-13-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeFormatTests.cs; Cinema_BE/MovieService/MovieService.Tests/Integration/ShowtimeTests.cs
WEB-FR-BOOK-01 ← FR-BOOK-01	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SUMMARY	WEB-API-055	WEB-FR-BOOK-01-AC-01, WEB-FR-BOOK-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-02 ← FR-BOOK-02	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-RESULT, WEB-SCR-TICKET, WEB-SCR-RECORD-BOOKINGS	WEB-API-046	WEB-FR-BOOK-02-AC-01, WEB-FR-BOOK-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-03 ← FR-BOOK-03	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-OPERATIONS	WEB-API-048, WEB-API-052, WEB-API-054	WEB-FR-BOOK-03-AC-01, WEB-FR-BOOK-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-04 ← FR-BOOK-04	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-TICKETS	WEB-API-053	WEB-FR-BOOK-04-AC-01, WEB-FR-BOOK-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-05 ← FR-BOOK-05	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-PAYMENT, WEB-SCR-RESULT, WEB-SCR-TICKET	WEB-API-086, WEB-API-087, WEB-API-089	WEB-FR-BOOK-05-AC-01, WEB-FR-BOOK-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-06 ← FR-BOOK-06	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-TICKET, WEB-SCR-RECORD-BOOKINGS	WEB-API-043	WEB-FR-BOOK-06-AC-01, WEB-FR-BOOK-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-07 ← FR-BOOK-07	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SEATS	WEB-API-049	WEB-FR-BOOK-07-AC-01, WEB-FR-BOOK-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-08 ← FR-BOOK-08	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SEATS	WEB-API-055	WEB-FR-BOOK-08-AC-01, WEB-FR-BOOK-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-09 ← FR-BOOK-09	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SEATS, WEB-SCR-SUMMARY	WEB-API-055	WEB-FR-BOOK-09-AC-01, WEB-FR-BOOK-09-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-10 ← FR-BOOK-10	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-PAYMENT	WEB-API-046	WEB-FR-BOOK-10-AC-01, WEB-FR-BOOK-10-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-11 ← FR-BOOK-11	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SEATS	WEB-API-013, WEB-API-049	WEB-FR-BOOK-11-AC-01, WEB-FR-BOOK-11-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-12 ← FR-BOOK-12	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-OPERATIONS, WEB-SCR-RECORD-SHOWTIMES	WEB-API-051	WEB-FR-BOOK-12-AC-01, WEB-FR-BOOK-12-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-13 ← FR-BOOK-13	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-COUNTER	WEB-API-055, WEB-API-057	WEB-FR-BOOK-13-AC-01, WEB-FR-BOOK-13-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-14 ← FR-BOOK-14	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-ADMIN-SNACKS	WEB-API-056, WEB-API-057, WEB-API-058	WEB-FR-BOOK-14-AC-01, WEB-FR-BOOK-14-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-15 ← FR-BOOK-15	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-COUNTER	WEB-API-055, WEB-API-099	WEB-FR-BOOK-15-AC-01, WEB-FR-BOOK-15-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-16 ← FR-BOOK-16	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-ADMIN-PROMOTIONS	WEB-API-098, WEB-API-100, WEB-API-101	WEB-FR-BOOK-16-AC-01, WEB-FR-BOOK-16-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-17 ← FR-BOOK-17	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-SUMMARY, WEB-SCR-TICKET, WEB-SCR-PROFILE, WEB-SCR-COUNTER	WEB-API-055, WEB-API-060	WEB-FR-BOOK-17-AC-01, WEB-FR-BOOK-17-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-BOOK-18 ← FR-BOOK-18	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-ADMIN-MEMBERS	WEB-API-030, WEB-API-059, WEB-API-061, WEB-API-062	WEB-FR-BOOK-18-AC-01, WEB-FR-BOOK-18-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-BOOK-19 ← FR-BOOK-19	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07	WEB-SCR-TICKET	WEB-API-017	WEB-FR-BOOK-19-AC-01, WEB-FR-BOOK-19-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-01 ← FR-PAY-01	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-PAYMENT, WEB-SCR-CHECKOUT	WEB-API-088	WEB-FR-PAY-01-AC-01, WEB-FR-PAY-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-PAY-02 ← FR-PAY-02	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-PAYMENT, WEB-SCR-RESULT, WEB-SCR-CHECKOUT	WEB-API-087	WEB-FR-PAY-02-AC-01, WEB-FR-PAY-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-03 ← FR-PAY-03	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-TICKET, WEB-SCR-PAYHISTORY, WEB-SCR-RECORD-PAYMENTS	WEB-API-084	WEB-FR-PAY-03-AC-01, WEB-FR-PAY-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-PAY-04 ← FR-PAY-04	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-PAYHISTORY	WEB-API-095	WEB-FR-PAY-04-AC-01, WEB-FR-PAY-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-05 ← FR-PAY-05	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-RECORD-PAYMENTS	WEB-API-085	WEB-FR-PAY-05-AC-01, WEB-FR-PAY-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-PAY-06 ← FR-PAY-06	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-TICKET	WEB-API-043, WEB-API-082, WEB-API-083	WEB-FR-PAY-06-AC-01, WEB-FR-PAY-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-07 ← FR-PAY-07	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08		WEB-API-089	WEB-FR-PAY-07-AC-01, WEB-FR-PAY-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-PAY-08 ← FR-PAY-08	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-RESULT	WEB-API-087, WEB-API-089	WEB-FR-PAY-08-AC-01, WEB-FR-PAY-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-09 ← FR-PAY-09	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-OPERATIONS, WEB-SCR-RECORD-PAYMENTS	WEB-API-087, WEB-API-096	WEB-FR-PAY-09-AC-01, WEB-FR-PAY-09-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-PAY-10 ← FR-PAY-10	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08		WEB-API-094	WEB-FR-PAY-10-AC-01, WEB-FR-PAY-10-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-PAY-11 ← FR-PAY-11	WEB-BR-09, WEB-BR-10, WEB-BR-11, WEB-BR-12, WEB-BR-13, WEB-BR-14, WEB-BR-20	UC-WEB-08	WEB-SCR-PAYMENT	WEB-API-046, WEB-API-087	WEB-FR-PAY-11-AC-01, WEB-FR-PAY-11-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/PayOsOrderCodeTests.cs
WEB-FR-REF-01 ← FR-REF-01	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-TICKET	WEB-API-043, WEB-API-083, WEB-API-089	WEB-FR-REF-01-AC-01, WEB-FR-REF-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart
WEB-FR-REF-02 ← FR-REF-02	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-OPERATIONS, WEB-SCR-RECORD-REFUNDS	WEB-API-092, WEB-API-093	WEB-FR-REF-02-AC-01, WEB-FR-REF-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-REF-03 ← FR-REF-03	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-RECORD-REFUNDS	WEB-API-090	WEB-FR-REF-03-AC-01, WEB-FR-REF-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart
WEB-FR-REF-04 ← FR-REF-04	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-RECORD-REFUNDS	WEB-API-091	WEB-FR-REF-04-AC-01, WEB-FR-REF-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart
WEB-FR-REF-05 ← FR-REF-05	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-TICKET, WEB-SCR-RECORD-REFUNDS	WEB-API-016, WEB-API-043	WEB-FR-REF-05-AC-01, WEB-FR-REF-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-REF-06 ← FR-REF-06	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-TICKET, WEB-SCR-RECORD-PAYMENTS	WEB-API-082	WEB-FR-REF-06-AC-01, WEB-FR-REF-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart
WEB-FR-REF-07 ← FR-REF-07	WEB-BR-17, WEB-BR-18, WEB-BR-19, WEB-BR-20	UC-WEB-09	WEB-SCR-RECORD-REFUNDS	WEB-API-043, WEB-API-082	WEB-FR-REF-07-AC-01, WEB-FR-REF-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/RefundTests.cs; Cinema_Mobile/test/booking_details_ui_test.dart; Cinema_Mobile/test/refund_details_ui_test.dart
WEB-FR-POS-01 ← FR-POS-01	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-COUNTER	WEB-API-013, WEB-API-023, WEB-API-030, WEB-API-055	WEB-FR-POS-01-AC-01, WEB-FR-POS-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-POS-02 ← FR-POS-02	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-046	WEB-FR-POS-02-AC-01, WEB-FR-POS-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-POS-03 ← FR-POS-03	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-086	WEB-FR-POS-03-AC-01, WEB-FR-POS-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-POS-04 ← FR-POS-04	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-087, WEB-API-088	WEB-FR-POS-04-AC-01, WEB-FR-POS-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-POS-05 ← FR-POS-05	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-086	WEB-FR-POS-05-AC-01, WEB-FR-POS-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-POS-06 ← FR-POS-06	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-055, WEB-API-086	WEB-FR-POS-06-AC-01, WEB-FR-POS-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-POS-07 ← FR-POS-07	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT	WEB-API-044, WEB-API-046	WEB-FR-POS-07-AC-01, WEB-FR-POS-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-POS-08 ← FR-POS-08	WEB-BR-08, WEB-BR-15, WEB-BR-16, WEB-BR-22, WEB-BR-23, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-10	WEB-SCR-CHECKOUT, WEB-SCR-RECORD-BOOKINGS	WEB-API-046	WEB-FR-POS-08-AC-01, WEB-FR-POS-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CounterSaleTests.cs; Cinema_Mobile/test/counter_booking_test.dart
WEB-FR-CHK-01 ← FR-CHK-01	WEB-BR-21	UC-WEB-11	WEB-SCR-TICKET, WEB-SCR-CHECKOUT	WEB-API-046	WEB-FR-CHK-01-AC-01, WEB-FR-CHK-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart
WEB-FR-CHK-02 ← FR-CHK-02	WEB-BR-21	UC-WEB-11	WEB-SCR-CHECKIN	WEB-API-047	WEB-FR-CHK-02-AC-01, WEB-FR-CHK-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-CHK-03 ← FR-CHK-03	WEB-BR-21	UC-WEB-11	WEB-SCR-CHECKIN, WEB-SCR-CHECKOUT	WEB-API-044	WEB-FR-CHK-03-AC-01, WEB-FR-CHK-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart
WEB-FR-CHK-04 ← FR-CHK-04	WEB-BR-21	UC-WEB-11	WEB-SCR-CHECKIN	WEB-API-044, WEB-API-047	WEB-FR-CHK-04-AC-01, WEB-FR-CHK-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart
WEB-FR-CHK-05 ← FR-CHK-05	WEB-BR-21	UC-WEB-11	WEB-SCR-TICKET, WEB-SCR-CHECKIN, WEB-SCR-RECORD-BOOKINGS	WEB-API-046, WEB-API-050	WEB-FR-CHK-05-AC-01, WEB-FR-CHK-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/CheckInTests.cs; Cinema_Mobile/test/check_in_test.dart
WEB-FR-NOTI-01 ← FR-NOTI-01	WEB-BR-11, WEB-BR-13	UC-WEB-12	WEB-SCR-NOTIFICATIONS	WEB-API-078, WEB-API-079	WEB-FR-NOTI-01-AC-01, WEB-FR-NOTI-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-NOTI-02 ← FR-NOTI-02	WEB-BR-11, WEB-BR-13	UC-WEB-12	WEB-SCR-COMPOSE	WEB-API-080	WEB-FR-NOTI-02-AC-01, WEB-FR-NOTI-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-03 ← FR-NOTI-03	WEB-BR-11, WEB-BR-13	UC-WEB-12	WEB-SCR-COMPOSE	WEB-API-073	WEB-FR-NOTI-03-AC-01, WEB-FR-NOTI-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-04 ← FR-NOTI-04	WEB-BR-11, WEB-BR-13	UC-WEB-12	WEB-SCR-NOTIFICATION	WEB-API-075	WEB-FR-NOTI-04-AC-01, WEB-FR-NOTI-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-05 ← FR-NOTI-05	WEB-BR-11, WEB-BR-13	UC-WEB-12		WEB-API-074	WEB-FR-NOTI-05-AC-01, WEB-FR-NOTI-05-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-NOTI-06 ← FR-NOTI-06	WEB-BR-11, WEB-BR-13	UC-WEB-12		WEB-API-078	WEB-FR-NOTI-06-AC-01, WEB-FR-NOTI-06-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-07 ← FR-NOTI-07	WEB-BR-11, WEB-BR-13	UC-WEB-12		WEB-API-026, WEB-API-033, WEB-API-042	WEB-FR-NOTI-07-AC-01, WEB-FR-NOTI-07-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-08 ← FR-NOTI-08	WEB-BR-11, WEB-BR-13	UC-WEB-12		WEB-API-078	WEB-FR-NOTI-08-AC-01, WEB-FR-NOTI-08-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart
WEB-FR-NOTI-09 ← FR-NOTI-09	WEB-BR-11, WEB-BR-13	UC-WEB-12	WEB-SCR-HOME, WEB-SCR-NOTIFICATIONS, WEB-SCR-NOTIFICATION	WEB-API-072, WEB-API-076, WEB-API-077	WEB-FR-NOTI-09-AC-01, WEB-FR-NOTI-09-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/ShowtimeReminderTests.cs; Cinema_Mobile/test/notification_read_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-DASH-01 ← FR-DASH-01	WEB-BR-09, WEB-BR-23	UC-WEB-13	WEB-SCR-STAFF-DASH, WEB-SCR-ADMIN-DASH	WEB-API-019, WEB-API-048, WEB-API-051	WEB-FR-DASH-01-AC-01, WEB-FR-DASH-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_Mobile/test/today_at_cinema_test.dart
WEB-FR-DASH-02 ← FR-DASH-02	WEB-BR-09, WEB-BR-23	UC-WEB-13	WEB-SCR-STAFF-DASH, WEB-SCR-ADMIN-DASH	WEB-API-019, WEB-API-051	WEB-FR-DASH-02-AC-01, WEB-FR-DASH-02-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_Mobile/test/today_at_cinema_test.dart
WEB-FR-DASH-03 ← FR-DASH-03	WEB-BR-09, WEB-BR-23	UC-WEB-13	WEB-SCR-STAFF-DASH	WEB-API-052, WEB-API-093	WEB-FR-DASH-03-AC-01, WEB-FR-DASH-03-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_Mobile/test/today_at_cinema_test.dart
WEB-FR-DASH-04 ← FR-DASH-04	WEB-BR-09, WEB-BR-23	UC-WEB-13	WEB-SCR-ADMIN-DASH	WEB-API-040, WEB-API-048, WEB-API-069, WEB-API-094	WEB-FR-DASH-04-AC-01, WEB-FR-DASH-04-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/PaymentSummaryTests.cs; Cinema_Mobile/test/today_at_cinema_test.dart

Continued on next page

Table 198 — continued

Requirement / source	Business rules	Use case	Screens	API	AC / test evidence
WEB-FR-NAV-01 ← Code-derived	WEB-BR-01, WEB-BR-02, WEB-BR-03, WEB-BR-24, WEB-BR-25	UC-WEB-01	WEB-SCR-LOGIN, WEB-SCR-STAFF-ONLY	—	WEB-FR-NAV-01-AC-01, WEB-FR-NAV-01-AC-02; Cinema_BE/AuthService/AuthService.Tests/Integration/AccountValidationTests.cs; Cinema_BE/AuthService/AuthService.Tests/Integration/AuthFlowTests.cs; Cinema_Mobile/test/account_lock_flow_test.dart
WEB-FR-STATE-01 ← Code-derived	WEB-BR-07, WEB-BR-08, WEB-BR-14, WEB-BR-15, WEB-BR-16, WEB-BR-17, WEB-BR-27, WEB-BR-28, WEB-BR-29	UC-WEB-07		—	WEB-FR-STATE-01-AC-01, WEB-FR-STATE-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/BookingFlowTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/ConcessionTests.cs; Cinema_BE/BookingService/BookingService.Tests/Integration/LoyaltyTests.cs
WEB-FR-MEDIA-01 ← Code-derived	WEB-BR-26	UC-WEB-03	WEB-SCR-HOME	WEB-API-064	WEB-FR-MEDIA-01-AC-01, WEB-FR-MEDIA-01-AC-02; Cinema_BE/BookingService/BookingService.Tests/Integration/MovieReviewTests.cs; Cinema_Mobile/test/age_rating_test.dart; Cinema_Mobile/test/coming_soon_test.dart

23. Constraints

Table 199: 23. Constraints

ID	Constraint / open item
WEB-CON-01	The .NET implementation is an API backend. A separate MVC/Razor/Blazor course UI is [TO BE CONFIRMED].
WEB-CON-02	DB submodule absent/uninitialized; exact SQL DDL/seed and live migration state [TO BE CONFIRMED]. Local EF/migrations are the available evidence.
WEB-CON-03	Live merchant keys, public webhook, SMTP and deployment URLs are environment dependencies. No financial transaction or email was initiated for documentation.
WEB-CON-04	Physical OS/device acceptance, signing/store release, SLA, backup retention and load targets [TO BE CONFIRMED]. Platform scaffolding is not a compatibility certificate.
WEB-CON-05	Android emulator gateway defaults to http://10.0.2.2:5000; other local targets http://localhost:5000. Phones need reachable API_BASE_URL; ADMIN_URL/CUSTOMER_URL/PAYMENT_RETURN_URL depend on deployment.

24. Out of Scope

Exclude native-only device work, store distribution and a hypothetical `.NET`-rendered `UI`. Native phone check-in belongs to Mobile; browser camera check-in belongs here.

No implemented `FCM` push or automatic bank-refund integration is claimed. Payroll, inventory and `AI` recommendations are not added. Existing concessions, promotions/voucher codes, membership/loyalty and reviews remain in scope where actually exposed by code.

25. Future Enhancements

Table 200: 25. Future Enhancements

ID	Future work, not a current FR	Reason
WEB-FUT-01	Load more owned-ticket pages	Only first 50 currently returned to UI
WEB-FUT-02	FCM push if approved	Current notifications are email/API history
WEB-FUT-03	Automated refund if a suitable provider is chosen	Current flow records an external manual transfer

26. Conclusion

This specification allocates one system to the course/client boundary, preserves language parity and makes `API`-only functions, shared dependencies and unverified gaps explicit. It supports test-case design and explanation of the actual implementation.

27. Implemented vs Requirement

Status vocabulary: Implemented; Partially Implemented; UI Only; Mock Data; Backend Pending; Not Implemented; To Be Confirmed. Only applicable statuses are used. Production dependencies use AppDependencies.remote; test fakes are not production mock data.

Table 201: 27. Implemented vs Requirement

Feature	SRS Requirement	Code Status	API Status	Notes
Register	WEB-FR-AUTH-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Sign in	WEB-FR-AUTH-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
JWT claims	WEB-FR-AUTH-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Validate JWT	WEB-FR-AUTH-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Authorize roles	WEB-FR-AUTH-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Unauthenticated and forbidden responses	WEB-FR-AUTH-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Verify email OTP	WEB-FR-AUTH-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Repeat unverified registration	WEB-FR-AUTH-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Recover password	WEB-FR-AUTH-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Refresh and sign out	WEB-FR-AUTH-10	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Current account	WEB-FR-AUTH-11	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Lock and role-change enforcement	WEB-FR-AUTH-12	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Account field validation	WEB-FR-AUTH-13	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Authentication rate limits	WEB-FR-AUTH-14	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Find member by phone	WEB-FR-AUTH-15	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Edit profile	WEB-FR-AUTH-16	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Change current password	WEB-FR-AUTH-17	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
List accounts	WEB-FR-USER-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Account details	WEB-FR-USER-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create account	WEB-FR-USER-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Change account roles	WEB-FR-USER-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Lock and unlock	WEB-FR-USER-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
List roles	WEB-FR-USER-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Paged catalogue	WEB-FR-MOV-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Movie details	WEB-FR-MOV-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Title search	WEB-FR-MOV-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Active catalogue	WEB-FR-MOV-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create movie	WEB-FR-MOV-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Edit and reactivate movie	WEB-FR-MOV-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Deactivate movie	WEB-FR-MOV-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Open trailer	WEB-FR-MOV-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Age rating	WEB-FR-MOV-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Now showing and coming soon	WEB-FR-MOV-10	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Director and cast	WEB-FR-MOV-11	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Movie reviews	WEB-FR-MOV-12	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Find rooms	WEB-FR-ROOM-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Room details	WEB-FR-ROOM-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create room and seats	WEB-FR-ROOM-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Rename room	WEB-FR-ROOM-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Delete eligible room	WEB-FR-ROOM-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Generate seats	WEB-FR-SEAT-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Seat details	WEB-FR-SEAT-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Read room map	WEB-FR-SEAT-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Delete seat API	WEB-FR-SEAT-04	Implemented	Controller exists; runtime unverified	API only; no dedicated current UI action.
Change seat type	WEB-FR-SEAT-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Paint seat types	WEB-FR-SEAT-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Change capacity	WEB-FR-SEAT-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Protect scheduled rooms	WEB-FR-SEAT-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Refresh management maps	WEB-FR-SEAT-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
List showtimes	WEB-FR-SHOW-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create showtime	WEB-FR-SHOW-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Showtime and prices	WEB-FR-SHOW-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Edit showtime	WEB-FR-SHOW-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Cancel showtime	WEB-FR-SHOW-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Open future showtimes	WEB-FR-SHOW-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Showtimes by movie	WEB-FR-SHOW-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Bookable movie showtimes	WEB-FR-SHOW-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Date range search	WEB-FR-SHOW-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Open and close sales	WEB-FR-SHOW-10	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Reject overlap	WEB-FR-SHOW-11	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Showtime seat/ customer map	WEB-FR-SHOW-12	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Screening format	WEB-FR-SHOW-13	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Hold booking	WEB-FR-BOOK-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Booking details	WEB-FR-BOOK-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Management booking search	WEB-FR-BOOK-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Own ticket history	WEB-FR-BOOK-04	Partially Implemented	Controller exists; runtime unverified	Only the first 50 tickets are loaded; older tickets are missing from the UI.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Confirm paid booking	WEB-FR-BOOK-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Cancel booking	WEB-FR-BOOK-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Occupied seat IDs	WEB-FR-BOOK-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Concurrent seat conflict	WEB-FR-BOOK-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Sweetbox pair	WEB-FR-BOOK-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Expire unpaid hold	WEB-FR-BOOK-10	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Refresh customer map	WEB-FR-BOOK-11	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Showtime booking statistics	WEB-FR-BOOK-12	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Concessions with tickets	WEB-FR-BOOK-13	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Concessions administration	WEB-FR-BOOK-14	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Promotion and voucher	WEB-FR-BOOK-15	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Promotion administration	WEB-FR-BOOK-16	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Membership points	WEB-FR-BOOK-17	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Member administration	WEB-FR-BOOK-18	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Share and calendar	WEB-FR-BOOK-19	Partially Implemented	Controller exists; runtime unverified	Calendar uses actual show end or a 150-minute fallback; sharing falls back to clipboard.
Create/reuse PayOS link	WEB-FR-PAY-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Verify PayOS	WEB-FR-PAY-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Payment details	WEB-FR-PAY-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Own payment history	WEB-FR-PAY-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Payments by booking	WEB-FR-PAY-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Refund access	WEB-FR-PAY-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Signed webhook	WEB-FR-PAY-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Late payment	WEB-FR-PAY-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Management payments	WEB-FR-PAY-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Revenue summary	WEB-FR-PAY-10	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Automatic payment polling	WEB-FR-PAY-11	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create refund request	WEB-FR-REF-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Management refund list	WEB-FR-REF-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Record completed transfer	WEB-FR-REF-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Reject refund	WEB-FR-REF-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Reopen rejected refund	WEB-FR-REF-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Own refund status	WEB-FR-REF-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Unique refund per payment	WEB-FR-REF-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Create counter sale	WEB-FR-POS-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Counter checkout	WEB-FR-POS-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Cash and change	WEB-FR-POS-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Bank-transfer QR	WEB-FR-POS-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Switch transfer to cash	WEB-FR-POS-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Retain counter email	WEB-FR-POS-06	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Print and admit	WEB-FR-POS-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Reopen paid ticket	WEB-FR-POS-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Ticket QR	WEB-FR-CHK-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Lookup ticket code	WEB-FR-CHK-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Check-in validation	WEB-FR-CHK-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Camera/manual scanner	WEB-FR-CHK-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Display check-in time	WEB-FR-CHK-05	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Notification history	WEB-FR-NOTI-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Compose booking message	WEB-FR-NOTI-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Send/retry message	WEB-FR-NOTI-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Notification details	WEB-FR-NOTI-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Delete notification API	WEB-FR-NOTI-05	Implemented	Controller exists; runtime unverified	API only; no dedicated current UI action.

Continued on next page

Table 201 — continued

Feature	SRS Requirement	Code Status	API Status	Notes
Event emails	WEB-FR-NOTI-06	Partially Implemented	Controller exists; runtime unverified	Event deduplication exists; SMTP/DB crash window prevents absolute exactly-once delivery.
OTP and welcome email	WEB-FR-NOTI-07	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Showtime reminder	WEB-FR-NOTI-08	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Read/unread notifications	WEB-FR-NOTI-09	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Today totals	WEB-FR-DASH-01	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Live showtime board	WEB-FR-DASH-02	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Staff attention queue	WEB-FR-DASH-03	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Admin monthly overview	WEB-FR-DASH-04	Implemented	Controller exists; runtime unverified	Static implementation evidence; runtime acceptance not executed.
Route guards and recovery	WEB-FR-NAV-01	Implemented	Client/platform; no dedicated API	Static implementation evidence; runtime acceptance not executed.
Loading, error and empty states	WEB-FR-STATE-01	Partially Implemented	Client/platform; no dedicated API	State widgets exist; exhaustive device/error-path verification was not run.
Platform-specific movie media	WEB-FR-MEDIA-01	Implemented	Controller exists; runtime unverified	Web preview and poster fallback exist; autoplay depends on browser.

Table 202: 27. Implemented vs Requirement

Feature	SRS Requirement	Code Status	API Status	Notes
.NET-rendered UI	WEB-CON-01	Not Implemented	Existing ASP.NET API	Flutter Web UI only; course necessity [TO BE CONFIRMED]
FCM push	WEB-FUT-02	Not Implemented	—	Future work, not accepted current FR
DB submodule SQL verification	WEB-CON-02	To Be Confirmed	Local EF mappings exist	Submodule uninitialized; no live DB check

28. Evidence and source reconciliation

Table 203: 28. Evidence and source reconciliation

Evidence	Use / reconciliation
docs/SRS_EN.md; docs/SRS_VI.md	125 legacy FR rows per language, scoped with suffix IDs; NAV/STATE/MEDIA refine code behaviour
Cinema_BE/Cinema_Microservice_SRS_Complete.md	Historical six-service intent; current three services take precedence
README.md; Cinema_Mobile/README.md; DEMO.md	Actual entry points/setup, no separate .NET UI
Cinema_Mobile/lib/app/routes/app_router.dart; Cinema_Mobile/lib/admin/admin_router.dart	Channel routes/role guards; native management only check-in
Cinema_Mobile/lib/app/app_dependencies.dart; Cinema_Mobile/lib/core/network/api_client.dart	Remote dependencies, 15-second timeout, refresh and public-session probe
Cinema_BE/ApiGateway/ApiGateway/appsettings.json	Actual route policies/rates/health; old prose route count is stale
.github/workflows/ci.yml; .github/workflows/flutter.yml	Declared CI; Flutter pin 3.47.4 supersedes older environment prose; old test counts are unverified current totals
Cinema_Mobile/lib/core/widgets/trailer_preview_stub.dart; Cinema_Mobile/lib/core/widgets/trailer_player_native.dart	Native preview stub is empty; full trailer opens externally
Cinema_Mobile/lib/features/booking/widgets/ticket_share_actions.dart	Calendar end fallback 150 minutes, not unconditional exact show end
Cinema_BE/Booking Service/Booking Service/Services/Implementations/NotificationService.cs	Event deduplication and non-atomic SMTP/DB delivery; no absolute exactly-once claim

Review inventory: 763 files enumerated; 594 relevant text/code/configuration files indexed, including approximately 64,397 C#/Dart lines. Generated build/cache/vendor folders, binary media, encrypted secret payloads and missing DB-submodule contents are excluded from functional evidence. No secret values are reproduced.